Counterparty Risk in a CDS Portfolio: CVA, Netting and Contagion

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Chapter 1

Introduction

1.1 Motivation

Consider the buyer of protection in a credit default swap. If the protection seller defaults before the contract matures, the buyer loses the protection and must replace it with a new contract at the credit spread of the reference entity prevailing at that date (McNeil et al., 2015, Section 17.1.1). If that spread has widened since the contract was entered, the new contract carries a higher spread than the one it replaces: the contract has a positive value for the buyer, who recovers only a fraction of it from the seller. The credit value adjustment is the value today of this loss, which arises only when the value of the contract is positive at the seller's default (Section 3.2). If the default of the seller itself raises the credit spread of the reference entity, the loss becomes larger exactly in the event that causes it: contagion "might lead to a substantial increase in the credit spread of the reference entity and hence in turn to a much higher replacement value for the CDS" (McNeil et al., 2015, Section 17.4.4). The exposure then increases in the event in which the counterparty defaults, which is wrong way risk (Definition 3.1).

After the crisis of 2007–2009, in which a large part of the counterparty credit losses were mark-to-market losses caused by the deterioration of the credit quality of counterparties, the Basel III framework introduced a capital charge against the variability of the adjustment. Banks must therefore model the joint evolution of their exposures and of the credit quality of their counterparties, and in such models wrong way risk becomes a critical factor (Ceci, 2025, Part 4, Section 1). The formula that treats the exposure and the default of the counterparty as independent needs only the law of the counterparty's default time and the expected exposure, and is much easier to evaluate than the adjustment itself; but the independence is often difficult to justify, and the formula can be misleading (McNeil et al., 2015, Section

17.2.1). Under wrong way risk it understates the adjustment, by an amount that depends on the model that generates the dependence (Section 3.3).

Contagion is one mechanism that generates such dependence (McNeil et al., 2015, Section 17.1.2), and in the situation above it acts through the credit spread of the reference entity. There is substantial evidence for contagion effects: the default of Lehman Brothers, together with the worsening crisis, sent credit spreads to unprecedented levels (McNeil et al., 2015, Section 17.1.2). In historical default patterns, on the other hand, Lando and Nielsen (2010) find no strong evidence of default contagion (McNeil et al., 2015, Notes and Comments to Section 17.1).

The thesis computes two things, in the model of Bo and Ceci (2020): how much contagion changes the adjustment of a portfolio of CDS over that of the same model without contagion, which is an adjustment under independence; and how contagion changes the reduction of the adjustment obtained by netting the portfolio.

1.2 A model of contagion

In models with interacting intensities the impact of the default of one firm on the default intensities of the surviving firms is specified explicitly (McNeil et al., 2015, Section 17.1.2). Bottom-up models of this kind are considered by Davis and Lo (2001), Jarrow and Yu (2001), Yu (2007), Frey and Backhaus (2008) and Herbertsson (2008) (McNeil et al., 2015, Notes and Comments to Section 17.1), and Frey and Rösler (2014) study the effect of contagion on the credit value adjustment of a single CDS, in a model with incomplete information (McNeil et al., 2015, Section 17.4.4). Bo and Ceci (2020, Section 1) propose a model with interacting intensities for a portfolio of credit derivatives traded between a default-free investor and a defaultable counterparty. At the default of name j the intensity of every surviving name i jumps upwards by a fixed weight w_{ij}, and in their square-root specification it then reverts to its long-run level (Bo and Ceci, 2020, Section 2.4). The model extends that of Bo et al. (2019), and the decay of the impact of a default over time is documented in historical default data by Azizpour et al. (2018) (Bo and Ceci, 2020, Section 1). Bo and Ceci (2020, Section 3) also derive a hedging strategy for the adjustment, which is not treated here.

The model has four properties that the two computations of Section 1.1 require. First, it produces wrong way risk through the event at which the adjustment is computed. At the default of the counterparty the intensities of the reference entities jump by the weights w_{iN+1}, and every contract of the netting set is valued at the raised intensities (5.33): the default that triggers the loss also changes its size. With

conditionally independent defaults, a default leaves the intensities of the other names unchanged, and the exposure has no such jump (Section 4.6).

Second, it remains computable. The intensities and the default state together form a Markov process (Proposition 5.3), and the prices and the adjustment solve recursive systems of Cauchy problems over the default states (Propositions 5.15 and 5.19), which are solved numerically at one and two reference entities.

Third, it contains its own benchmark. Each intensity is driven by its own Brownian motion (5.11), so that between defaults the intensities move independently and the names are linked only by the jumps. Correlated Brownian motions would link the names as well, but through a common noise acting at every instant rather than at a default (Remark 5.6). With the contagion weights set to zero the default times are independent, and with constant loss rates the adjustment is the one computed under independence (Proposition 6.2). The effect of contagion is then measured by comparing two configurations of the same model.

Fourth, it values the netting set as a whole. The contracts are revalued together at the raised intensities, so that contagion changes the offset between values of opposite sign on which netting acts; with a single contract netting has no effect (Remark 5.22).

1.3 Outline and results

The thesis specifies the model, derives the equations solved by the prices and by the adjustment, and solves them numerically. No market data are used: the parameters of the experiments are illustrative, and the contagion weights are varied rather than estimated.

Chapter 2 presents reduced-form credit risk and the CDS, Chapter 3 the credit value adjustment and wrong way risk, and Chapter 4 the adjustment of a CDS portfolio with netting when defaults are conditionally independent.

Chapter 5 presents the model with interacting intensities (Section 5.1) and specializes its defaultable claims to the portfolio of CDS (Section 5.2). The prices solve recursive Cauchy systems over the default states, in which the functions after a default are read at the intensities shifted by the contagion weights (Proposition 5.15). The chapter writes the exposure at the default of the counterparty through the pricing functions (5.33), identifies the adjustment with the solution of a Cauchy problem of its own (Proposition 5.19), and bounds the netted adjustment by the adjustment computed contract by contract (5.37). At one reference entity it gives the prices in closed form (Corollary 5.20) and proves that the adjustment of a bought

position is non-decreasing in the weight by which the default of the counterparty raises the intensity of the reference entity (Proposition 5.21).

Chapter 6 solves the systems by an explicit finite-difference scheme built on the positive coefficient condition of Forsyth and Labahn (2007, Condition 4.1) and Wang and Forsyth (2008, Condition 3.1). Its monotonicity is proved under a bound on the time step, and since the convergence results of those articles do not cover it, its error is measured (Section 6.1). The chapter validates the scheme at one reference entity against the closed forms and a Monte Carlo simulation (Section 6.3). At two reference entities (Section 6.4) it compares the adjustment with contagion against the benchmark of Proposition 6.2, measures separately the two channels of wrong way risk of the model, contagion in the intensities and loss rates that depend on the default state (Remark 5.11), measures the effect of contagion on the reduction due to netting (Section 6.6), and varies the contagion weights, as a whole and block by block (Section 6.7). At the parameters of the experiment, for protection bought from the counterparty on both reference entities, contagion raises the adjustment by 150.5% (Table 6.4), and the reduction due to netting falls from 26.6% to 7.5% (Table 6.6). The direction of the effect is a theorem at one reference entity and a property of the configurations computed at two.

The exposure formula and the Cauchy problem of the adjustment are in Bo and Ceci (2020, Lemma 3.6 and Sections 3.5 and 4.1). The problem is obtained there for the conditional expectation of the loss at the counterparty's default under their measure for hedging, an expectation that, with zero risk premia and before that default, is the adjustment. The derivation of that problem from the definition of the adjustment, the comparison with the adjustment computed contract by contract, the direction of the contagion effect, the case without contagion and the numerical solution are not treated in Bo and Ceci (2020) and are presented in this thesis.

Chapter 7 states what the results establish and what they do not.

1.4 Scope

The adjustment is unilateral: the investor is default-free, and no debit value adjustment is computed (Remark 4.2). The positions are uncollateralized, and the collateralized adjustments of McNeil et al. (2015, Section 17.2.2) are not treated. The model is specified directly under a pricing measure $\mathbb{Q}$, as in martingale modelling, in which the parameters of a pricing model are determined by calibration to observed prices (McNeil et al., 2015, Section 10.4.2). As in the applications of Bo and Ceci (2020, Section 4), the interest rate, the market price of diffusion risk and the default

risk premium are zero; with the last two at zero, the real-world and the pricing measure coincide (Bo and Ceci, 2020, Section 3.3). The model and its pricing systems are stated for any number N of reference entities and solved numerically for $N = 1$ and $N = 2$: there are 2^{N+1} default states, and the scheme carries at each of them a grid in $N + 1$ dimensions.

Chapter 2

Reduced-Form Credit Risk and Credit Default Swaps

In the *reduced-form* approach to credit risk the mechanism leading to default is left unspecified, and the default time of a firm is modelled directly as a non-negative random variable whose law is described by a hazard rate (McNeil et al., 2015, Section 10.4.1); see also Bielecki and Rutkowski (2002) and, for a general introduction to survival analysis, Cox and Oakes (1984).

2.1 Credit-risky instruments and the anatomy of a credit loss

Credit risk is the risk of a loss arising from the failure of a counterparty to honour its contractual obligations. It arises in loans and in defaultable bonds, where the exposure is essentially the principal at stake, but also — and in a considerably less transparent way — in over-the-counter (OTC) derivative transactions, where no central clearing counterparty guarantees the fulfilment of the contractual obligations.

The following example, taken from McNeil et al. (2015, Section 10.1.3), isolates two features of counterparty credit exposure. Two parties A and B enter an interest rate swap, A receiving fixed and paying floating. Suppose A defaults at some time τ_A before the maturity of the contract. If interest rates have risen since inception, the fixed payments have lost value and the swap is worth more to B; the default of A then constitutes a loss for B, of a size determined by the term structure prevailing at τ_A. If instead rates have fallen, the swap has negative value for B, who must still pay that value into the bankruptcy pool: there is no upside for B in the default of A. If B defaults first, the situation is reversed.

Two conclusions follow. The size of the counterparty credit exposure is not known a priori, since it depends jointly on the default time and on the evolution of the value of the contract up to that time; and it is not even clear which of the two parties carries the exposure. Counterparty risk has therefore to be taken into account in the valuation itself, which is what a credit value adjustment does, and to be controlled by risk-mitigation techniques such as netting and collateralization (McNeil et al., 2015, Section 10.1.3).

Credit default swaps

A credit default swap (CDS) is a contract between a protection buyer and a protection seller whose pay-offs are tied to the default of a third party, the *reference entity*. If the reference entity experiences a default event before the maturity date T of the contract, the protection seller makes a default payment to the protection buyer which mimics the loss due to the default of a bond issued by the reference entity; this is the *default payment leg*. As compensation, and until default or maturity, whichever comes first, the protection buyer makes premium payments to the protection seller; this is the *premium payment leg*. There is no initial payment, and the premium is quoted as an annualized percentage x^* of the notional of the reference asset, called the *fair* or *market-quoted* CDS spread.

A CDS is traded over the counter and is not guaranteed by a clearing house, so a CDS position can itself carry a substantial amount of counterparty risk: the protection buyer holds a credit derivative that is subject to the default of the party who sold the protection. The bailout of AIG, which had sold many protection positions, is the case in point given by McNeil et al. (2015, Section 10.1.4); on the size of the speculative CDS market and the role of these instruments in the crisis see Cont (2010) and Stulz (2010), and on the effect of collateral management on contagion in CDS markets Brunnermeier et al. (2013).

The market for CDSs written on larger corporations is nevertheless fairly liquid, and models for pricing portfolio-related credit derivatives are usually calibrated to quoted CDS spreads (McNeil et al., 2015, Section 10.1.4).

PD, LGD and EAD

Whatever the instrument, the risk of a credit loss is governed by three quantities (McNeil et al., 2015, Section 10.1.5): the *probability of default* (PD) of the obligor by a given horizon, the *loss given default* (LGD), the proportion of the exposure actually lost, the recovery being its converse $1 - \text{LGD}$, and the *exposure at default*

(EAD), the amount at stake at the default time. For a loan or a bond the exposure is essentially the principal; for an OTC derivative it is a random variable depending jointly on the default time and on the evolution of the value of the contract up to that time, as the swap example above shows. When the loss depends on the exact timing of default, the risk of default is described by the whole distribution of possible default times and not by the probability of default by a fixed horizon.

These three quantities are dependent, and it is unrealistic to model them as independent (McNeil et al., 2015, Section 10.1.5). In a period of financial distress, when default probabilities are high, asset values are depressed and recoveries are correspondingly low, so that PD and LGD are positively dependent.

2.2 Hazard rate models

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let τ be a random time on it, that is, an $\mathcal{F}$-measurable random variable with values in $[0, \infty]$, interpreted as the default time of a company. Write $F(t) = \mathbb{P}(\tau \leqslant t)$ for its distribution function and $\bar{F}(t) = 1 - F(t)$ for its survival function, and assume throughout that $F(0) = 0$ and that $\bar{F}(t) > 0$ for every $t < \infty$. The associated *default indicator process* is

$$H_t = \mathbf{1}_{\{\tau \leqslant t\}}, \qquad t \geqslant 0, \tag{2.1}$$

a right-continuous process which jumps from 0 to 1 at τ; correspondingly $1 - H_t = \mathbf{1}_{\{\tau > t\}}$ is the survival indicator.

Definition 2.1 (cumulative hazard function and hazard function)**.** The function $\Gamma(t) := -\ln \bar{F}(t)$ is the *cumulative hazard function* of τ. If F is absolutely continuous with density f, the function $\gamma(t) := f(t)/\bar{F}(t)$ is the *hazard function* of τ.

By construction $\bar{F}(t) = e^{-\Gamma(t)}$, and if F admits a density then $\Gamma'(t) = \gamma(t)$ for almost every t, so that the survival function is recovered from the hazard function by

$$\bar{F}(t) = \exp\Big(-\int_0^t \gamma(s)\, ds\Big). \tag{2.2}$$

The hazard rate measures the instantaneous risk of default given survival. Being absolutely continuous, F is differentiable with $F' = f$ outside a Lebesgue-null set, and at every t outside it

$$\lim_{h \downarrow 0} \frac{1}{h}\, \mathbb{P}(\tau \leqslant t + h \mid \tau > t) = \lim_{h \downarrow 0} \frac{F(t+h) - F(t)}{h\, \bar{F}(t)} = \gamma(t). \tag{2.3}$$

For the exponential distribution the hazard rate is time independent and equal to the parameter of the distribution (McNeil et al., 2015, Section 10.4.1).

The default filtration

In this section the only observable quantity is the default event itself, so the relevant filtration is the one generated by the default indicator process,

$$\mathcal{H}_t = \sigma\big(\{H_u : u \leqslant t\}\big), \tag{2.4}$$

the default history up to and including t. By construction τ is an $\mathbb{H}$-stopping time, since $\{\tau \leqslant t\} = \{H_t = 1\} \in \mathcal{H}_t$, and $\mathbb{H}$ is the smallest filtration with this property. Pricing requires conditional expectations with respect to $\mathcal{H}_t$.

Lemma 2.2. *Every $\mathcal{H}_t$-measurable random variable is of the form $h(\tau)\mathbf{1}_{\{\tau\leqslant t\}} + c\,\mathbf{1}_{\{\tau>t\}}$ for a measurable function $h : [0,t] \to \mathbb{R}$ and a constant $c \in \mathbb{R}$. Consequently, for every integrable random variable X and every $t \geqslant 0$,*

$$\mathbb{E}\big(\mathbf{1}_{\{\tau>t\}} X \mid \mathcal{H}_t\big) = \mathbf{1}_{\{\tau>t\}}\, \frac{\mathbb{E}\big(\mathbf{1}_{\{\tau>t\}} X\big)}{\mathbb{P}(\tau > t)}. \tag{2.5}$$

Proof. The σ-algebra $\mathcal{H}_t$ is generated by the events $\{\tau \leqslant u\}$ for $u < t$ and by $\{\tau > t\}$, hence by the random variables $\tau \wedge t$ and $\mathbf{1}_{\{\tau>t\}}$; every $\mathcal{H}_t$-measurable random variable is therefore of the form $g(\tau\wedge t, \mathbf{1}_{\{\tau>t\}})$ for some measurable g, and the first claim follows on setting $h(u) := g(u,0)$ and $c := g(t,1)$. For the second claim, $\mathbb{E}(\mathbf{1}_{\{\tau>t\}} X \mid \mathcal{H}_t)$ is $\mathcal{H}_t$-measurable and vanishes on $\{\tau \leqslant t\}$, so by the first part it equals $\mathbf{1}_{\{\tau>t\}} c$ for a constant c; taking expectations gives $\mathbb{E}(\mathbf{1}_{\{\tau>t\}} X) = c\,\mathbb{P}(\tau > t)$. □

Applying (2.5) with $X = \mathbf{1}_{\{\tau>T\}}$ yields the conditional survival probability

$$\mathbb{P}(\tau > T \mid \mathcal{H}_t) = \mathbf{1}_{\{\tau>t\}}\, \frac{\bar{F}(T)}{\bar{F}(t)} = \mathbf{1}_{\{\tau>t\}} \exp\Big(-\int_t^T \gamma(s)\, ds\Big), \qquad t < T, \tag{2.6}$$

the second equality holding whenever τ admits a hazard function.

The default indicator process is non-decreasing — it is 0 before τ and 1 after — and bounded, hence a submartingale: $\mathbb{E}(H_s \mid \mathcal{H}_t) \geqslant H_t$ for $s \geqslant t$. The Doob–Meyer decomposition writes such a process uniquely as $H_t = M_t + A_t$, with (M_t) an $\mathbb{H}$-martingale and (A_t) predictable, increasing and null at 0 (Karatzas and Shreve, 1988); the process (A_t) is the *compensator* of (H_t). The following result identifies it when τ admits a hazard function, and shows it to be absolutely continuous with density $\mathbf{1}_{\{\tau>u\}}\,\gamma(u)$.

Proposition 2.3. *Let τ be a random time with absolutely continuous distribution function and hazard function γ. Then*

$$M_t := H_t - \int_0^t \mathbf{1}_{\{\tau>u\}}\,\gamma(u)\,du, \qquad t \geqslant 0, \tag{2.7}$$

is an $\mathbb{H}$-martingale.

Proof. Let $s > t$. By (2.6),

$$\mathbb{E}(H_s - H_t \mid \mathcal{H}_t) = \mathbf{1}_{\{\tau>t\}}\Big(1 - \frac{\bar F(s)}{\bar F(t)}\Big) = \mathbf{1}_{\{\tau>t\}}\,\frac{\bar F(t) - \bar F(s)}{\bar F(t)}.$$

The random variable $X := \int_t^s \gamma(u)\mathbf{1}_{\{u<\tau\}}\,du$ vanishes on $\{\tau \leqslant t\}$, so $X = X\mathbf{1}_{\{\tau>t\}}$. Using Lemma 2.2, Fubini's theorem and the identity $\bar F'(u) = -f(u) = -\gamma(u)\bar F(u)$,

$$\mathbb{E}(X \mid \mathcal{H}_t) = \mathbf{1}_{\{\tau>t\}}\,\frac{\int_t^s \gamma(u)\bar F(u)\,du}{\bar F(t)} = \mathbf{1}_{\{\tau>t\}}\,\frac{\bar F(t) - \bar F(s)}{\bar F(t)},$$

and the two displays coincide, which is the martingale property. □

2.3 Risk-neutral pricing and martingale modelling

All prices in this thesis are computed under a risk-neutral measure: a probability measure $\mathbb{Q}$, equivalent to the historical measure $\mathbb{P}$, under which discounted prices of traded securities are martingales. The risk-neutral pricing rule gives the price at time t of a claim with pay-off H and maturity T as

$$\mathbb{E}^{\mathbb{Q}}\Big(\exp\Big(-\int_t^T r_s\,ds\Big)H \;\Big|\; \mathcal{G}_t\Big), \tag{2.8}$$

where r_s is the continuously compounded default-free short rate and $\mathcal{G}_t$ is the information available to investors at time t.

Two justifications support (2.8), and both rest on the theorem below, stated and proved by Björk: its first part licenses (2.8) (Björk, 2009, Theorem 10.22), its second settles whether the price (2.8) returns is the only one consistent with absence of arbitrage (Björk, 2009, Theorem 10.24).

Theorem 2.4 (fundamental theorems of asset pricing)**.** *Consider a frictionless security market model on $[0, T]$.*

1. *The model is arbitrage free if and, essentially, only if it admits at least one equivalent martingale measure $\mathbb{Q}$.*

2. *An arbitrage-free model is* complete — *every claim is attainable by dynamic trading in the available assets — if and only if the equivalent martingale measure is unique.*

The first justification is Theorem 2.4(i): if a product is to be priced in accordance with no-arbitrage principles, its price must be of the form (2.8) for some $\mathbb{Q}$ (Björk, 2009, Proposition 10.25); the fundamental theorems and the conceptual underpinnings of risk-neutral pricing are treated in most textbooks on mathematical finance, see for example Delbaen and Schachermayer (2006). The second is by hedging: when the pay-off can be replicated by dynamic trading in the available assets, the cost of carrying out that hedge is given by (2.8).

Following McNeil et al. (2015, Section 10.4.2) we adopt throughout the *martingale-modelling* approach, a term coined by Björk in the context of default-free short-rate models (McNeil et al., 2015, Notes and Comments to Section 10.4). There, specifying the short-rate dynamics directly under an equivalent martingale measure $\mathbb{Q}$ determines the term structure, and with it the price of every other interest-rate derivative; specifying the dynamics under the objective measure $\mathbb{P}$ instead requires, in addition, the market price of risk connecting the two (Björk, 2009, Section 24.1). The same move is made here: the objects of interest — default times, intensities and prices of traded bonds — are modelled directly under an exogenously specified martingale measure $\mathbb{Q}$, whose parameters are then calibrated so that model prices reproduce quoted market prices. This ensures that the resulting model is arbitrage free, and it is frequently adopted in reduced-form models for credit-risky securities.

Martingale modelling has a cost (McNeil et al., 2015, Section 10.4.2). Realistic models for pricing credit derivatives are typically *incomplete*: not every claim is attainable, so that by Theorem 2.4(ii) the equivalent martingale measure is not unique and a whole interval of prices is consistent with absence of arbitrage. McNeil et al. (2015, Example 10.8) exhibits the phenomenon in a one-period model in which the loss given default may take two values: the set of equivalent martingale measures is a one-parameter family, the admissible values of a default put form an interval, and the replicating system has three equations and two unknowns, hence no solution.

In an incomplete market the risk carried by a sold claim cannot be eliminated by trading, and one has to settle for a strategy that makes the residual risk small in some prescribed sense. Quadratic hedging criteria, and local risk minimization in particular, are the route taken by Bo and Ceci (2020); they were first developed by Föllmer and Sondermann (1986) and Föllmer and Schweizer (1991), and Schweizer (2001) surveys the resulting literature. McNeil et al. (2015, Notes and Comments to Section 10.4) record a second, utility-based branch of the same literature.

2.4 Credit default swaps

The premium leg of a credit default swap is taken throughout to be a stream paid *continuously* in time rather than at discrete dates; a detailed analysis of CDS pricing is given in Schönbucher (2003). The valuation formulas below are those of Ceci (2025, Part 3, Section 1.2). We take the notional to be one, so that percentage and absolute loss given default coincide, and we write $\delta \in (0,1)$ for the loss given default of the reference entity. No specific law is assumed for τ; only the survival function $\bar{F}(t) = \mathbb{Q}(\tau > t)$ under the pricing measure is used.

Let x denote the spread understood as a fee rate paid continuously over time. The value at $t \in [0,T]$ of the premium leg is

$$V_t^{\mathrm{prem}}(x) = \mathbb{E}^{\mathbb{Q}}\Big(\int_t^T x\, e^{-r(s-t)} \mathbf{1}_{\{\tau > s\}}\, ds \;\Big|\; \mathcal{H}_t \Big) = \mathbf{1}_{\{\tau > t\}} \int_t^T x\, e^{-r(s-t)} \frac{\bar{F}(s)}{\bar{F}(t)}\, ds, \quad (2.9)$$

where the second equality uses $\mathbb{E}^{\mathbb{Q}}(\mathbf{1}_{\{\tau>s\}} \mid \mathcal{H}_t) = \mathbf{1}_{\{\tau>t\}} \bar{F}(s)/\bar{F}(t)$, that is (2.6). The value of the default leg at time 0 is

$$V_0^{\mathrm{def}} = \mathbb{E}^{\mathbb{Q}}\big(\delta\, e^{-r\tau} \mathbf{1}_{\{\tau \leqslant T\}} \big) = \delta \int_0^T e^{-rs} f(s)\, ds. \quad (2.10)$$

The fair spread x_0^* at inception equates the two legs, so that

$$x_0^* = \frac{\delta \int_0^T e^{-rs} f(s)\, ds}{\int_0^T e^{-rs} \bar{F}(s)\, ds}. \quad (2.11)$$

For $\tau > t$, the value of a CDS entered at time 0 with spread x_0^* is

$$p_{\mathrm{CDS}}(t,T) = V_t^{\mathrm{def}} - V_t^{\mathrm{prem}}(x_0^*) = \big(x_t^* - x_0^*\big) \int_t^T e^{-r(s-t)} \frac{\bar{F}(s)}{\bar{F}(t)}\, ds\, \mathbf{1}_{\{\tau>t\}}, \quad (2.12)$$

where x_t^* is the fair spread quoted at t, characterized by $V_t^{\mathrm{prem}}(x_t^*) = V_t^{\mathrm{def}}$. The position is therefore worth a positive amount to the protection buyer precisely when the spread has widened since inception, and a negative amount when it has tightened.

2.5 Doubly stochastic default times

Definition 2.1 takes the hazard function deterministic. Letting it be random requires a background process — interest rates, indicators of economic activity, or a latent factor — generating a filtration $\mathbb{F} = (\mathcal{F}_t)$, and an information structure combining

that background information with the default history,

$$\mathcal{G}_t = \mathcal{F}_t \vee \mathcal{H}_t, \qquad t \geqslant 0, \tag{2.13}$$

the smallest σ-algebra containing both. The default time is a $\mathbb{G}$-stopping time; it is not assumed to be an $\mathbb{F}$-stopping time, since default is not predictable from the background information alone.

Definition 2.5 (doubly stochastic random time)**.** A random time τ is *doubly stochastic* if there is a positive $\mathbb{F}$-adapted process (γ_t) such that $\Gamma_t = \int_0^t \gamma_s \, ds$ is strictly increasing and finite for every $t > 0$ and such that

$$\mathbb{P}(\tau > t \mid \mathcal{F}_\infty) = \exp\Big(-\int_0^t \gamma_s \, ds\Big), \qquad t \geqslant 0. \tag{2.14}$$

The process (γ_t) is the $\mathbb{F}$-conditional hazard process of τ.

Conditioning on $\mathcal{F}_\infty$ means knowing the entire trajectory of the background process, past and future. Condition (2.14) therefore says that, given the economic environment, τ is an ordinary random time with deterministic hazard function $s \mapsto \gamma_s(\omega)$: all the randomness beyond that of the environment is the exogenous randomness of a single exponential threshold. This is made explicit by the following construction, which is also the one used for simulation (McNeil et al., 2015, Section 10.5.1).

Definition 2.6 (generalized inverse)**.** For a nondecreasing, right-continuous function $g : [0, \infty) \to \mathbb{R}_+$, the *generalized inverse* $g^{\leftarrow}$ is

$$g^{\leftarrow}(y) := \inf\{t \geqslant 0 : g_t \geqslant y\}, \qquad y \geqslant 0, \tag{2.15}$$

with $\inf \emptyset := \infty$.

Lemma 2.7. *Let X be a standard exponentially distributed random variable independent of $\mathcal{F}_\infty$ and let (γ_t) be positive, $\mathbb{F}$-adapted, with Γ strictly increasing and finite. Then*

$$\tau := \Gamma^{\leftarrow}(X) = \inf\{t \geqslant 0 : \Gamma_t \geqslant X\} \tag{2.16}$$

is doubly stochastic with conditional hazard process (γ_t). Conversely, if τ is doubly stochastic then Γ_τ is standard exponential and independent of $\mathcal{F}_\infty$.[1]

[1]The converse implicitly assumes $\Gamma_\infty = \infty$ almost surely, equivalently $\tau < \infty$ almost surely: absent that, Γ_τ is exponential censored at Γ_∞, not standard exponential. The square-root diffusions of Section 2.7 satisfy it automatically when used as a hazard rate, being mean reverting and strictly positive by the Feller condition (2.24), so that $\int_0^t \Psi_s \, ds \to \infty$ almost surely.

Proof. By construction $\{\tau > t\} = \{\Gamma_t < X\}$. Since Γ_t is $\mathcal{F}_\infty$-measurable and X is independent of $\mathcal{F}_\infty$, $\mathbb{P}(\tau > t \mid \mathcal{F}_\infty) = \mathbb{P}(\Gamma_t < X \mid \mathcal{F}_\infty) = e^{-\Gamma_t}$, which is (2.14). The converse is McNeil et al. (2015, Lemma 10.12). □

Lemma 2.7 translates directly into a simulation algorithm: generate a trajectory of the hazard process, generate an independent unit exponential threshold, and set $\tau = \Gamma^{\leftarrow}(X)$ (McNeil et al., 2015, Algorithm 10.13); for simulation techniques for stochastic processes see Glasserman (2003).

The compensator result of Proposition 2.3 extends to this setting and leads to the notion of default intensity.

Proposition 2.8. *Let τ be doubly stochastic with conditional hazard process (γ_t). Then $M_t := H_t - \int_0^{t\wedge\tau} \gamma_s \, ds$ is a $\mathbb{G}$-martingale. More generally, a non-negative $\mathbb{G}$-adapted process (λ_t) is called a* $\mathbb{G}$-intensity *of τ if $H_t - \int_0^{t\wedge\tau} \lambda_s \, ds$ is a $\mathbb{G}$-martingale; the intensity is uniquely defined on $\{t < \tau\}$.*

The martingale property is proved by conditioning on $\mathcal{F}_\infty$: given the background trajectory, τ is a random time with deterministic hazard function, so Proposition 2.3 makes (M_t) a martingale for the filtration $\mathcal{F}_\infty \vee \mathcal{H}_t$, and (M_t) is $\mathbb{G}$-adapted with $\mathcal{G}_t \subset \mathcal{F}_\infty \vee \mathcal{H}_t$ (McNeil et al., 2015, Proposition 10.14); the notion of intensity is McNeil et al. (2015, Definition 10.15). The $\mathbb{G}$-intensity of a doubly stochastic random time is therefore its $\mathbb{F}$-conditional hazard process. More general reduced-form models, in which the default time is not doubly stochastic, are discussed for example in Kusuoka (1999), Elliott et al. (2000) and Collin-Dufresne et al. (2004).

Remark 2.9 (intensity does not imply doubly stochastic)**.** There exist random times that admit an intensity in the sense of Proposition 2.8 but are not doubly stochastic, and for which the pricing formulas of Theorem 2.12 do not hold.

2.6 General credit derivatives: three building blocks

Section 2.4 priced a CDS by decomposing its cash flows into elementary pieces. That decomposition is general. Following McNeil et al. (2015, Definition 10.18), the pay-off of a large class of credit-risky securities is built out of three blocks:

Definition 2.10 (building blocks)**.** Let T be a maturity date.

1. A *survival claim* is an $\mathcal{F}_T$-measurable promised payment X made at T if no default has occurred; its actual pay-off is $X\mathbf{1}_{\{\tau>T\}}$.

2. A *risky dividend stream* is a promised dividend stream with $\mathbb{F}$-adapted rate process ν_s, $0 \leqslant s \leqslant T$, whose payments stop at default, so that the actual payments occur at rate $\nu_t \mathbf{1}_{\{\tau > t\}}$.

3. A *payment-at-default claim* is a payment $Z_\tau \mathbf{1}_{\{\tau \leqslant T\}}$ made directly at the default time, where Z is an $\mathbb{F}$-adapted process.

A CDS is a risky dividend stream, paid by the protection buyer at rate x, against a payment-at-default claim with $Z \equiv \delta$, paid by the protection seller; a defaultable zero-coupon bond is a survival claim plus a payment-at-default claim (McNeil et al., 2015, Section 10.5.2).

The valuation of all three blocks reduces to a computation in a default-free market, at the cost of replacing the short rate by the *adjusted rate* $R := r + \gamma$. This is the content of the theorem below, stated for a doubly stochastic default time in the sense of Definition 2.5; with γ deterministic, $\mathbb{F}$ is trivial and $\mathbb{G} = \mathbb{H}$.

One ingredient is needed first. Lemma 2.2, which computed conditional expectations given $\mathcal{H}_t$, has to be replaced by its counterpart for the investor filtration (2.13). Before default all information is generated by the background filtration, so a conditional expectation given $\mathcal{G}_t$ collapses to one given $\mathcal{F}_t$, renormalized by the conditional survival probability.

Proposition 2.11 (Dellacherie formula)**.** *Let τ be an arbitrary random time — not necessarily doubly stochastic — with $\mathbb{P}(\tau > t \mid \mathcal{F}_t) > 0$ for every $t \geqslant 0$. Then, for every integrable random variable X,*

$$\mathbb{E}\big(\mathbf{1}_{\{\tau > t\}} X \;\big|\; \mathcal{G}_t\big) = \mathbf{1}_{\{\tau > t\}} \, \frac{\mathbb{E}\big(\mathbf{1}_{\{\tau > t\}} X \;\big|\; \mathcal{F}_t\big)}{\mathbb{P}\big(\tau > t \;\big|\; \mathcal{F}_t\big)}. \tag{2.17}$$

Proof. The left-hand side is $\mathcal{G}_t$-measurable and vanishes on $\{\tau \leqslant t\}$. Since prior to default all information is generated by $\mathbb{F}$, for every $\mathcal{G}_t$-measurable random variable there is an $\mathcal{F}_t$-measurable one agreeing with it on $\{\tau > t\}$; this is the one measure-theoretic step of the argument; a formal proof is given in Bielecki and Rutkowski (2002, Section 5.1.1), the reference to which McNeil et al. (2015, proof of Proposition 10.16) point. Hence there is an $\mathcal{F}_t$-measurable $\widetilde{Z}$ with $\mathbb{E}(\mathbf{1}_{\{\tau > t\}} X \mid \mathcal{G}_t) = \mathbf{1}_{\{\tau > t\}} \widetilde{Z}$. Taking conditional expectations given $\mathcal{F}_t \subset \mathcal{G}_t$ gives $\mathbb{E}(\mathbf{1}_{\{\tau > t\}} X \mid \mathcal{F}_t) = \mathbb{P}(\tau > t \mid \mathcal{F}_t)\, \widetilde{Z}$, and solving for $\widetilde{Z}$ yields (2.17). □

Taking $\mathbb{F}$ trivial in (2.17) returns (2.5). The theorem below is what one obtains by applying Proposition 2.11 to each of the three building blocks in turn.

Theorem 2.12. *Suppose that, under $\mathbb{Q}$, τ is doubly stochastic with background filtration $\mathbb{F}$ and hazard process γ, and set $R_s := r_s + \gamma_s$. Assume that the random variables*

$$e^{-\int_t^T r_s ds}\,|X|, \qquad \int_t^T |\nu_s|\, e^{-\int_t^s r_u du}\, ds, \qquad \int_t^T |Z_s\gamma_s|\, e^{-\int_t^s R_u du}\, ds$$

are $\mathbb{Q}$-integrable. Then

$$\mathbb{E}^{\mathbb{Q}}\Big(e^{-\int_t^T r_s ds}\mathbf{1}_{\{\tau>T\}}X \;\Big|\; \mathcal{G}_t\Big) = \mathbf{1}_{\{\tau>t\}}\,\mathbb{E}^{\mathbb{Q}}\Big(e^{-\int_t^T R_s ds}X \;\Big|\; \mathcal{F}_t\Big), \tag{2.18}$$

$$\mathbb{E}^{\mathbb{Q}}\Big(\int_t^T \nu_s \mathbf{1}_{\{\tau>s\}} e^{-\int_t^s r_u du}\, ds \;\Big|\; \mathcal{G}_t\Big) = \mathbf{1}_{\{\tau>t\}}\,\mathbb{E}^{\mathbb{Q}}\Big(\int_t^T \nu_s e^{-\int_t^s R_u du}\, ds \;\Big|\; \mathcal{F}_t\Big), \tag{2.19}$$

$$\mathbb{E}^{\mathbb{Q}}\Big(\mathbf{1}_{\{t<\tau\leqslant T\}} e^{-\int_t^\tau r_s ds} Z_\tau \;\Big|\; \mathcal{G}_t\Big) = \mathbf{1}_{\{\tau>t\}}\,\mathbb{E}^{\mathbb{Q}}\Big(\int_t^T Z_s\gamma_s e^{-\int_t^s R_u du}\, ds \;\Big|\; \mathcal{F}_t\Big). \tag{2.20}$$

The integrability assumptions are what makes the three conditional expectations well defined. For the first identity, apply Proposition 2.11 to $X' := e^{-\int_t^T r_s ds} X \mathbf{1}_{\{\tau>T\}}$ and use (2.14) to evaluate the conditional survival probability; the second follows from the first by Fubini's theorem for conditional expectations, and the third by the same argument applied to the law of τ on $(t, T]$. The details are in McNeil et al. (2015, Theorem 10.19). The theorem is originally due to Lando (1998); related results were obtained by Jarrow and Turnbull (1995) and Jarrow et al. (1997). The three identities say that pricing a credit-risky claim amounts to pricing a corresponding default-free claim after the substitution $r \rightsquigarrow r + \gamma$, together with the passage from the investor filtration $\mathbb{G}$ to the background filtration $\mathbb{F}$ — the indicator $\mathbf{1}_{\{\tau>t\}}$ being the only trace left of the default history.

2.7 Affine intensities and the CIR square-root diffusion

To use Theorem 2.12 one must be able to evaluate conditional expectations of the form

$$\mathbb{E}\Big(e^{-\int_t^T R(\Psi_s) ds} g(\Psi_T) + \int_t^T h(\Psi_s) e^{-\int_t^s R(\Psi_u) du}\, ds \;\Big|\; \mathcal{F}_t\Big), \tag{2.21}$$

where the short rate and the hazard rate are functions of a Markovian state variable process (Ψ_t) generating $\mathbb{F}$. By the Markov property, (2.21) is a function $f(t, \Psi_t)$, and under regularity conditions f solves a parabolic terminal-value problem.

Lemma 2.13 (Feynman–Kac)**.** *Let (Ψ_t) solve $d\Psi_t = \mu(\Psi_t)\, dt + \sigma(\Psi_t)\, dW_t$ on a*

domain $D \subseteq \mathbb{R}$, with $\mu : D \to \mathbb{R}$ and $\sigma : D \to [0,\infty)$ continuous, let $R, g, h : D \to [0,\infty)$, and let f be continuous on $[0,T] \times D$, once continuously differentiable in t and twice continuously differentiable in ψ on $[0,T) \times D$, solving

$$f_t + \mu(\psi) f_\psi + \tfrac{1}{2}\sigma^2(\psi) f_{\psi\psi} + h(\psi) = R(\psi) f, \qquad f(T,\psi) = g(\psi). \tag{2.22}$$

If f satisfies the growth condition $\max_{0 \leqslant t \leqslant T} f(t,\psi) \leqslant C(1+\psi^2)$, then $f(t,\Psi_t)$ equals (2.21).

The statement is McNeil et al. (2015, Lemma 10.21)[2], where it is stated without proof. The Feynman–Kac formula is discussed, for example, in Björk (2009, Section 5.5) and, at a more technical level, in Karatzas and Shreve (1988, Section 5.7). In Björk (2009, Section 5.5), for a constant discount rate and no running source, it is proved by applying Itô's formula to the discounted solution along the diffusion and requiring the integrand of the resulting stochastic integral to be square integrable, so that the integral, a local martingale in general, is a martingale with zero expectation.

Within the class of affine models this terminal-value problem can be solved in closed form up to a system of ordinary differential equations (McNeil et al., 2015, Section 10.6); the original papers on affine models in term structure modelling are Duffie and Kan (1996) for diffusion models and Duffie et al. (2000) for jump diffusions.

A much used member of the affine class is the square-root diffusion proposed by Cox et al. (1985) as a model for the short rate of interest,

$$d\Psi_t = \kappa(\bar{\theta} - \Psi_t)\, dt + \sigma \sqrt{\Psi_t}\, dW_t, \qquad \Psi_0 = \psi > 0, \tag{2.23}$$

with $\kappa, \bar{\theta}, \sigma > 0$ and state space $D = [0,\infty)$. Two of its properties matter here (McNeil et al., 2015, Section 10.6.2). It is *mean reverting*: when Ψ_t deviates from the level $\bar{\theta}$ it is pulled back towards it, at a speed governed by κ. And it stays strictly positive provided the mean reversion is strong enough relative to the volatility: writing $\tau_0(\Psi) := \inf\{t \geqslant 0 : \Psi_t = 0\}$, one has $\mathbb{P}(\tau_0(\Psi) < \infty) = 0$ if and only if

$$\kappa\bar{\theta} \geqslant \tfrac{1}{2}\sigma^2, \tag{2.24}$$

which is the Feller condition; the mathematical properties of the model are discussed in, for example, Lamberton and Lapeyre (1996, Chapter 6.2). The condition is

[2]The lemma assumes μ and σ continuous but no growth bound on σ, and its growth condition bounds f itself rather than $|f|$; without a further hypothesis on σ a solution satisfying it can still fail to make the local martingale a true one, a strict local martingale arising already for a superlinear diffusion coefficient such as $\sigma(\psi) = \psi^2$. Chapter 5 does not invoke this lemma: Proposition 5.15 proves the martingale property directly, with its own localizing sequence.

imposed here because the process models an intensity.

Chapter 3

Counterparty Credit Risk and the Credit Value Adjustment

The previous chapter priced credit-risky claims under the assumption that the party who has promised the payments will actually make them. This chapter removes that assumption. The resulting correction to the price is the credit value adjustment.

3.1 Counterparty credit risk and the need for a dynamic model

Counterparty credit risk is the risk that a party to an over-the-counter transaction defaults before the transaction has run its course, so that the cash flows actually received differ from those promised. As Section 2.1 showed, the exposure it generates is neither known in advance nor attached a priori to one of the two parties.

Following the crisis of 2007–2009, supervisors observed that a large fraction of counterparty credit losses had been caused not by actual defaults but by mark-to-market losses associated with the deterioration of the credit quality of counterparties (Ceci, 2025, Part 4, Section 1). The Basel III framework consequently introduced a capital charge for CVA risk, requiring capital to be held against the variability of the CVA induced by movements in the credit spreads of counterparties (Basel Committee on Banking Supervision, 2011). Even in the absence of default, a widening spread generates a loss through the increase of the adjustment. From a modelling point of view the consequence is stated in Ceci (2025, Part 4, Section 1): banks must use models describing the joint evolution of exposure profiles and counterparty credit quality over time.

A model that specifies the joint distribution of the default times at a fixed date —

a copula model, say — leaves the evolution of that distribution unmodelled, and with it the dynamics of the prices it returns (McNeil et al., 2015, Section 17.1.1). What this excludes is the pricing of options on credit-risky instruments, and the adjustment for counterparty risk is one. Consider a protection buyer who has entered a CDS with some protection seller, and suppose the seller defaults during the lifetime of the contract at a moment when the credit spread of the reference entity has widened. The buyer then suffers a loss, since renewing the protection requires entering a new contract at a higher spread. Accounting for that loss means valuing

> an option on the value of the future cash flows of the CDS, with maturity date equal to the default time of the protection seller

(McNeil et al., 2015, Section 17.1.1). It is an option, so its value depends on the whole distribution of the future value of the contract and not merely on its expectation; and its maturity is a random time, the default time of a party whose credit quality is itself part of the model.

3.2 Exposure, expected exposure and the shape of the CVA

Consider a general position between an institution — which we take to be the party computing the adjustment — and a counterparty S whose default time is τ_S and whose loss given default is δ_S. Let (V_t) denote the value of the position ignoring counterparty risk, so that V_t may be of either sign.

If the counterparty defaults at $\tau_S \leqslant T$, the position is settled at that date. When $V_{\tau_S} > 0$ the institution is a creditor and recovers only the fraction $1 - \delta_S$ of what it is owed; when $V_{\tau_S} < 0$ it is a debtor and must pay the full amount. The loss caused by the default of the counterparty is therefore

$$\delta_S \, (V_{\tau_S})^+ , \tag{3.1}$$

and the random variable $(V_{\tau_S})^+$ is the *exposure at default*: a position of negative value produces no loss at the default of the counterparty, being settled in full. The *expected exposure* at a date t is

$$\mathrm{EE}(t) = \mathbb{E}^{\mathbb{Q}}\big[(V_t)^+\big]. \tag{3.2}$$

The loss (3.1) is incurred on the event $\{\tau_S \leqslant T\}$, at the date τ_S. Discounting it back to $t = 0$ and taking the risk-neutral expectation defines the *credit value*

adjustment

$$\mathrm{CVA} = \mathbb{E}^{\mathbb{Q}}\Big[\mathbf{1}_{\{\tau_S \leqslant T\}}\, e^{-\int_0^{\tau_S} r_s\, ds}\, \delta_S\, (V_{\tau_S})^+\Big], \tag{3.3}$$

whose four factors are the default indicator of the counterparty before the position has run its course, the discount factor over the random horizon τ_S, the loss given default of the counterparty and the exposure at default (McNeil et al., 2015, Section 17.2.1). For a CDS, with in addition the indicator of the event that the seller defaults before the reference entity, it is Proposition 1.1 of Ceci (2025, Part 4, Section 1.2). Nothing has been assumed about the joint law of τ_S and (V_t), and in this form the adjustment is the price of a call on (V_t) with strike 0 and maturity the random time τ_S.

Suppose now that τ_S has density f_S under $\mathbb{Q}$ and that the short rate is deterministic. Conditioning on τ_S and integrating against its density gives

$$\mathrm{CVA} = \delta_S \int_0^T e^{-rt}\, \mathbb{E}^{\mathbb{Q}}\big[(V_t)^+ \;\big|\; \tau_S = t\big]\, f_S(t)\, dt, \tag{3.4}$$

each occurrence of τ_S inside the expectation having been replaced by the value t on which one conditions. No assumption on the dependence between τ_S and (V_t) has been used so far; one is used now. If τ_S is independent of the process (V_t), the conditioning may be dropped, and (3.4) becomes

$$\mathrm{CVA}^{\mathrm{indep}} = \delta_S \int_0^T e^{-rt}\, \mathbb{E}^{\mathbb{Q}}\big[(V_t)^+\big]\, f_S(t)\, dt = \delta_S \int_0^T e^{-rt}\, \mathrm{EE}(t)\, f_S(t)\, dt. \tag{3.5}$$

The expected exposure (3.2) therefore appears as a factor of (3.5) separate from the law of τ_S only because that last step could be taken. For a credit default swap on a reference entity R with a deterministic hazard rate, as in Ceci (2025, Part 4, Section 1.2), the counterparty-risk-free price (V_t) is a function of τ_R, and the independence reduces to that of τ_R and τ_S; with a stochastic hazard rate (V_t) depends on it as well, and τ_S has to be independent of the whole process. Without the independence the integrand of (3.4) does not factorize, and the marginal law of τ_S together with the profile $t \mapsto \mathrm{EE}(t)$ no longer determines the adjustment.

On a time grid $t_0 < t_1 < \cdots < t_n = T$, the adjustment (3.5) is discretized as

$$\mathrm{CVA}^{\mathrm{indep}} \approx \delta_S \sum_i e^{-rt_i}\, \mathbb{E}^{\mathbb{Q}}\big[(V_{t_i})^+\big]\, \mathbb{Q}\big(\tau_S \in (t_{i-1}, t_i]\big), \tag{3.6}$$

a sum, over time intervals, of the discounted expected positive exposure, times the probability of default in the interval, times the loss given default (Ceci, 2025, Part 4, Section 1.2).

The dependence that the assumption rules out has a mechanism. In a model where the default of one name raises the default intensity of the surviving names — the protection seller among them, with respect to the reference entity — (Bo and Ceci, 2020, Section 2.4), the intensity of R jumps upward at τ_S, its default probability rises, the protection bought on R becomes more valuable, and the exposure $(V_{\tau_S})^+$ exceeds the exposure expected at that date without the information that S has defaulted. Exposure and counterparty default are then positively linked, which is wrong way risk.

3.3 Wrong way risk

The independence used in the step from (3.4) to (3.5) is often difficult to justify, and the simplified adjustment it produces can be misleading (McNeil et al., 2015, Section 17.2.1).

Definition 3.1 (wrong way risk)**.** *Wrong way risk* is present when the exposure to a counterparty tends to increase precisely in those states of the world in which the credit quality of that counterparty deteriorates. In the terminology of the Basel framework it refers to "cases in which exposure increases when the credit quality of the counterparty deteriorates", as quoted by Ceci (2025, Part 4, Section 1.3) from Basel Committee on Banking Supervision (2011).

In CDS markets the dependence runs in one direction. When the reference entity and the protection seller belong to the same economic sector, an adverse macroeconomic shock raises at once the default probability of the reference entity, the market value of the protection and the default probability of the seller (Ceci, 2025, Part 4, Section 1.3). If the reference entity R and the protection seller S are financial institutions and the seller defaults before maturity, that default is likely to fall in a period in which financial institutions face adverse conditions, so that the credit spread of the reference entity, and with it the market value of the protection, is comparatively high (McNeil et al., 2015, Section 17.2.1). Formally one expects

$$\mathbb{E}^{\mathbb{Q}}\big[(V_{\tau_S})^+ \,\big|\, \tau_S = t\big] > \mathbb{E}^{\mathbb{Q}}\big[(V_t)^+\big], \tag{3.7}$$

and therefore

$$\mathrm{CVA} > \mathrm{CVA}^{\mathrm{indep}}, \tag{3.8}$$

where CVA is (3.4), the adjustment computed with the conditional expected exposure (McNeil et al., 2015, Section 17.2.1); see also Brigo and Chourdakis (2009), Brigo

et al. (2014) and Frey and Rösler (2014). A related way of expressing the same unfavourable dependence, used in Ceci (2025, Part 4, Section 1.3), replaces the pointwise conditioning on $\{\tau_S = t\}$ of (3.7) by a covariance condition on the different event $\{\tau_S \leqslant t\}$:

$$\operatorname{Cov}\big((V_t)^+, \mathbf{1}_{\{\tau_S \leqslant t\}}\big) > 0. \tag{3.9}$$

The credit value adjustment computed under the independence assumption therefore systematically underestimates counterparty risk (Ceci, 2025, Part 4, Section 1.3). The inequality (3.8) does not say by how much: the size of the understatement is a property of the model that generates the dependence.

3.4 Conditionally independent doubly stochastic default times

The simplest reduced-form models of portfolio credit risk make the default times independent conditionally on a background process (McNeil et al., 2015, Section 17.1.2), extending Definition 2.5 to several names.

Consider a portfolio of m obligors with default times $\tau_1, \dots, \tau_m$ and default indicators $H_{t,i} = \mathbf{1}_{\{\tau_i \leqslant t\}}$. Denote by $T_1 < \cdots < T_m$ the ordered default times and by ξ_1 the identity of the firm defaulting at T_1. The default history of the portfolio is generated by

$$\mathcal{H}_t^i = \sigma\big(\{H_{s,i} : s \leqslant t\}\big), \qquad \mathcal{H}_t = \mathcal{H}_t^1 \vee \cdots \vee \mathcal{H}_t^m, \tag{3.10}$$

and, as in (2.13), the information available to investors is $\mathcal{G}_t = \mathcal{F}_t \vee \mathcal{H}_t$, where $\mathbb{F}$ is the filtration of a background process.

Definition 3.2 (conditionally independent doubly stochastic default times)**.** The default times $\tau_1, \dots, \tau_m$ are *conditionally independent doubly stochastic random times* if there are positive $\mathbb{F}$-adapted processes $(\gamma_{t,i})$, $1 \leqslant i \leqslant m$, with $\Gamma_{t,i} = \int_0^t \gamma_{s,i}\, ds$ strictly increasing and finite for every $t > 0$, such that for all $t_1, \dots, t_m \geqslant 0$

$$\mathbb{P}\big(\tau_1 > t_1, \dots, \tau_m > t_m \mid \mathcal{F}_\infty\big) = \prod_{i=1}^{m} \exp\Big(-\int_0^{t_i} \gamma_{s,i}\, ds\Big). \tag{3.11}$$

Each τ_i is then doubly stochastic in the sense of Definition 2.5 with conditional hazard process $(\gamma_{t,i})$, and the default times are conditionally independent given $\mathcal{F}_\infty$: all the dependence between defaults is carried by the common dependence of the intensities on the background process. The intensity of one name does not react to

the default of another, the processes $(\gamma_{t,i})$ being adapted to $\mathbb{F}$, which does not carry the default history.

The threshold construction of Lemma 2.7 extends verbatim and gives both an existence result and a simulation algorithm.

Lemma 3.3. *Let $(\gamma_{t,i})$, $1 \leqslant i \leqslant m$, be positive $\mathbb{F}$-adapted processes with $\Gamma_{t,i}$ strictly increasing and finite, and let $X_1, \dots, X_m$ be independent standard exponential random variables, independent of $\mathcal{F}_\infty$. Then the random times $\tau_i := \Gamma_i^{\leftarrow}(X_i)$ are conditionally independent doubly stochastic with hazard processes $(\gamma_{t,i})$. Conversely, if the τ_i are conditionally independent doubly stochastic, then $(\Gamma_1(\tau_1), \dots, \Gamma_m(\tau_m))$ is a vector of independent standard exponential random variables, independent of $\mathcal{F}_\infty$.*

The proof is that of Lemma 2.7 carried out coordinatewise, the mutual independence of the X_i turning the single exponential factor into the product (3.11); the converse is McNeil et al. (2015, Lemma 17.5).

For an alternative textbook-level treatment of models with conditionally independent defaults see Bielecki and Rutkowski (2002, Chapter 9); the existence of default intensities is discussed in Aven (1985) and in Janson et al. (2011, Section 2).

The credit value adjustment is a payment made at the first default among the parties to a contract. Two properties of the first default time follow from Definition 3.2.

Lemma 3.4. *Under Definition 3.2, $T_1 = \min_i \tau_i$ is a doubly stochastic random time with $\mathbb{F}$-conditional hazard process $\bar{\gamma}_t := \sum_{i=1}^m \gamma_{t,i}$.*

Proof. By conditional independence, $\mathbb{P}(T_1 > t \mid \mathcal{F}_\infty) = \prod_{i=1}^m \exp(-\int_0^t \gamma_{s,i}\, ds) = \exp(-\int_0^t \bar{\gamma}_s\, ds)$, and this expression is $\mathcal{F}_t$-measurable. □

Proposition 3.5. *Under the assumptions of Lemma 3.4,*

$$\mathbb{P}\big(\xi_1 = i \,\big|\, \mathcal{F}_\infty \vee \sigma(T_1)\big) = \frac{\gamma_{T_1,i}}{\bar{\gamma}_{T_1}}, \qquad i \in \{1, \dots, m\}. \tag{3.12}$$

The proof, given in McNeil et al. (2015, Proposition 17.7), conditions on $\mathcal{F}_\infty$ — which reduces the statement to one about independent random times with deterministic hazard functions — and then compares the probability of a default of i in a short interval $(t-h, t]$ with that of a default of any name.

3.5 Modelling wrong way risk through a common factor

Definition 3.2 becomes a model of wrong way risk as soon as the conditional hazard processes of the counterparty and of the reference entity are driven by the same background factor. The specification used in Ceci (2025, Part 4, Section 1.4) is affine: let $(\Psi_t)_{t\in[0,T]}$ be a non-negative common macroeconomic factor and set

$$\gamma_t^R = \gamma_0^R + \beta\Psi_t, \qquad \gamma_t^S = \gamma_0^S + \alpha\Psi_t, \tag{3.13}$$

with $\gamma_0^R, \gamma_0^S, \beta, \alpha$ positive constants. The joint survival probability is then

$$\begin{aligned}\mathbb{Q}(\tau_S > t, \tau_R > t) &= \mathbb{E}^{\mathbb{Q}}\big[\mathbb{Q}(\tau_S > t, \tau_R > t \mid \mathcal{F}_t)\big] = \mathbb{E}^{\mathbb{Q}}\Big[e^{-\int_0^t(\gamma_s^R+\gamma_s^S)\,ds}\Big] \\ &= e^{-(\gamma_0^R+\gamma_0^S)t}\,\mathbb{E}^{\mathbb{Q}}\Big[e^{-(\beta+\alpha)\int_0^t \Psi_s\,ds}\Big],\end{aligned} \tag{3.14}$$

so that default correlation is generated entirely through the common factor. If (Ψ_t) is taken to be an affine diffusion — a square-root process as in Section 2.7, say — the expectation in (3.14) is available in closed form.

Within the class the adjustment is computed by simulation, combining the multivariate threshold construction of Lemma 3.3 with the discretized adjustment (3.6). On a grid $0 = t_0 < t_1 < \cdots < t_n = T$ and for each of M replications:

1. simulate a trajectory of the common factor (Ψ_t), for instance by an Euler scheme;
2. evaluate the conditional hazard rates (3.13) along the trajectory and form the conditional survival probabilities $\exp(-\sum_{k<i}\gamma_{t_k}^S\Delta t)$;
3. draw an independent uniform threshold and set τ_S equal to the first grid point at which the survival probability falls below it, which is Lemma 3.3 in discrete form;
4. simulate the value process (V_t) from the same trajectory of (Ψ_t) drawn in step 1 — so that the dependence the scheme models is actually present in the simulation — forcing $V_t = 0$ once R has defaulted, and record the exposure $(V_t)^+$;
5. if default occurs at $\tau_S = t_i$, record the discounted loss $\delta_S e^{-rt_i}\,(V_{t_i})^+$.

The Monte Carlo estimator of the adjustment is the average of the recorded losses over the M replications (adapted from Ceci, 2025, Part 4, Section 1.4).

Chapter 4

CVA for a CDS Portfolio and Netting

A credit default swap is itself exposed to the default of the party that sold the protection. The adjustment carried by that exposure is computed below for a single swap, and then for a portfolio of swaps facing one counterparty.

Two conventions hold for the rest of the thesis. The adjustment is computed *unilaterally*, from the point of view of the protection buyer towards the protection seller: the buyer is the institution holding the portfolio, the seller is the counterparty, and the buyer neglects the possibility of its own default (Remark 4.2). The positions are *uncollateralized*.

4.1 The contract and its cash flows

We work on a filtered probability space $(\Omega, \mathcal{G}, \mathbb{G}, \mathbb{Q})$, where $\mathbb{Q}$ is the risk-neutral measure and $\mathbb{G}$ represents the information available to investors. Three parties are involved in a CDS on a reference entity R: the protection seller S, the protection buyer B and R itself. Their default times τ_S, τ_B, τ_R are $\mathbb{G}$-stopping times and $\delta_R, \delta_S, \delta_B$ denote their losses given default, taken constant. Since the buyer neglects its own default, τ_B enters nothing that follows — it reappears only in Remark 4.2 — and we write

$$T_1 = \min\{\tau_R, \tau_S\} \tag{4.1}$$

for the first default time and $\xi_1 \in \{R, S\}$ for the identity of the firm that defaults first, the two defaults not occurring together, $\mathbb{Q}(\tau_R = \tau_S) = 0$, under the doubly stochastic construction of Chapter 2 used from Section 4.4 on. The default-free short rate is a $\mathbb{G}$-adapted process (r_t), and we abbreviate

$$D(s_1, s_2) = \exp\Big(-\int_{s_1}^{s_2} r_u \, du\Big), \qquad 0 \leqslant s_1 \leqslant s_2 \leqslant T. \tag{4.2}$$

Following Section 2.4, the premium is paid continuously in time at the fixed rate x until the earliest of the default of R and the maturity T.

The *promised* cash flow of a protection-buyer position between two dates, discounted back to the first, is

$$\Pi(s_1, s_2) = \int_{s_1}^{s_2} D(s_1, u)\, \delta_R \, dH_{u,R} - x \int_{s_1}^{s_2} D(s_1, u)\big(1 - H_{u,R}\big)\, du, \tag{4.3}$$

the first term being the discounted default payment and the second the discounted premium payment. The value at a stopping time $\tau \geqslant t$ of the promised stream is

$$V_\tau := \mathbb{E}^{\mathbb{Q}}\big(\Pi(\tau, T) \;\big|\; \mathcal{G}_\tau\big), \tag{4.4}$$

which we call the *counterparty-risk-free CDS price*. The cash flows actually made and received by B are denoted $\Pi^{\text{real}}(s_1, s_2)$; they differ from the promised ones because S may default before T. Two scenarios arise (McNeil et al., 2015, Section 17.2.1); see also Brigo and Chourdakis (2009), Brigo et al. (2014) and Frey and Rösler (2014).

- If $T_1 > T$, or if $T_1 \leqslant T$ and $\xi_1 = R$ — that is, if the reference entity defaults no later than the seller — actual and promised streams coincide, $\Pi^{\text{real}}(\cdot, T) = \Pi(\cdot, T)$.

- If $T_1 < T$ and $\xi_1 = S$, the two streams coincide before T_1. At T_1, if $V_{T_1} > 0$, the buyer is entitled to a close-out amount, taken equal to V_{T_1}, but the seller can pay only the recovery $(1 - \delta_S)V_{T_1}$; if $V_{T_1} < 0$, the buyer must pay $|V_{T_1}|$ in full. Hence, on $\{T_1 < T\} \cap \{\xi_1 = S\}$,

$$\Pi^{\text{real}}(t, T) = \Pi(t, T_1) + D(t, T_1)\Big((1 - \delta_S)\,(V_{T_1})^+ - (V_{T_1})^-\Big). \tag{4.5}$$

The correct value of the protection-buyer position in the presence of counterparty risk is $\mathbb{E}^{\mathbb{Q}}(\Pi^{\text{real}}(t, T) \mid \mathcal{G}_t)$, and for $t < T_1$ the difference

$$\text{UCVA}_t := \mathbb{E}^{\mathbb{Q}}\big(\Pi(t, T) \;\big|\; \mathcal{G}_t\big) - \mathbb{E}^{\mathbb{Q}}\big(\Pi^{\text{real}}(t, T) \;\big|\; \mathcal{G}_t\big) \tag{4.6}$$

is the *unilateral credit value adjustment*. Equivalently $\mathbb{E}^{\mathbb{Q}}(\Pi^{\text{real}}(t, T) \mid \mathcal{G}_t) = V_t - \text{UCVA}_t$: the adjustment is what must be subtracted from the counterparty-risk-free price to obtain the value of the stream actually received.

4.2 The unilateral credit value adjustment

Proposition 4.1. *For $t < T_1$,*

$$\mathrm{UCVA}_t = \mathbb{E}^{\mathbb{Q}}\Big(\mathbf{1}_{\{T_1 \leqslant T\}}\mathbf{1}_{\{\xi_1 = S\}} D(t,T_1)\,\delta_S\,(V_{T_1})^+ \;\Big|\; \mathcal{G}_t\Big). \tag{4.7}$$

Proof. For $t < T_1 \leqslant s$ the discount factor factorizes, $D(t,s) = D(t,T_1)D(T_1,s)$, so that on $\{T_1 \leqslant T\}$ we may write

$$\Pi(t,T) = \Pi(t,T_1) + D(t,T_1)\Pi(T_1,T).$$

Subtracting (4.5) from this identity gives

$$\Pi(t,T) - \Pi^{\mathrm{real}}(t,T) = \mathbf{1}_{\{T_1 \leqslant T\}}\, D(t,T_1)\Big(\Pi(T_1,T) - \mathbf{1}_{\{\xi_1 = S\}}\big((1-\delta_S)\,(V_{T_1})^+ - (V_{T_1})^-\big)\Big), \tag{4.8}$$

the two streams coinciding on $\{\xi_1 = R\}$. Take conditional expectations given $\mathcal{G}_{T_1}$ and use the tower property. By (4.4), $\mathbb{E}^{\mathbb{Q}}(\Pi(T_1,T) \mid \mathcal{G}_{T_1}) = V_{T_1}$, and the decomposition $V_{T_1} = (V_{T_1})^+ - (V_{T_1})^-$ holds with both terms $\mathcal{G}_{T_1}$-measurable. On $\{\xi_1 = S\}$ the bracket in (4.8) therefore reduces to

$$(V_{T_1})^+ - (V_{T_1})^- - (1-\delta_S)\,(V_{T_1})^+ + (V_{T_1})^- = \delta_S\,(V_{T_1})^+,$$

and taking conditional expectations given $\mathcal{G}_t$ yields (4.7). □

The right-hand side of (4.7) is the price of a call on (V_t) with strike $K = 0$ and random maturity T_1. Evaluating it calls for the joint law of V_{T_1} and T_1, and the marginal law of either is not enough.

Remark 4.2 (why the adjustment is taken unilaterally)**.** If the default of the buyer is not neglected, the same argument applied to the third scenario $\{\xi_1 = B\}$ produces a second term, the *debt value adjustment*

$$\mathrm{DVA}_t = \mathbb{E}^{\mathbb{Q}}\Big(\mathbf{1}_{\{T_1 \leqslant T\}}\mathbf{1}_{\{\xi_1 = B\}} D(t,T_1)\,\delta_B\,(V_{T_1})^- \;\Big|\; \mathcal{G}_t\Big),$$

and the bilateral adjustment is $\mathrm{CVA}_t - \mathrm{DVA}_t$ (McNeil et al., 2015, Proposition 17.1). The derivation of the bilateral value adjustments is based on Brigo and Chourdakis (2009) and Brigo et al. (2014), see also Frey and Rösler (2014). Its use is not uncontroversial: a deterioration in the credit quality of B increases the probability that B defaults first and hence increases DVA_t, which is then reported as a profit for B, and it is unclear how such an accounting profit could ever be converted into a

cash flow for the shareholders of B (McNeil et al., 2015, Section 17.2.1). Detailed introductions to counterparty risk management are given in the textbooks of Gregory (2010), Cesari et al. (2009) and Brigo et al. (2013), and a non-technical introduction to the computation of value adjustments in Hull and White (2012) and Hull and White (2013). The unilateral convention avoids the paradox, at the price that B and S will in general disagree about the value of a given deal. All the exposures below are exposures of the buyer.

Writing (4.7) with the first-default structure made explicit — the seller must default before the reference entity, since on $\{\tau_R \leqslant t\}$ the CDS is worthless and $V_t = 0$ — gives the form in which the result appears in Ceci (2025, Part 4, Section 1.2):

$$\mathrm{UCVA}_t = \mathbb{E}^{\mathbb{Q}}\Big(\mathbf{1}_{\{\tau_S<\tau_R\}}\mathbf{1}_{\{t<\tau_S\leqslant T\}}\, D(t,\tau_S)\,\delta_S\,\left(V_{\tau_S}\right)^+ \,\Big|\, \mathcal{G}_t\Big)\mathbf{1}_{\{\tau_S>t\}}. \tag{4.9}$$

The adjustment depends on the position only through the exposure at default $(V_{\tau_S})^+$ and on the counterparty only through δ_S and the law of τ_S — jointly with that of V, which is the whole difficulty.

4.3 Evaluation under independence

Assume that the default indicator process of S and the counterparty-risk-free price (V_t) are independent, and that the risk-free rate is deterministic. Conditioning on $\{\tau_S = t\}$ then lets the expectation factorize — the indicator $\mathbf{1}_{\{\tau_S<\tau_R\}}$ being absorbed by the identity $V_s \equiv 0$ on $\{\tau_R \leqslant s\}$ — and the adjustment at $t = 0$ reduces to (3.5) of Section 3.2 with V the CDS price:

$$\mathrm{CVA}^{\mathrm{indep}} = \delta_S \int_0^T D(0,t)\,\mathbb{E}^{\mathbb{Q}}\big[(V_t)^+\big]\, f_S(t)\,dt, \tag{4.10}$$

which is the formula of Ceci (2025, Part 4, Section 1.2).

Formula (4.10) uses the marginal law of τ_S and the expected exposure profile, and nothing of their joint behaviour. Under wrong way risk it underestimates the adjustment, by (3.7).

4.4 Evaluation in a conditionally independent model

A model with conditionally independent default times, in the sense of Definition 3.2, retains dependence between the exposure and the credit quality of the counterparty and still admits a closed formula. Let τ_S, τ_R be conditionally independent doubly

stochastic under $\mathbb{Q}$ with $\mathbb{F}$-conditional hazard processes γ^S, γ^R, and let the short rate be $\mathbb{F}$-adapted.

The first step is to strip the default indicator of the reference entity out of the price. The promised stream (4.3) is, by Definition 2.10, a payment-at-default claim with $Z \equiv \delta_R$ against a risky dividend stream with rate x, both attached to the default time τ_R. Applying (2.20) and (2.19) of Theorem 2.12 to (4.4) therefore gives $V_t = \mathbf{1}_{\{\tau_R > t\}}\,\widetilde{V}_t$, the indicator being the only trace of the default history left by the theorem, with the $\mathbb{F}$-adapted process $\widetilde{V}$ given by

$$\widetilde{V}_t = \delta_R\,\mathbb{E}^{\mathbb{Q}}\Big(\int_t^T \gamma_s^R e^{-\int_t^s (r_u+\gamma_u^R)\,du}\,ds \;\Big|\; \mathcal{F}_t\Big) - x\,\mathbb{E}^{\mathbb{Q}}\Big(\int_t^T e^{-\int_t^s (r_u+\gamma_u^R)\,du}\,ds \;\Big|\; \mathcal{F}_t\Big), \quad (4.11)$$

that is, the default leg minus the premium leg of Section 2.4, evaluated at the adjusted rate $r + \gamma^R$. This is the stochastic-hazard counterpart of the valuation of Section 2.4, which assumed the hazard rate of the reference entity deterministic; the same computation is carried out in McNeil et al. (2015, Section 10.5.3).

The second step integrates out the identity of the first defaulter. By Proposition 4.1 and since $V_{T_1} = \widetilde{V}_{T_1}$ on $\{\xi_1 = S\}$, the adjustment at $t = 0$ is UCVA $= \mathbb{E}^{\mathbb{Q}}(\mathbf{1}_{\{T_1<T\}}\mathbf{1}_{\{\xi_1=S\}}D(0,T_1)\delta_S\left(\widetilde{V}_{T_1}\right)^+)$. Conditioning on $\sigma(T_1) \vee \mathcal{F}_\infty$ and applying Proposition 3.5 replaces the indicator $\mathbf{1}_{\{\xi_1=S\}}$ by the ratio $\gamma^S_{T_1}/(\gamma^S_{T_1} + \gamma^R_{T_1})$. What remains is a payment-at-default claim with payment at the stopping time T_1, which by Lemma 3.4 is doubly stochastic with hazard rate $\gamma^S + \gamma^R$; applying the third identity (2.20) of Theorem 2.12, the ratio cancels against that total hazard rate and leaves the closed form

$$\text{UCVA} = \mathbb{E}^{\mathbb{Q}}\Big(\int_0^T \gamma_t^S\,\delta_S\,\left(\widetilde{V}_t\right)^+ e^{-\int_0^t (r_s+\gamma_s^S+\gamma_s^R)\,ds}\,dt\Big). \quad (4.12)$$

In (4.12) the adjustment is an integral over possible default dates of the exposure $\left(\widetilde{V}_t\right)^+$, weighted by the intensity γ_t^S of the counterparty and discounted at the rate r augmented by the total intensity of the names involved. Dependence between the exposure and the credit quality of the counterparty enters only through the joint law of $\widetilde{V}$ and γ^S induced by the common background process of Section 3.5.

4.5 Netting

Under a legally enforceable netting agreement, the market values of all transactions between B and S are aggregated at the default of S and only the aggregate is subject to the bankruptcy procedure; perfectly offsetting transactions cancel each other out. Let there be n transactions covered by the agreement — a *netting set* — indexed by $i = 1, \dots, n$, each written by B against S on a reference entity R_i, not necessarily distinct across i, and let $V_{t,i}$ be the value at t of the i-th transaction, given by (2.12): in particular $V_{t,i} = 0$ once R_i has defaulted, the factor $\mathbf{1}_{\{\tau_{R_i} > t\}}$ already carried by that formula. The netting set has no single reference entity to pair with S in (4.1), but the unilateral adjustment does not need one: the promised and actual aggregate streams differ only through the settlement triggered by S's own default, any R_i that has already defaulted by then contributing nothing to either stream from that point on. An argument identical to the proof of Proposition 4.1, applied to the aggregate cash-flow stream and conditioned on τ_S alone, gives

$$\mathrm{UCVA}_t = \mathbb{E}^{\mathbb{Q}}\Big(\mathbf{1}_{\{t<\tau_S\leqslant T\}}\, D(t,\tau_S)\, \delta_S\, \big(\textstyle\sum_{i=1}^{n} V_{\tau_S,i}\big)^{+} \;\Big|\; \mathcal{G}_t\Big). \tag{4.13}$$

In (4.13) the positive part is applied to the sum of the values and not to each value separately. Since $x \mapsto x^{+}$ is subadditive,

$$\left(\sum_{i=1}^{n} V_{\tau_S,i}\right)^{+} \leqslant \sum_{i=1}^{n} \left(V_{\tau_S,i}\right)^{+}, \tag{4.14}$$

and therefore

$$\mathrm{UCVA}_t^{\text{netting}} \leqslant \sum_{i=1}^{n} \mathrm{UCVA}_t^{i}, \tag{4.15}$$

where UCVA_t^{i} is the adjustment of the i-th contract taken in isolation. The inequality is strict whenever the netting set holds values of both signs, which it typically does: by (2.12) a position is worth a positive amount exactly when the spread of its own reference entity has widened since inception, so contracts written on different reference entities, or bought and sold positions on the same one, compensate one another. In the illustration of Ceci (2025, Part 4, Section 1.5), with $V_{\tau_S,1} = 100$, $V_{\tau_S,2} = -30$, $V_{\tau_S,3} = 10$, the netted exposure is 80 whereas the sum of the individual exposures is 110.

By (4.13) the adjustment is the price of an option on a sum of value processes. For a portfolio of n credit default swaps, each written on a different reference entity, the pay-off depends on $n + 1$ default times jointly and does not decompose into

n one-dimensional problems: the individual options can be priced separately, the option on their sum cannot (McNeil et al., 2015, Section 17.2.1).

4.6 What conditional independence cannot deliver

In a model with conditionally independent defaults, all dependence between default events is generated by the joint dependence of the intensities on a background process (Ψ_t). The default event of one name has no effect on the intensity of the others: by Definition 3.2 the processes $\gamma_{t,i}$ are $\mathbb{F}$-adapted, and $\mathbb{F}$ does not contain the default history. The empirical record describes the opposite: there is substantial evidence for contagion effects, the default of Lehman Brothers in 2008 being an example (McNeil et al., 2015, Section 17.1.2), and when an institution defaults the default probabilities of the others increase (Ceci, 2025, Part 4, Section 1.3). Two further observations run the same way. The default correlations attainable in this class are widely held to be too low relative to those observed empirically (Ceci, 2025, Part 4, Section 1.4); and the close-out amount of a contract may jump at a default, for instance through contagion effects (McNeil et al., 2015, Section 17.2.2).

McNeil et al. (2015, Section 17.1.2) classify the reduced-form models of portfolio credit risk and name the class that does produce this behaviour: *models with interacting intensities*, in which the impact of the default of one firm on the intensities of the surviving firms is specified explicitly. The common feature of such models is *default contagion* — the intensity of a surviving firm jumps, usually upwards, on the information that another firm has defaulted, so that the credit spreads of the survivors widen at default events. Contagion may arise through direct economic links between firms, a channel which, broadly speaking, is linked to counterparty risk, or through information effects. McNeil et al. (2015, Section 17.1.2 and Notes and Comments) list the literature without developing these models, for reasons of space. Bottom-up models with interacting intensities are considered by Davis and Lo (2001), Jarrow and Yu (2001), Yu (2007), Frey and Backhaus (2008) and Herbertsson (2008). The empirical evidence runs both ways: Jarrow and Yu (2001) provide anecdotal evidence for counterparty-risk-related contagion in small portfolios, and Collin-Dufresne et al. (2010) find that the returns of large corporate bond indices are significantly lower in months in which one or more large firms experience a significant widening in credit spreads, whereas Lando and Nielsen (2010) find no strong empirical evidence for default contagion in historical default patterns.

In such a model the default times admit intensities but are not doubly stochastic (Remark 2.9). Definition 3.2 fails, Lemma 3.4 and Proposition 3.5 are unavailable,

and with them the derivation of (4.12); the valuation formulas of Theorem 2.12 have to be replaced. Since the default of a name shifts the intensities of the survivors, the state of the portfolio — which names have defaulted and which have not — enters the state variable, and prices solve a system indexed by the default states rather than a single equation.

Chapter 5

A Contagion Model for Counterparty Risk

In the models of Chapters 3 and 4 the default of one name leaves the default intensities of the surviving names unchanged. This chapter drops that assumption: the intensity of every surviving name jumps at the default of any other name, by an amount fixed in advance. The model is that of Bo and Ceci (2020), built on the construction of Bo et al. (2019).

Conventions

Two conventions are in force for the whole chapter.

The model is specified directly under a pricing measure $\mathbb{Q}$, as in the martingale-modelling approach adopted in Section 2.3 (McNeil et al., 2015, Section 10.4.2): the dynamics below are the $\mathbb{Q}$-dynamics and the intensities are the $\mathbb{Q}$-intensities. The reference paper starts instead from a real-world measure and characterizes $\mathbb{Q}$ through a market price of diffusion risk and a default risk premium (Bo and Ceci, 2020, Section 2.5); both are taken to be zero throughout, as in the applications of the paper (Bo and Ceci, 2020, Section 4), and the two measures then coincide (Bo and Ceci, 2020, Section 3.3).

Quoted CDS spreads and bond prices are observed under the historical measure $\mathbb{P}$, but a pricing model is calibrated to them under $\mathbb{Q}$: risk-neutral pricing reproduces prices quoted in the market and is the method of choice where the underlying instruments are actively traded, whereas the traditional alternative — pricing from $\mathbb{P}$ and historical default data — suits illiquid markets instead (McNeil et al., 2015, Section 10.4.2).

The interest rate is zero. The whole analysis generalizes to a non-zero interest

rate (Bo and Ceci, 2020, Section 2.5), and no discount factor appears below.

Notation

Chapters 2 to 4 follow the notation of McNeil et al. (2015); the present chapter follows that of Bo and Ceci (2020). Table 5.1 gives the correspondence.

Table 5.1: The symbols of Chapters 2 to 4 and their counterparts in the present chapter.

Object	Chapters 2–4	Chapter 5
reference entity	R	names $1, \dots, N$
counterparty (seller)	S	name $N+1$
default time	τ_R, τ_S	τ_i, $i = 1, \dots, N+1$
default indicator	H_t, $H_{t,i}$	$H_i(t)$, vector $H(t)$
default state	—	$z \in \mathcal{S}$
hazard rate / intensity	γ_t, $\lambda_{t,i}$	$X_i(t)$, vector $X(t)$
background state variable	Ψ_t	$X(t)$ itself
background filtration	$\mathbb{F}$	$\mathbb{F}$
market filtration	$\mathbb{G}$	$\mathbb{G}$
loss given default	δ_S, constant	$L_i(H(t))$, state-dependent
CDS premium	spread x, fair x_t^*	contractual rate ε_i
dividend rate of a stream	ν_s	$a_i(t)$
payment at default	Z	$Z_i(t)$
counterparty-risk-free price	V_t	$S_i(t,T)$
netted exposure	$\sum_i V_{T_1,i}$	$\varepsilon_{\bar{N}}(\tau_{N+1}, T)$
the adjustment	UCVA_t	$\mathrm{CVA}_{\bar{N}}(t,T)$

Five letters carry a second meaning here. The X of Chapter 2 — a standard exponential threshold, or the promised payment of a survival claim — does not occur in this chapter, and X denotes a default intensity throughout. So do ξ, the identity of the first defaulter in Chapters 3 and 4 and the promised terminal payoff of a defaultable claim here; ν, the rate of a risky dividend stream in Chapter 2 and a mean-reversion speed here; x, the CDS spread there and a point of $\mathbb{R}_+^{N+1}$ here; and ε, which carries both the premium rate ε_i of a single contract and the aggregate exposure $\varepsilon_{\bar{N}}$ of the netting set. The last collision is internal to the reference paper, and the bar over $\bar{N}$ is what separates the two.

5.1 The model: interacting default intensities

There are $N \geqslant 1$ risky entities, referred to as name 1, name 2, …, name N, and the counterparty of the investor is the name $N+1$ (Bo and Ceci, 2020, Section 2).

The two filtrations

Let $(\Omega, \mathcal{G}, \mathbb{Q})$ be a probability space carrying a d-dimensional Brownian motion $W(t) = (W_k(t))_{k=1,\dots,d}^{\top}$, $t \geqslant 0$, and $N+1$ square-integrable positive random variables $\chi_1, \dots, \chi_{N+1}$ independent of W, which are the initial levels of the intensities. The *background filtration* is

$$\mathbb{F} = (\mathcal{F}(t))_{t \geqslant 0}, \qquad \mathcal{F}(t) = \sigma\big(W(s);\, s \leqslant t\big) \vee \sigma\big(\chi_i;\, i = 1, \dots, N+1\big), \tag{5.1}$$

so that $\mathcal{F}(0)$ is not trivial. Let $H(t) = (H_1(t), \dots, H_{N+1}(t))$ be the $(N+1)$-dimensional default indicator process, $H_i(t) = 1$ if name i has defaulted before or at t and zero otherwise, so that the state space of H is $\mathcal{S} = \{0,1\}^{N+1}$. With $\mathcal{H}_i(t) = \sigma(H_i(s);\, s \leqslant t)$ and $\mathbb{H}_i = (\mathcal{H}_i(t))_{t \geqslant 0}$, the *market filtration* is

$$\mathbb{G} = (\mathcal{G}(t))_{t \geqslant 0} = \mathbb{F} \vee \mathbb{H}_1 \vee \cdots \vee \mathbb{H}_{N+1}, \tag{5.2}$$

augmented by all $\mathbb{Q}$-null sets so as to satisfy the usual conditions.

Two filtrations are carried rather than one because the default times are not $\mathbb{F}$-stopping times, as in Section 2.2: conditionally on $\mathcal{F}(\infty)$, the paths of the Brownian motion together with the initial intensities, a name may or may not have defaulted, and it is this residual randomness that gives the default times an intensity. The enlargement leaves the diffusive part of the model undistorted, since W remains a $(\mathbb{Q}, \mathbb{G})$-Brownian motion (Bo and Capponi, 2018).

Default states

Write $\mathbb{R}_+ := (0, \infty)$, an open half-line, and $\mathcal{S} := \{0,1\}^{N+1}$. The vector $z = (z_1, \dots, z_{N+1}) \in \mathcal{S}$ is the *default state* of the portfolio with counterparty, $z_i = 0$ if name i is alive and $z_i = 1$ if it has defaulted. For $z \in \mathcal{S}$ with $z_j = 0$,

$$z^j := (z_1, \dots, z_{j-1}, 1, z_{j+1}, \dots, z_{N+1}), \qquad j = 1, \dots, N+1, \tag{5.3}$$

is the state obtained from z by setting its j-th component to 1. The operation is defined only where $z_j = 0$, default being absorbing. Iterating it, $z^{j_1,\dots,j_l}$ is the state obtained by setting l distinct components $j_1, \dots, j_l$ to 1, with $z^{j_1,\dots,j_l} = z$ when $l = 0$; the order of the indices does not enter, since the symbol records which set of names has defaulted and not the order in which they did (Bo and Ceci, 2020, Section 2.1).

Contagion

The impact on name i of all defaults occurring before or at t is carried by the pure-jump process

$$J_i(t) := \sum_{j=1}^{N+1} w_{ij}\, H_j(t), \tag{5.4}$$

whose weights form the vectors $w_j = (w_{ij})_{i=1,\dots,N+1} \in [0,\infty)^{N+1}$. Thus w_{ij} measures the extent to which the default of name j impacts the default intensity of name i: in (5.4) it multiplies H_j and enters the intensity of i.

The process J_i is piecewise constant and non-decreasing, jumps by w_{ij} at τ_j, and has at most $N+1$ jumps, since each H_j jumps once; $J_i(0) = 0$, all names being alive at time zero. The contagion that name i can receive over $[0,T]$ is bounded by $\sum_j w_{ij}$, so that the weights fix how much pressure may accumulate on a name and the defaults fix only when it arrives. And (5.4) contains no decay mechanism: once H_j has switched to 1 the contribution w_{ij} stays, so any attenuation over time of the effect of the default of j has to come from the drift of the intensity.

Under $\mathbb{Q}$ the default intensities satisfy a system of interacting stochastic differential equations: for $i = 1,\dots,N+1$,

$$dX_i(t) = \mu_i(X(t))\,dt + \sum_{k=1}^{d} \sigma_{ik}(X(t))\,dW_k(t) + dJ_i(t), \qquad X_i(0) = \chi_i, \tag{5.5}$$

with $X(t) = (X_i(t))_{i=1,\dots,N+1}^{\top}$. The names interact through two separate channels. The coefficients μ_i and σ_{ik} are functions of the whole vector $X(t)$ and not of $X_i(t)$ alone: this is the *diffusive channel*, and it operates in continuous time with no default involved. The term $dJ_i(t)$ operates only at default times: this is the *contagion channel*.

Between two defaults X is a diffusion, and at τ_j it jumps as a block,

$$X(\tau_j) = X(\tau_j-) + w_j. \tag{5.6}$$

The default of j is one event that strikes every intensity at once, and w_j — the j-th column of the contagion matrix — is the increment the whole vector receives. If the weight w_{ij} is high, the default of name j increases substantially the default intensity of name i; if the w_{ij} are high for sufficiently many i, the probability of several firms defaulting within a short time after the default of j is high, which is the *default clustering* documented in Azizpour et al. (2018).

The process X is the intensity of H, and H enters the dynamics of X through J.

The two processes therefore have to be constructed together, which is the content of Proposition 5.3 below. In particular X is $\mathbb{G}$-adapted and not $\mathbb{F}$-adapted.

Assumptions

Assumption 5.1. The coefficients

$$\mu(x) = (\mu_i(x))_{i=1,\dots,N+1}^{\top}, \qquad \sigma(x) = (\sigma_{ik}(x))_{i=1,\dots,N+1;\,k=1,\dots,d},$$

are locally Lipschitz continuous with linear growth in $x \in \mathbb{R}_+^{N+1}$. Additionally, $\det((\sigma\sigma^{\top})(x)) \neq 0$ for $x \in \mathbb{R}_+^{N+1}$.

Assumption 5.2. For $(t,x) \in [0,T] \times \mathbb{R}_+^{N+1}$, let $\widetilde{X}^{t,x}(s) = (\widetilde{X}_i^{t,x}(s))_{i=1,\dots,N+1}^{\top}$ satisfy $\widetilde{X}^{t,x}(t) = x$ and, for $s \in [t,T]$,

$$d\widetilde{X}^{t,x}(s) = \mu(\widetilde{X}^{t,x}(s))\,ds + \sigma(\widetilde{X}^{t,x}(s))\,dW(s). \tag{5.7}$$

Then $\mathbb{Q}(\widetilde{X}^{t,x}(s) \in \mathbb{R}_+^{N+1} \text{ for all } s \in [t,T]) = 1$.

The two halves of Assumption 5.1 do separate work. The regularity of the coefficients gives (5.7) a unique strong solution (Protter, 2005, Theorem V.38). The non-degeneracy condition is not needed for that: it makes the infinitesimal generator of $\widetilde{X}^{t,x}$ elliptic (Heath and Schweizer, 2000, Lemma 3), uniformly so on every compact subset of $\mathbb{R}_+^{N+1}$ rather than with a single constant over the whole open, unbounded domain — which is exactly the form the localisation argument in the proof of Proposition 5.15 uses, and what Proposition 5.14's existence result requires of it.

Assumption 5.2 is a positivity requirement on the process $\widetilde{X}^{t,x}$ of (5.7), which is (5.5) without its jump term, and is $\mathbb{F}$-adapted where X is only $\mathbb{G}$-adapted. Positivity of the diffusive part suffices for X itself, the jumps being non-negative, and it is strict positivity that is required, $\mathbb{R}_+$ being open and X_i having to serve as an intensity. The assumption is imposed at every pair (t,x) and not at the initial datum alone because it is reapplied after every default: between defaults X evolves as some $\widetilde{X}$, and at each default it restarts from a new point at a new instant.

Existence of the model, and the compensator

Proposition 5.3. *Under Assumptions 5.1 and 5.2 there exists a unique $\mathbb{R}_+^{N+1} \times \mathcal{S}$-valued and $\mathbb{G}$-adapted Markov process (X,H) satisfying* (5.4) *and* (5.5) *and such that*

$$M_i(t) := H_i(t) - \int_0^t \big(1 - H_i(s)\big) X_i(s)\,ds, \qquad t \geqslant 0, \tag{5.8}$$

is a $(\mathbb{Q}, \mathbb{G})$*-martingale for* $i = 1, \dots, N+1$.

Bo and Ceci (2020, Proposition 2.1) omit the proof, which follows that of Bo et al. (2019). From the same construction, simultaneous defaults do not occur,

$$\mathbb{Q}(\tau_i = \tau_j) = 0 \qquad \text{for all } i \neq j, \tag{5.9}$$

where $\tau_i := \inf\{t > 0;\ H_i(t) = 1\}$ with $\inf \emptyset = +\infty$. Hence the shift in (5.6) involves a single column w_j at a time.

Were X $\mathbb{F}$-adapted, Proposition 5.3 would reduce to the construction of Lemma 3.3: one builds the path of the intensities, then draws the defaults with those intensities. Here that order is unavailable, since (5.5) contains dJ and hence depends on H, whose arrival rate is X. What the proposition supplies is a pair satisfying (5.4), (5.5) and (5.8) at once.

It is the pair that is Markov. The intensity vector alone is not, its jumps being driven by H; the default indicator alone is not, its intensity being stochastic. Prices may therefore be sought as deterministic functions $F(t, x, z)$ of time, of the current intensities and of the current default state, evaluated along (X, H).

Equation (5.8) identifies the compensator of H_i. The Doob–Meyer decomposition of Section 2.2, applied to each coordinate, writes $H_i = M_i + A_i$ with M_i a $\mathbb{G}$-martingale and A_i predictable and increasing; (5.8) states that A_i is absolutely continuous with density $(1 - H_i(s))X_i(s)$, which is what makes X_i the intensity of name i. The factor $1 - H_i$ reduces A_i to $\int_0^{t\wedge\tau_i} X_i(s)\,ds$: after τ_i the indicator can no longer rise, and with the compensator $\int_0^t X_i(s)\,ds$ the difference $M_i(t) = 1 - \int_0^t X_i(s)\,ds$ would decrease continuously and predictably. The truncation stops A_i, not X_i, which goes on jumping at the defaults of the other names while ceasing to be the intensity of anything.

Remark 5.4 (the default times are not doubly stochastic)**.** Each τ_i admits an intensity, namely X_i on $\{t < \tau_i\}$ by (5.8), but X_i is not $\mathbb{F}$-adapted, since (5.5) makes it jump at the default times of the other names. This model therefore does not admit the filtration $\mathbb{F}$ of (5.1) as one making the defaults conditionally independent in the sense of Definition 3.2, and neither Lemma 3.4 nor Proposition 3.5 applies to it. Equation (4.12), which rests on both, does not apply to the present model.

5.2 Contagion, and the CDS portfolio as a defaultable claim

The intensities

For $i = 1, \dots, N+1$ the default intensity of the i-th name follows

$$dX_i(t) = \big(\kappa_i - \nu_i X_i(t)\big)dt + \sigma_i \sqrt{X_i(t)}\, dW_i(t) + \sum_{j \neq i} w_{ij}\, dH_j(t), \qquad X_i(0) = \chi_i, \tag{5.10}$$

where κ_i, ν_i and σ_i are positive constants and $W_1, \dots, W_{N+1}$ are independent one-dimensional Brownian motions, so that $d = N + 1$ and

$$\sigma_{ik}(x) = \sigma_i \sqrt{x_i}\, \delta_{ik}, \tag{5.11}$$

where δ_{ik} is the Kronecker delta, equal to 1 if $i = k$ and 0 otherwise. The constants satisfy Feller's boundary classification condition

$$2\kappa_i \geqslant \sigma_i^2, \qquad i = 1, \dots, N+1. \tag{5.12}$$

Equation (5.10) without its jump term is the square-root diffusion (2.23), written in the parametrization of the reference paper. Factoring the speed out of the drift,

$$\kappa_i - \nu_i x = \nu_i\big(\kappa_i/\nu_i - x\big),$$

exhibits the long-run level $\kappa_i/\nu_i > 0$ and the speed ν_i. Below the long-run level the drift is positive and the intensity is pushed up, above it the drift is negative and the intensity is pushed down. Condition (5.12) is (2.24) written for these parameters, $2\nu_i(\kappa_i/\nu_i) \geqslant \sigma_i^2$, and it implies Assumption 5.2 (Bo and Ceci, 2020, Section 2.4): the diffusion is strictly positive and never reaches zero, as an intensity must be. Assumption 5.1 holds as well, since by (5.11) $\sigma\sigma^\top(x) = \operatorname{diag}(\sigma_i^2 x_i)$ has determinant $\prod_i \sigma_i^2 x_i$, which is positive on $\mathbb{R}_+^{N+1}$ and vanishes only on its boundary.

Since H_j is non-decreasing with a single unit jump at τ_j, the jump term of (5.10) is $dJ_i(t) = \sum_j w_{ij}\, dH_j(t)$, and the intensity of name i passes through three regimes: between two defaults J_i is constant and X_i reverts to κ_i/ν_i at speed ν_i; at τ_j it jumps upward, $X_i(\tau_j) = X_i(\tau_j-) + w_{ij}$; after τ_j the drift pulls it back and the excess decays at the exponential rate ν_i (Bo and Ceci, 2020, Section 2.4). Contagion moves the state and not the long-run level, which is κ_i/ν_i before and after the default of j: the effect is an impulse that is reabsorbed rather than a change of regime.

Remark 5.5 (a name does not infect itself)**.** The sum in (5.10) runs over $j \neq i$, whereas (5.4) includes the term $j = i$, whose weight the reference paper does not fix. We set $w_{ii} = 0$. Self-exciting intensities, of Hawkes type, model repeated events, each arrival raising the probability of the next; a default is terminal, and the same name does not default twice, so only cross-excitation applies here. The factor $1 - H_i$ in (5.8) already detaches X_i from its own default at that instant, but X_i remains a coordinate of the state vector at which prices are evaluated, and (5.10) would otherwise carry an unspecified jump.

Remark 5.6 (contagion as a shock)**.** Specification (5.11) closes the diffusive channel of (5.5): the intensities are independent between defaults and coupled only through the jumps. The reference paper is more general, driving every intensity with the same d-dimensional Brownian motion through a full matrix $\sigma(x)$. The two devices represent different things. A correlation between the Brownian motions is a common random disturbance, a dependence that shows up over time; a jump at a default time is a shock. Default contagion is a shock, and the diagonal specification isolates it.

Defaultable claims

Definition 5.7 (defaultable claim)**.** A *defaultable claim* maturing at $T > 0$ is a quadruple (ξ, a, Z, K), where the random variable $\xi := \xi(H(T))$ and the processes $a(t) := a(H(t))$ and $Z(t) := Z(H(t))$, $t \in [0, T]$. The process $K(t) := K(H(t))$, $t \in [0, T]$, is the indicator function of a positive $\mathbb{G}$-stopping time $\bar{\tau}$, that is $K(t) = \mathbf{1}_{\{\bar{\tau} \leqslant t\}}$. Here, with a slight abuse of notation, $\xi(z)$, $a(z)$, $Z(z)$ and $K(z)$ denote deterministic functions on $z \in \mathcal{S}$.

Definition 5.8 (dividend process)**.** The *dividend process* $D = (D(t))_{t \geqslant 0}$ associated with the defaultable claim (ξ, a, Z, K) maturing at T equals, for every $t \geqslant 0$,

$$D(t) = \xi\big(1 - K(T)\big)\mathbf{1}_{\{t=T\}} + \int_0^{t\wedge T} \big(1 - K(u)\big)a(u)\,du + \int_0^{t\wedge T} Z(u)\,dK(u). \quad (5.13)$$

The process D is of finite variation and collects the cash flows generated by the claim over $(0, T]$, that is after the contract is initiated at time zero. Its three terms are the promised payoff ξ, paid at T if default has not happened before or at T; the promised dividends a, paid until the earliest of T and default; and the payoff Z delivered at the default time if it occurs before maturity (Bo and Ceci, 2020, Section 2.3).

Remark 5.9 (the recovery reads the post-default state)**.** Definition 5.7 does not assume Z to be $\mathbb{G}$-predictable, which departs from Bielecki et al. (2008) and admits

claims whose recovery depends on a totally inaccessible stopping time (Bo and Ceci, 2020, Section 2.3). Since $Z(u) = Z(H(u))$ and H jumps at $\bar{\tau}$, the process $u \mapsto Z(u)$ need not be $\mathbb{G}$-predictable: its value at $\bar{\tau}$ can differ from its left limit there, which reads the pre-default state.

The CDS portfolio

The reference entities of the portfolio are the names $1, \dots, N$ and the counterparty of the investor is the name $N+1$. The notional amount of every contract is one, so that percentage and absolute losses coincide, and all contracts mature at T. For $i = 1, \dots, N$, the CDS on entity i is a contract between the protection buyer, who is the investor, and the protection seller, who is the counterparty: the buyer pays a contractually specified premium $\varepsilon_i > 0$ continuously in time until the earlier of the default time τ_i of the reference entity and the maturity T, and the seller pays the loss rate $L_i(t) := L_i(H(t)) \in (0, 1]$ at τ_i if the reference entity defaults before maturity (Bo and Ceci, 2020, Section 2.4). Every position is written from the point of view of the protection buyer.

In terms of Definition 5.7, the quadruple of the i-th contract is

$$\xi_i = 0, \qquad a_i(t) = -\varepsilon_i, \qquad Z_i(t) = L_i(t), \qquad K_i(t) = H_i(t), \tag{5.14}$$

for $i = 1, \dots, N+1$, so that $\bar{\tau}_i = \tau_i$ and each contract is switched off by the default of the name it is written on. Of the three payment components, ξ_i and a_i are constants, and the whole dependence of a CDS on the default state passes through its loss rate.

Substituting (5.14) into (5.13) term by term gives the dividend process of the i-th CDS. The first term vanishes, ξ_i being zero. In the second, $1 - H_i(u) = \mathbf{1}_{\{u < \tau_i\}}$, so that

$$\int_0^{t \wedge T} \big(1 - H_i(u)\big)(-\varepsilon_i)\, du = -\varepsilon_i \int_0^{t \wedge T} \mathbf{1}_{\{u < \tau_i\}}\, du = -\varepsilon_i\, (t \wedge T \wedge \tau_i),$$

the premium accumulated up to the earlier of default and maturity. In the third, H_i is non-decreasing with a single jump of size one at τ_i, so that dH_i is a unit mass at τ_i and

$$\int_0^{t \wedge T} L_i(u)\, dH_i(u) = L_i(\tau_i) \mathbf{1}_{\{\tau_i \leqslant t \wedge T\}},$$

the inequality being weak because $H_i(t) = 1$ if default occurs before or at t. Hence

$$\begin{aligned} D_i(t) &= -\varepsilon_i \int_0^{t\wedge T} \big(1 - H_i(u)\big)\,du + \int_0^{t\wedge T} L_i(u)\,dH_i(u) \\ &= -\varepsilon_i\,(t \wedge T \wedge \tau_i) + L_i(\tau_i)\mathbf{1}_{\{\tau_i \leqslant t\wedge T\}}. \end{aligned} \tag{5.15}$$

The process has two pieces: an absolutely continuous part of slope $-\varepsilon_i$ up to $\tau_i \wedge T$, and a single jump of size $L_i(H(\tau_i))$ at τ_i if the reference entity defaults before maturity. With a zero interest rate the two terms are the premium payment leg (2.9) and the default payment leg (2.10) of Section 2.4, the constant loss given default δ of that section being replaced by the function L_i of the default state.

Remark 5.10 (the dividend process carries no counterparty risk)**.** Equation (5.15) involves τ_i, T, ε_i and L_i, and never τ_{N+1}: the stream runs to $\tau_i \wedge T$ whatever the counterparty does. The quadruple (5.14) therefore specifies the promised stream, as $\Pi(t, T)$ does in (4.3), and not the stream actually received. The correction enters through the adjustment of Section 5.4.

Remark 5.11 (two channels of wrong way risk)**.** Wrong way risk in the sense of Definition 3.1 arises here through two mechanisms. In the intensities, the default of name j raises the probability that name i defaults, by the weight w_{ij} of (5.10). In the loss rates, it raises the severity, where the loss rates are specified so that $L_i(z^j) > L_i(z)$, a worse recovery in the state in which j has defaulted. That ordering is a property of the chosen L_i and not of the model: a constant L_i removes the second channel and leaves the first.

The portfolio

Definition 5.12 (defaultable claim portfolio)**.** Let $\bar{N} \geqslant 1$ and let (ξ_i, a_i, Z_i, K_i), $i = 1, \ldots, \bar{N} + 1$, be defaultable claims in the sense of Definition 5.7, with $K_i(t) = K_i(H(t)) = \mathbf{1}_{\{\bar{\tau}_i \leqslant t\}}$ for positive $\mathbb{G}$-stopping times $\bar{\tau}_i$. The family is a *defaultable claim portfolio* provided the processes $K_1, \ldots, K_{\bar{N}+1}$ do not jump simultaneously.

The CDS portfolio satisfies the hypothesis of Definition 5.12 without further assumption. Here $\bar{N} = N$ and $K_i = H_i$ by (5.14), so the killing times are the default times of distinct names, which do not coincide by (5.9). The portfolio is

$$\big(0,\, -\varepsilon_i,\, L_i,\, H_i\big)_{i=1,\ldots,N+1}, \qquad \bar{N} = N, \qquad \bar{\tau}_i = \tau_i. \tag{5.16}$$

The investor holds $b_i \in \mathbb{R}$ contracts of the i-th kind, $b_i > 0$ for protection bought from the counterparty and $b_i < 0$ for protection sold to it (Bo and Ceci, 2020, Section

2.6). The premium rate ε_i is contractual and positive, and the direction of a position is carried by its weight. The *netting set* is $i = 1, \dots, N$; the remaining claim of (5.16), written on the counterparty itself, is excluded from it.

5.3 Risk-neutral pricing: the recursive Cauchy systems

The *ex-dividend price* at t of a defaultable claim (ξ, a, Z, K) maturing at T is

$$S(t,T) := \mathbb{E}^{\mathbb{Q}}\big[D(T) - D(t) \;\big|\; \mathcal{G}(t)\big], \tag{5.17}$$

the value of the dividends the claim has still to pay over $(t, T]$ (Bo and Ceci, 2020, Section 2.5). On $\{\tau_i > t\}$ the dividend process (5.15) of the i-th CDS gives

$$D_i(T) - D_i(t) = -\varepsilon_i\big(T \wedge \tau_i - t\big) + L_i(\tau_i)\mathbf{1}_{\{t<\tau_i\leqslant T\}},$$

and $T \wedge \tau_i - t = \int_t^T \mathbf{1}_{\{\tau_i > s\}}\, ds$ there, so that

$$S_i(t,T) = \mathbb{E}^{\mathbb{Q}}\big[L_i(\tau_i)\mathbf{1}_{\{t<\tau_i\leqslant T\}} \;\big|\; \mathcal{G}(t)\big] - \varepsilon_i\, \mathbb{E}^{\mathbb{Q}}\Big[\int_t^T \mathbf{1}_{\{\tau_i > s\}}\, ds \;\Big|\; \mathcal{G}(t)\Big]. \tag{5.18}$$

Equation (5.18) is the default leg minus the premium leg of Section 2.4, and neither leg is an integral against a survival function here. The conditioning is on $\mathcal{G}(t)$, which carries the default history of all the names, and the intensity of name i depends on that history through (5.10), so the survival probability of i does not factor out of the second term; and L_i is a function of the default state of the whole portfolio, so it does not leave the first. The price is therefore characterized as the solution of a system of equations rather than evaluated as an integral.

The price as a function of the state

Proposition 5.13 (the pricing function)**.** *For $i = 1, \dots, N+1$ and $(t, x, z) \in [0,T] \times \mathbb{R}_+^{N+1} \times \mathcal{S}$ let*

$$\begin{aligned} F_i^{\mathrm{cds}}(t,x,z) := \mathbb{E}^{\mathbb{Q}}_{t,x,z}\Bigg[& L_i\big(H(T)\big)H_i(T) - \varepsilon_i \int_t^T \big(1 - H_i(u)\big)du \\ & - \int_t^T H_i(u) \sum_{j\,:\,H_j(u)=0} \big[L_i\big(H^j(u)\big) - L_i\big(H(u)\big)\big]\, X_j(u)\, du\Bigg], \end{aligned} \tag{5.19}$$

where $\mathbb{E}^{\mathbb{Q}}_{t,x,z}[\,\cdot\,] := \mathbb{E}^{\mathbb{Q}}[\,\cdot \mid X(t) = x,\ H(t) = z]$ *and, for* j *with* $H_j(u) = 0$, $H^j(u)$ *is the vector* $H(u)$ *with its* j*-th component set to* 1*, as in* (5.3)*; the sum inside the integral ranges at each instant* u *over the names still alive there, matching the convention of Proposition 5.15 below. Then, for every* $t \in [0, T]$*,*

$$S_i(t,T) = F_i^{\mathrm{cds}}\big(t, X(t), H(t)\big) - L_i(t)H_i(t). \tag{5.20}$$

Proposition 5.13 is Bo and Ceci (2020, Proposition 2.3) evaluated at the quadruple (5.14); the proof is in their Appendix. The representation stated there carries a second function, the expectation of the promised terminal payoff, which vanishes identically here because $\xi_i = 0$, and with it the indicator $\mathbf{1}_{\{t\neq T\}}$ that multiplies it. One unknown function per contract is left.

The three terms of (5.19) are the loss rate at maturity in the states where the reference entity has already defaulted, the premium paid while it is alive, and a compensator for the jumps of the loss rate at the defaults of the other names. The second and third are mutually exclusive: the premium carries $1 - H_i$ and the compensator carries H_i, so before the default of i one pays the premium and after it one compensates the loss rate, never both.

Equation (5.20) is price plus recovery already owed. The process H_i jumps at τ_i, so S_i jumps there as well and is not a continuous function of $(t, X(t), H(t))$; a classical solution of a parabolic problem is. $F_i^{\mathrm{cds}}(\cdot,\cdot,z)$, for each fixed default state z, is the function required to be a classical solution of a parabolic problem on $[0,T) \times \mathbb{R}^{N+1}_+$ (Proposition 5.14); S_i is not a function of (t,x) alone, its recovery term depending on the default state directly rather than through a Cauchy problem, so it is F_i^{cds}, not S_i, that (5.23) is written for. The composed process $F_i^{\mathrm{cds}}(t, X(t), H(t))$ still jumps at every default, including at τ_i itself: the point is regularity in (t,x) at fixed z, not continuity of the path.

The generator, and the system

The generator of the Markov process (X, H) of Proposition 5.3 under $\mathbb{Q}$ is, for $f(\cdot, z) \in C^2(\mathbb{R}^{N+1}_+)$,

$$\mathcal{A}f(x,z) := \widetilde{\mathcal{A}}f(x,z) + \sum_{j=1}^{N+1} \big[f(x + w_j, z^j) - f(x,z)\big](1 - z_j)\, x_j, \tag{5.21}$$

where $\widetilde{\mathcal{A}}$ is the generator of the auxiliary diffusion (5.7),

$$\widetilde{\mathcal{A}}f(x,z) := \mu(x)^\top D_x f(x,z) + \tfrac{1}{2}\operatorname{tr}\big[(\sigma\sigma^\top)(x)\, D_{xx} f(x,z)\big], \tag{5.22}$$

with $D_x f := (\partial f/\partial x_i)^{\top}_{i=1,\dots,N+1}$ and $D_{xx} f := (\partial^2 f/\partial x_i \partial x_j)_{i,j=1,\dots,N+1}$ (Bo and Ceci, 2020, Section 2.5). The jump part of (5.21) moves the two arguments of f together: the default state from z to z^j and the point from x to $x + w_j$. This is (5.6) read on functions — the default of j and the contagion it transmits are one event — and it is the only place in the pricing problem where the contagion weights appear.

The function F_i^{cds} of (5.19) solves, on $[0,T) \times \mathbb{R}_+^{N+1} \times \mathcal{S}$,

$$\Big(\frac{\partial}{\partial t} + \mathcal{A}\Big) F_i^{\mathrm{cds}}(t,x,z) - (1 - z_i)\,\varepsilon_i - z_i \sum_{j=1}^{N+1} \big[L_i(z^j) - L_i(z)\big](1 - z_j)\,x_j = 0, \quad (5.23)$$

with the terminal condition

$$F_i^{\mathrm{cds}}(T,x,z) = L_i(z)\,z_i, \qquad (x,z) \in \mathbb{R}_+^{N+1} \times \mathcal{S}, \quad (5.24)$$

which is (Bo and Ceci, 2020, eqs. (37)–(38)). The two sources of (5.23) are the two mutually exclusive terms of (5.19). If the loss rates do not depend on the default state the second disappears and (5.23) reduces to $(\partial_t + \mathcal{A})F_i^{\mathrm{cds}} - (1 - z_i)\varepsilon_i = 0$ (Bo and Ceci, 2020, Remark 3.2).

The recursion

Equation (5.23) is one equation per default state, and the 2^{N+1} of them form a single system, coupled through the jump part of (5.21). At the state in which every name has defaulted that sum is empty, the problem is $(\partial_t + \widetilde{\mathcal{A}})F_i^{\mathrm{cds}} = 0$ with a constant terminal datum, and the solution is that constant. At every other state the surviving names contribute a killing term and a source built out of the states with one more default.

Proposition 5.14. *Let Assumptions 5.1 and 5.2 hold and let $z \in \mathcal{S}$ have at least one surviving name. If at every state z^j with $z_j = 0$ the problem* (5.23)–(5.24) *admits a unique bounded classical solution on $[0,T] \times \mathbb{R}_+^{N+1}$, then it admits one at z as well, in $C^{1,2}\big([0,T) \times \mathbb{R}_+^{N+1}\big) \cap C^0\big([0,T] \times \mathbb{R}_+^{N+1}\big)$.*

The proof is in the Appendix of Bo and Ceci (2020) and rests on the ellipticity of $\widetilde{\mathcal{A}}$, uniform on each of the bounded domains $(1/n, n)^{N+1}$ that exhaust $\mathbb{R}_+^{N+1}$, which is the non-degeneracy half of Assumption 5.1. The hypothesis of Proposition 5.14 is at the states with one default more than its conclusion: the system (5.23) is triangular in the number of defaults, and is solved from the fully defaulted state upwards in the number of survivors. The triangularity is a property of the model and not a device:

default is absorbing, so the jump part of (5.21) couples a state only to states with one more default.

Proposition 5.15 (transport form and Feynman–Kac representation)**.** *Let Assumptions 5.1 and 5.2 hold, fix i and a default state z with $z_j = 0$ for at least one j, and suppose* (5.23)–(5.24) *has a unique bounded classical solution at every state z^j with $z_j = 0$. Put*

$$c_z(x) := \sum_{j\,:\,z_j=0} x_j, \tag{5.25}$$

$$f_{i,z}(t,x) := \sum_{j\,:\,z_j=0} F_i^{\mathrm{cds}}(t, x+w_j, z^j)\, x_j - (1-z_i)\varepsilon_i - z_i \sum_{j\,:\,z_j=0} \big[L_i(z^j) - L_i(z)\big] x_j. \tag{5.26}$$

Then $u(t,x) := F_i^{\mathrm{cds}}(t,x,z)$ solves the linear parabolic problem with killing rate $c_z \geqslant 0$ and source $f_{i,z}$,

$$\Big(\frac{\partial}{\partial t} + \widetilde{\mathcal{A}}\Big) u(t,x) - c_z(x)\, u(t,x) + f_{i,z}(t,x) = 0, \qquad u(T,x) = L_i(z) z_i, \tag{5.27}$$

and admits, for $(t,x) \in [0,T] \times \mathbb{R}^{N+1}_+$, the representation

$$F_i^{\mathrm{cds}}(t,x,z) = \mathbb{E}\bigg[L_i(z) z_i\, e^{-\int_t^T c_z(\widetilde{X}^{t,x}(u))\,du} + \int_t^T f_{i,z}\big(s, \widetilde{X}^{t,x}(s)\big)\, e^{-\int_t^s c_z(\widetilde{X}^{t,x}(u))\,du}\, ds \bigg], \tag{5.28}$$

with $\widetilde{X}^{t,x}$ the auxiliary diffusion (5.7)*.*

Representation (5.28) is a Feynman–Kac formula, for the auxiliary diffusion (5.7) with killing rate c_z and source $f_{i,z}$; Bo and Ceci (2020, eq. (27)) obtain the identification of the solutions of the system with the pricing functions from their Proposition 2.4, which is Proposition 5.14, and the Feynman–Kac formula. Lemma 2.13 does not cover this case: it is stated for a scalar diffusion and a source that does not depend on time, whereas $\widetilde{X}^{t,x}$ takes values in $\mathbb{R}^{N+1}_+$ and $f_{i,z}$ depends on t through the values $F_i^{\mathrm{cds}}(t,\cdot,z^j)$ at the successor states.

Proof. In the state z the sum in (5.21) runs over the surviving names alone, the factor $1 - z_j$ vanishing otherwise. Substituting (5.21) into (5.23) and splitting the bracket,

$$\sum_{j\,:\,z_j=0} \big[F_i^{\mathrm{cds}}(t, x+w_j, z^j) - u(t,x)\big] x_j = \sum_{j\,:\,z_j=0} F_i^{\mathrm{cds}}(t, x+w_j, z^j)\, x_j - c_z(x)\, u(t,x),$$

gives (5.27) with the source (5.26); the constraint $j \neq i$ of the last sum in (5.23) has been absorbed into $z_j = 0$, that term carrying the factor z_i and so being present

only when $z_i = 1$, in which case $j = i$ is excluded already. The terminal condition is (5.24).

For the representation, fix $t < T$, the case $t = T$ being the terminal condition. Write $\widetilde{X} := \widetilde{X}^{t,x}$ and $\Theta(s) := e^{-\int_t^s c_z(\widetilde{X}(u))du}$, and for $n > 1/(T-t)$ let

$$\rho_n := \inf\big\{s \geqslant t : \widetilde{X}(s) \notin (1/n, n)^{N+1}\big\} \wedge \big(T - 1/n\big).$$

By Assumption 5.2 the diffusion stays in $\mathbb{R}_+^{N+1}$ and does not explode, so $\rho_n \uparrow T$ almost surely. The bound $T - 1/n$ keeps $[t, \rho_n]$ inside $[0, T)$, where u is continuously differentiable in t and twice in x (Proposition 5.14). Itô's formula applied to $s \mapsto \Theta(s)u(s, \widetilde{X}(s))$ on $[t, \rho_n]$, with $d\Theta(s) = -c_z(\widetilde{X}(s))\Theta(s)\,ds$, gives

$$\begin{aligned}\Theta(\rho_n)u(\rho_n, \widetilde{X}(\rho_n)) = u(t,x) &- \int_t^{\rho_n} \Theta(s)\, f_{i,z}(s, \widetilde{X}(s))\, ds \\ &+ \int_t^{\rho_n} \Theta(s)\, D_x u(s, \widetilde{X}(s))^\top \sigma(\widetilde{X}(s))\, dW(s),\end{aligned}$$

the drift term having been identified through (5.27). On $[t, \rho_n]$ the integrand of the stochastic integral is bounded, $D_x u$ and σ being continuous on the compact set $[t, T - 1/n] \times [1/n, n]^{N+1}$, so that integral is a martingale and has zero expectation. Taking expectations and letting $n \to \infty$: in the first term $0 < \Theta \leqslant 1$, since $c_z \geqslant 0$ on $\mathbb{R}_+^{N+1}$, and u is bounded and continuous on $[0, T] \times \mathbb{R}_+^{N+1}$, so $\Theta(\rho_n)u(\rho_n, \widetilde{X}(\rho_n)) \to \Theta(T)u(T, \widetilde{X}(T))$ along the continuous paths of $\widetilde{X}$, and dominated convergence gives $\mathbb{E}[\Theta(T)L_i(z)z_i]$. In the second, $f_{i,z}(s,x)$ is bounded by $C(1 + \sum_j x_j)$, the $F_i^{\mathrm{cds}}(\cdot,\cdot,z^j)$ and the L_i being bounded and ε_i a constant, and $\int_t^T \mathbb{E}[\sum_j \widetilde{X}_j(s)]\, ds < \infty$ by the linear growth of the coefficients in Assumption 5.1; dominated convergence applies again and yields (5.28). □

In (5.28) the discount factor is the probability that no surviving name defaults between t and s, that is the probability of still being in the state z; the source is the value of the successor states times the rate of transition into them, less the premium. The shift w_j sits inside $f_{i,z}$, in the first argument of F_i^{cds}, while the multiplier x_j inherited from the compensator is evaluated at $\widetilde{X}_j^{t,x}(s)$ unshifted.

Remark 5.16 (the shift is not a reparametrization)**.** Evaluating F_i^{cds} at $\widetilde{X}^{t,x}(s)$ rather than at $\widetilde{X}^{t,x}(s) + w_j$ would amount to a reparametrization of the diffusion if $\widetilde{X}^{t,x}$ itself jumped by w_j at the default of j. It does not: the default of j is not an event of $\widetilde{X}^{t,x}$, which is the jump-free process (5.7) used to represent the solution of a parabolic problem and not the intensity process X of the model. The jump of X at τ_j is recorded once, in (5.21), as the pair consisting of the shift of the argument and

the change of the state, and the two halves travel together. Keeping $z \mapsto z^j$ while dropping $x \mapsto x + w_j$ retains which names have defaulted and discards the size of the contagion.

Half of the lattice is constant

Proposition 5.17 (the price after the default of the reference entity)**.** *Under the hypotheses of Proposition 5.15, for every i and every $z \in \mathcal{S}$ with $z_i = 1$,*

$$F_i^{\mathrm{cds}}(t, x, z) = L_i(z), \qquad (t, x) \in [0, T] \times \mathbb{R}_+^{N+1}. \tag{5.29}$$

Proof. By downward induction on the number of surviving names. If z has none, the equation is $(\partial_t + \widetilde{\mathcal{A}})u = 0$ with the constant terminal datum $L_i(z)$, solved by that constant. Let $z_i = 1$, let z have at least one surviving name, and assume (5.29) at every z^j with $z_j = 0$, which also satisfies $(z^j)_i = 1$. Substitute the constant $L_i(z)$ into (5.27). Its time and space derivatives vanish; the killing term contributes $-L_i(z)\sum_{j:z_j=0} x_j$; the premium term of (5.26) vanishes, z_i being 1; the successor term contributes $\sum_{j:z_j=0} L_i(z^j)x_j$ by the inductive hypothesis, the shift being immaterial against a constant; and the compensator term contributes $-\sum_{j:z_j=0}[L_i(z^j) - L_i(z)]x_j$. Each surviving name contributes $x_j[-L_i(z) + L_i(z^j) - L_i(z^j) + L_i(z)] = 0$, and the terminal condition (5.24) reads $L_i(z)z_i = L_i(z)$. The constant solves the problem, and the solution is unique by Proposition 5.14. □

By (5.20), on $\{H_i(t) = 1\}$ the price is $S_i(t, T) = L_i(H(t)) - L_i(t) = 0$: a CDS whose reference entity has defaulted is worth nothing, the recovery having been paid at τ_i and recorded in the dividend process (5.15). And the recursion has to be run only on the 2^N states with $z_i = 0$, where the terminal datum (5.24) is zero, the remaining half of the lattice being given by (5.29).

The two propositions together are an algorithm. Initialize F_i^{cds} at the constant $L_i(z)$ on the states with $z_i = 1$; climb the states with $z_i = 0$ by decreasing number of defaults, each one given by (5.28) in terms of states already computed; and read the price off (5.20). At the state z the successor $j = i$ reaches into the constant half of the lattice, which is where the default of the reference entity enters the price of its own CDS, and the successors $j \neq i$ stay in the half being computed and carry the contagion coming from the other names.

5.4 CVA with netting under contagion

The investor is default-free and cannot itself fail to pay, so the adjustment computed here is unilateral in the sense of Remark 4.2 (Bo and Ceci, 2020, Section 2.6). The weights b_i of (5.16) are constants: the portfolio is held fixed and the identity of the counterparty is given, neither being an object of choice (Bo and Ceci, 2020, Remark 2.6). The adjustment is then a function of the state of the model and not of a strategy.

The exposure

The *exposure* of the investor to the counterparty at a date $t \in [0, \tau_{N+1}]$ is

$$\varepsilon_{\bar{N}}(t,T) := \sum_{i=1}^{N} b_i \, S_i(t,T) \, \mathbf{1}_{\{\tau_i \geqslant t\}}, \tag{5.30}$$

the value of the netting set at t, each contract weighted by the number of units held (Bo and Ceci, 2020, eq. (30)). The sum runs to N and not to $N+1$: the claim of (5.16) written on the counterparty is excluded, as Section 5.2 has it. The contracts are aggregated before any truncation is applied, the positive part of (5.34) acting on $\varepsilon_{\bar{N}}$ and not on the individual terms. And the indicator carries a weak inequality, so a contract remains inside the exposure at the instant of its own default, while the interval $[0, \tau_{N+1}]$ is closed on the right because the exposure is evaluated exactly at τ_{N+1}.

Since $H_i(t) = \mathbf{1}_{\{\tau_i \leqslant t\}}$, splitting $\mathbf{1}_{\{\tau_i \geqslant t\}} = (1 - H_i(t)) + \mathbf{1}_{\{\tau_i = t\}}$ writes (5.30) as

$$\varepsilon_{\bar{N}}(t,T) = \sum_{i=1}^{N} b_i \Big[\big(1 - H_i(t)\big) S_i(t,T) + S_i(\tau_i,T) \mathbf{1}_{\{\tau_i = t\}} \Big], \tag{5.31}$$

the value of the contracts still alive plus whatever one of them is worth at the very instant of its own default. Evaluating at $t = \tau_{N+1}$ kills the second term: the processes $H_1, \ldots, H_{N+1}$ do not jump simultaneously, by (5.9), so $\mathbb{Q}(\tau_i = \tau_{N+1}) = 0$ for $i = 1, \ldots, N$ and

$$\varepsilon_{\bar{N}}(\tau_{N+1},T) = \sum_{i=1}^{N} b_i \big(1 - H_i(\tau_{N+1})\big) \, S_i(\tau_{N+1},T) \qquad \text{a.s.} \tag{5.32}$$

(Bo and Ceci, 2020, eq. (36)). This is the one step at which the hypothesis of Definition 5.12 is used.

Write $x := X(\tau_{N+1}-)$ and $z := H(\tau_{N+1}-)$ for the intensity vector and the default

state just before the counterparty defaults. The two factors of (5.32) read that instant differently. The indicator does not jump at τ_{N+1} for $i \leqslant N$, so $1 - H_i(\tau_{N+1}) = 1 - z_i$, a quantity of the state before the jump. The price is read after it: by (5.6) the state moves to z^{N+1} and the intensity vector to $x + w_{N+1}$, and on $\{z_i = 0\}$ the representation (5.20) reduces to $S_i = F_i^{\mathrm{cds}}$. Hence

$$\varepsilon_{\bar{N}}(\tau_{N+1}, T) = \sum_{i=1}^{N} b_i\big(1 - z_i\big)\, F_i^{\mathrm{cds}}\big(\tau_{N+1},\, x + w_{N+1},\, z^{N+1}\big), \tag{5.33}$$

which is the exposure in the integrand of Bo and Ceci (2020, Lemma 3.6), read at the CDS portfolio.

Who is alive is read before the jump, what it is worth after it. That asymmetry is the one point at which contagion enters the exposure: the weights w_{iN+1}, $i = 1, \ldots, N$, raise the intensities of the surviving reference entities at the moment the counterparty defaults, and so change the prices at which the netting set is aggregated. The factor $1 - z_i$ is also where the indicator $\mathbf{1}_{\{\tau_S < \tau_R\}}$ of (4.9) has gone: it switches off the contracts whose reference entity has defaulted first, and with a portfolio it has to do so contract by contract, so it can no longer stand outside the expectation.

The adjustment

The loss rate of the counterparty is $L_{N+1}(t) := L_{N+1}(H(t))$, the percentage of the exposure that the investor fails to recover when the counterparty defaults; it is a deterministic function on $\mathcal{S}$, and a different object from the loss rates L_i, $i \leqslant N$, of the reference entities, with which it shares a letter (Bo and Ceci, 2020, Section 2.6). The *credit value adjustment* of the CDS portfolio is the market value of that loss, the replacement cost incurred at τ_{N+1}:

$$\mathrm{CVA}_{\bar{N}}(t, T) = \mathbb{E}^{\mathbb{Q}}\Big[L_{N+1}(\tau_{N+1})\, \mathbf{1}_{\{t < \tau_{N+1} \leqslant T\}}\, \big(\varepsilon_{\bar{N}}(\tau_{N+1}, T)\big)^{+} \,\Big|\, \mathcal{G}(t)\Big], \tag{5.34}$$

as in Bo and Ceci (2020, eq. (31)), following Brigo et al. (2014). No discount factor appears, the interest rate being zero; with $r \neq 0$ the factor $e^{-r(\tau_{N+1} - t)}$ would enter and nothing else would change.

The three factors of (5.34) are the three channels of counterparty risk, and only the last is moved by contagion. The indicator is the trigger, and makes the adjustment vanish on $\{\tau_{N+1} \leqslant t\}$. The loss rate is the severity, read — like the price — in the state after the jump, so that $L_{N+1}(\tau_{N+1}) = L_{N+1}(z^{N+1})$. The positive part of the exposure is the amount at risk, and selects the states in which the counterparty

owes the investor rather than the reverse. Substituting (5.33),

$$\mathrm{CVA}_{\bar{N}}(t,T) = \mathbb{E}^{\mathbb{Q}}\Bigg[L_{N+1}(z^{N+1})\,\mathbf{1}_{\{t<\tau_{N+1}\leqslant T\}} \times \left(\sum_{i=1}^{N} b_i(1-z_i)\,F_i^{\mathrm{cds}}\big(\tau_{N+1}, x+w_{N+1}, z^{N+1}\big)\right)^{+} \Bigg|\ \mathcal{G}(t)\Bigg], \tag{5.35}$$

with x and z as above.

The premium rates do not appear in (5.35). They have not left the problem: each sits inside the corresponding F_i^{cds}, as the constant source of (5.26). The adjustment depends on the contractual premiums only through the prices. The positive part, on the other hand, is the only non-linearity in the construction: the pricing systems of Section 5.3 are linear, and $(\cdot)^+$ enters exactly once, here.

Netting

Let

$$\mathrm{CVA}_{\bar{N}}^{\mathrm{gross}}(t,T) := \sum_{i=1}^{N} \mathbb{E}^{\mathbb{Q}}\Big[L_{N+1}(\tau_{N+1})\,\mathbf{1}_{\{t<\tau_{N+1}\leqslant T\}} \times \big(b_i\,S_i(\tau_{N+1},T)\,\mathbf{1}_{\{\tau_i\geqslant\tau_{N+1}\}}\big)^{+} \Big|\ \mathcal{G}(t)\Big] \tag{5.36}$$

be the adjustment computed contract by contract, as it would be with no netting agreement in force. By the subadditivity (4.14) of the positive part, applied to the summands of (5.30) at $t=\tau_{N+1}$, and by the monotonicity of the conditional expectation against the non-negative factor $L_{N+1}\mathbf{1}_{\{t<\tau_{N+1}\leqslant T\}}$,

$$\mathrm{CVA}_{\bar{N}}(t,T) \leqslant \mathrm{CVA}_{\bar{N}}^{\mathrm{gross}}(t,T) \qquad \text{a.s.} \tag{5.37}$$

Equality holds exactly when the quantities $b_iS_i(\tau_{N+1},T)$ have a common sign on the triggering event: the reduction requires the *values* $b_iS_i(\tau_{N+1},T)$, and not merely the weights b_i, to offset one another. Two positions bought from the counterparty, $b_i, b_k > 0$, still leave room for $S_i(\tau_{N+1},T) > 0 > S_k(\tau_{N+1},T)$ if the spread of one reference entity has widened and the other's tightened since the two contracts were entered, by (2.12), and netting reduces the adjustment all the same. This is the content of (4.15) restated for the present model.

Remark 5.18 (there is no expected exposure to factor out)**.** The step that produced (3.5), conditioning on $\{\tau_S = t\}$ and integrating the expected exposure (3.2) against

the law of the counterparty's default time, is unavailable here, and fails twice. The exposure $\varepsilon_{\bar{N}}(\tau_{N+1}, T)$ depends on τ_{N+1} not only through its time argument but through the state, which the jump has displaced by w_{N+1}; and the row of the counterparty in the contagion matrix raises its intensity at each default of a reference entity, so that τ_{N+1} depends on the default history of the netting set even when $w_{N+1} = 0$. Hence, in general,

$$\mathbb{E}^{\mathbb{Q}}\big[(\varepsilon_{\bar{N}}(\tau_{N+1}, T))^{+} \;\big|\; \tau_{N+1} = s\big] \neq \mathbb{E}^{\mathbb{Q}}\big[(\varepsilon_{\bar{N}}(s, T))^{+}\big], \tag{5.38}$$

and the adjustment is not determined by the marginal law of τ_{N+1} together with an exposure profile.

The adjustment as a valuation equation

The adjustment is not the price of a defaultable claim in the sense of Definition 5.7. Formally it is a claim paying nothing at maturity, nothing along its life, and the single amount $L_{N+1}\,(\varepsilon_{\bar{N}}(\tau_{N+1}, T))^{+}$ at τ_{N+1}; but Definition 5.7 requires the payment at default to be a function of the default state alone, and this one depends on the intensities as well, through the prices F_i^{cds}. The systems of Section 5.3 do not apply to it, and it carries a Cauchy problem of its own.

On $[0, T) \times \mathbb{R}_{+}^{N+1} \times \mathcal{S}$, let

$$\Phi(t, x, z) := L_{N+1}(z^{N+1}) \left(\sum_{i=1}^{N} b_i (1 - z_i)\, F_i^{\mathrm{cds}}\big(t, x + w_{N+1}, z^{N+1}\big) \right)^{+} (1 - z_{N+1})\, x_{N+1}, \tag{5.39}$$

and consider the Cauchy problem of Bo and Ceci (2020, Section 4.1),

$$\Big(\frac{\partial}{\partial t} + \mathcal{A}\Big) g^{\mathrm{cds}}(t, x, z) + \Phi(t, x, z) = 0, \qquad g^{\mathrm{cds}}(T, x, z) = 0. \tag{5.40}$$

The three factors of (5.39) are the three factors of (5.34): the loss rate of the counterparty read in the state after the jump, the aggregate exposure (5.33) at that same state, and the counterparty's default intensity, which is the trigger in intensity form and is alive only while the counterparty is.

Proposition 5.19 (the adjustment is the solution)**.** *Let Assumptions 5.1 and 5.2 hold, let the functions F_i^{cds}, $i = 1, \dots, N$, be the bounded classical solutions of* (5.23)–(5.24)*, and suppose* (5.40) *admits a unique bounded classical solution g^{cds}. Then*

$$\mathrm{CVA}_{\bar{N}}(t, T) = g^{\mathrm{cds}}\big(t, X(t), H(t)\big), \qquad t \in [0, T]. \tag{5.41}$$

In Bo and Ceci (2020, proof of Proposition 3.7, eqs. (66)–(68)) the same problem characterizes, before the counterparty's default, the conditional expectation of the loss $L_{N+1}(\tau_{N+1})\mathbf{1}_{\{\tau_{N+1}\leqslant T\}}\,(\varepsilon_{\bar{N}}(\tau_{N+1},T))^+$ under the measure used there for hedging; with zero risk premia that measure is $\mathbb{Q}$ (Bo and Ceci, 2020, Section 3.3), and the expectation is (5.34).

The identification is obtained as Bo and Ceci (2020, eq. (27)) obtain that of the prices, through the Feynman–Kac formula. The jump of H_{N+1} has $\mathbb{Q}$-intensity $(1-H_{N+1}(s))X_{N+1}(s)$ by (5.8), and compensating it writes (5.35) as the $\mathcal{G}(t)$-conditional expectation of $\int_t^T \Phi(s,X(s),H(s))\,ds$. The pair (X,H) is a Markov process under $\mathbb{Q}$ with generator (5.21) by Proposition 5.3, and the Feynman–Kac formula, with the localization used in the proof of Proposition 5.15, identifies the bounded classical solution of (5.40), evaluated along that pair, with this expectation.

The pricing functions enter (5.40) through Φ alone. The CVA system is therefore not coupled to the pricing systems but driven by them: the F_i^{cds} are solved first, everywhere on the shifted lattice that (5.39) reaches, and (5.40) is then one further problem of the same kind. It is linear in g^{cds}, the positive part acting on data and not on the unknown.

5.5 The explicit case $N = 1$

Take $N = 1$: one reference entity, name 1, and the counterparty, name 2. The default state space $\mathcal{S} = \{0,1\}^2$ has four elements, the netting set is the single contract $i = 1$, and the contagion weights form the two columns $w_1 = (0, w_{21})^\top$ and $w_2 = (w_{12}, 0)^\top$, the diagonal entries vanishing by Remark 5.5. In this section $\widetilde{X}$ abbreviates the auxiliary diffusion $\widetilde{X}^{t,x}$ of (5.7), started at x at time t; under (5.11) its two components are independent scalar square-root diffusions. Figure 5.1 shows the four states, the default that carries the model from each one to the next, and the shift the intensity vector receives on the way.

Corollary 5.20 (the four states)**.** *Under Assumptions 5.1 and 5.2, with $N = 1$ and*

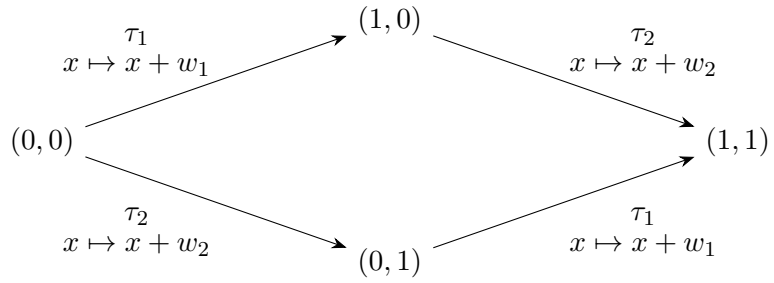


Figure 5.1: The default states at $N = 1$ and the transitions between them. Each arrow is the default of one name: it sets that name's component of the default state to 1 and shifts the intensity vector by the corresponding column of the contagion matrix. The recursion of Proposition 5.15 runs against the arrows, from $(1, 1)$ back to $(0, 0)$.

$i = 1$,

$$F_1^{\mathrm{cds}}\big(t, x, (1,1)\big) = L_1\big((1,1)\big), \tag{5.42}$$

$$\begin{aligned} F_1^{\mathrm{cds}}\big(t, x, (1,0)\big) &= L_1\big((1,0)\big)\mathbb{E}\Big[e^{-\int_t^T \widetilde{X}_2(s)\,ds}\Big] \\ &\quad + L_1\big((1,0)\big)\mathbb{E}\Big[\int_t^T \widetilde{X}_2(s)\, e^{-\int_t^s \widetilde{X}_2(u)\,du}\,ds\Big], \end{aligned} \tag{5.43}$$

$$F_1^{\mathrm{cds}}\big(t, x, (0,1)\big) = \mathbb{E}\Big[\int_t^T \big\{L_1\big((1,1)\big)\widetilde{X}_1(s) - \varepsilon_1\big\}\, e^{-\int_t^s \widetilde{X}_1(u)\,du}\,ds\Big], \tag{5.44}$$

and

$$\begin{aligned} F_1^{\mathrm{cds}}\big(t, x, (0,0)\big) = \mathbb{E}\Big[\int_t^T \Big\{ & L_1\big((1,0)\big)\widetilde{X}_1(s) + F_1^{\mathrm{cds}}\big(s, \widetilde{X}(s) + w_2, (0,1)\big)\widetilde{X}_2(s) \\ & - \varepsilon_1\Big\}\, e^{-\int_t^s (\widetilde{X}_1 + \widetilde{X}_2)(u)\,du}\,ds\Big]. \end{aligned} \tag{5.45}$$

Proof. Each state is (5.28) with the surviving names and the successors written out. At $(1, 1)$ no name survives and (5.29) applies. At $(1, 0)$ the survivor is name 2; the premium term of (5.26) is switched off by $z_1 = 1$ and the compensator term is switched on, the single successor is $(1, 1)$, where F_1^{cds} is the constant $L_1((1, 1))$ by (5.42) and the shift w_2 is therefore immaterial, and the two terms combine into $L_1((1, 0))x_2$; the terminal datum is $L_1((1, 0))$. At $(0, 1)$ the survivor is name 1; the premium is on and the compensator off, the single successor is again $(1, 1)$, and the terminal datum is zero. At $(0, 0)$ both names survive, the killing rate is $x_1 + x_2$, the successors are $(1, 0)$ — where F_1^{cds} is constant, so the shift w_1 is immaterial — and $(0, 1)$, and the terminal datum is zero. □

The formulas (5.42)–(5.44) are those printed in Bo and Ceci (2020, Section 4.1). Equation (5.43) is a constant: for any non-negative $\widetilde{X}_2$,

$$\int_t^T \widetilde{X}_2(s)\, e^{-\int_t^s \widetilde{X}_2(u)\,du}\, ds = 1 - e^{-\int_t^T \widetilde{X}_2(u)\,du} \qquad \text{a.s.}, \tag{5.46}$$

so the two terms of (5.43) sum to $L_1((1,0))$, as Proposition 5.17 requires, and the price of the contract after the default of its reference entity is zero. A sign error in the killing rate, in the source or in the terminal datum breaks the collapse, so (5.43) checks (5.28) at a state where the answer is known independently.

In (5.44) the killing rate and the source involve $\widetilde{X}_1$ alone, so $F_1^{\mathrm{cds}}(\cdot,\cdot,(0,1))$ depends on x through x_1 only, the two components of $\widetilde{X}$ being independent. The shift in (5.45) therefore acts as $x_1 \mapsto x_1 + w_{12}$, and w_{12} — the impact on the reference entity of the default of the counterparty — is the one entry of the contagion matrix that reaches the price of the traded contract.

The adjustment at $N = 1$

At $N = 1$ the adjustment vanishes in three of the four states. Where $z_2 = 1$ the factor $1 - z_2$ of (5.39) is zero, so $\Phi \equiv 0$; with the zero terminal datum this gives $g^{\mathrm{cds}}(\cdot,\cdot,(1,1)) = 0$, and then $g^{\mathrm{cds}}(\cdot,\cdot,(0,1)) = 0$, its only successor being $(1,1)$. At $(1,0)$ the counterparty is alive but the reference entity has defaulted, so the sum inside the positive part of (5.39) carries the factor $1 - z_1 = 0$, $\Phi \equiv 0$ again, and the only successor is once more $(1,1)$: the netting set is empty and there is nothing further to lose on the counterparty. In the remaining state both successors vanish and the same computation that gave Proposition 5.15, applied to (5.40), leaves a killing rate $x_1 + x_2$, a zero terminal datum and the single source $\Phi(\cdot,\cdot,(0,0))$:

$$\begin{aligned} g^{\mathrm{cds}}\big(t,x,(0,0)\big) = L_2\big((0,1)\big)\mathbb{E}\Bigg[\int_t^T \widetilde{X}_2(s)\, \Big(b_1 F_1^{\mathrm{cds}}\big(s,\widetilde{X}(s)+w_2,(0,1)\big)\Big)^+ \\ \times\, e^{-\int_t^s (\widetilde{X}_1+\widetilde{X}_2)(u)\,du}\, ds\Bigg], \end{aligned} \tag{5.47}$$

which is the formula printed in Bo and Ceci (2020, Section 4.1) and, by Proposition 5.19, the adjustment itself.

The discount factor of (5.47) is the joint survival of both names, and not of the counterparty alone: if the reference entity defaults first the state moves to $(1,0)$, where the adjustment is zero, so the integral has to stop counting at the first default of either name. And the exposure is evaluated at $\widetilde{X}(s) + w_2$ in the state $(0,1)$, after the jump in both arguments, while the multiplier $\widetilde{X}_2(s)$ — the intensity at which

the counterparty defaults — is read before it. The whole adjustment rests on the single scalar function (5.44); the pre-default price (5.45) does not enter it.

Proposition 5.21 (the direction of the contagion effect)**.** *Let $N = 1$, $L_1((1,1)) \geqslant 0$ and $\varepsilon_1 \geqslant 0$. If $b_1 > 0$, the map $w_{12} \mapsto g^{\mathrm{cds}}(t, x, (0,0))$ is non-decreasing on $[0, \infty)$; if $b_1 < 0$ it is non-increasing.*

Proof. By (5.46) applied to $\widetilde{X}_1$, (5.44) may be written as

$$F_1^{\mathrm{cds}}\big(t, x, (0,1)\big) = L_1\big((1,1)\big)\Big(1 - \mathbb{E}\big[e^{-\int_t^T \widetilde{X}_1(u)\,du}\big]\Big) - \varepsilon_1 \mathbb{E}\Big[\int_t^T e^{-\int_t^s \widetilde{X}_1(u)\,du}\,ds\Big], \tag{5.48}$$

the default leg minus the premium leg. Under (5.11) the component $\widetilde{X}_1$ solves a scalar equation driven by W_1 alone: the square-root diffusion (2.23), with speed ν_1, long-run level κ_1/ν_1 and volatility σ_1, started at x_1 at time t. For $s \in [t, T]$ the expectation $\mathbb{E}[e^{-\int_t^s \widetilde{X}_1(u)\,du}]$ is therefore the exponential of an affine function of x_1, whose slope is given in closed form by McNeil et al. (2015, Section 10.6.2, eq. (10.69)), the rate integrated being the diffusion itself. The denominator of that formula is positive, so the slope has the sign of $1 - e^{(s-t)(\nu_1^2 + 2\sigma_1^2)^{1/2}} \leqslant 0$. Both expectations in (5.48) are therefore non-increasing in x_1, the second after exchanging expectation and ds-integral, its integrand being non-negative. The first enters with a minus sign inside a bracket multiplied by $L_1((1,1)) \geqslant 0$ and the second is multiplied by $-\varepsilon_1 \leqslant 0$, so (5.48) is non-decreasing in x_1. In (5.47) the weight w_{12} enters only through the first argument of $F_1^{\mathrm{cds}}(s, \widetilde{X}(s) + w_2, (0,1))$, which depends on it as $x_1 \mapsto x_1 + w_{12}$; the map $u \mapsto (b_1 u)^+$ is non-decreasing for $b_1 > 0$ and non-increasing for $b_1 < 0$; and the remaining factors $L_2((0,1)) \geqslant 0$, $\widetilde{X}_2(s) \geqslant 0$ and the discount factor do not involve w_{12}, $\widetilde{X}$ having no jumps. Monotonicity passes through the expectation. □

Proposition 5.21 compares the adjustment across values of w_{12}, within one family of models. It is the counterpart of (3.8) in that family, not that inequality itself: there the adjustment under independence keeps the law of τ_S and the expected exposure of the model, whereas w_{12} also moves the law of τ_1 and the price of the contract before the counterparty's default. At $w_{12} = w_{21} = 0$ the two default times are independent, the Brownian motions being independent already by (5.11), and the adjustment computed there is a lower bound for a protection buyer and an upper bound for a protection seller. Equation (5.47) itself depends on w_{12} alone: w_{21} can only act after τ_1, where Proposition 5.17 has already fixed the netting-set CDS at zero, so setting $w_{12} = 0$ alone already isolates the value used below. The sign of b_1 decides which bound applies: contagion raises the value of the protection the investor

holds exactly when the counterparty defaults, and that is a loss only for the party owed the protection. What the proposition does not give is the size of the effect.

Remark 5.22 (what one contract cannot show)**.** The positive part in (5.47) truncates a single term, so at $N = 1$ the inequality (5.37) holds with equality and the netting agreement does nothing. The order of summation and truncation that (5.30) imposes, and the offset between positions of opposite sign that it produces, are invisible with one contract, as is the contagion between reference entities: the weights w_{ij} with $i, j \leqslant N$ do not exist at $N = 1$.

Chapter 6

Numerical Experiments

By Proposition 5.19, the adjustment at time zero of a portfolio in which no name has defaulted, with deterministic initial intensities $\chi = (\chi_1, \ldots, \chi_{N+1})$, is

$$\mathrm{CVA}_{\bar{N}}(0, T) = g^{\mathrm{cds}}(0, \chi, \mathbf{0}), \tag{6.1}$$

where $\mathbf{0} \in \mathcal{S}$ is the default state whose components are all zero. The function g^{cds} solves (5.40), whose source (5.39) is built from the pricing functions F_i^{cds}, $i = 1, \ldots, N$, which solve (5.23)–(5.24). At $N = 1$ the price after the counterparty's default involves the reference entity's intensity alone: by (5.48) it is expressed through the expectations $\mathbb{E}[e^{-\int_t^s \widetilde{X}_1(u)\,du}]$, which are known in closed form for the square-root diffusion (McNeil et al., 2015, Section 10.6.2). The price at $(0, 0)$, (5.45), and the adjustment, (5.47), contain that price inside a further expectation along the auxiliary diffusion, and reduce no further.

Two methods apply to such quantities: simulation, and the numerical solution of the Cauchy problems. The adjustment is estimated by simulation in the algorithm of Section 3.5, and according to McNeil et al. (2015, Section 17.4.4) credit value adjustments are mostly computed by Monte Carlo simulation. Bo et al. (2019, Section 5 and Appendix B) solve by a Crank–Nicolson finite-difference scheme the Hamilton–Jacobi–Bellman equations of an optimal allocation problem with two names, square-root intensities and contagion, one equation for each default state with a surviving name. The pricing and the adjustment systems are solved here by finite differences, and a simulation serves as a check, at $N = 1$ and, when the reference entities do not act on each other's intensities, at $N = 2$.

6.1 The numerical problem and the scheme

The states to be solved

Substituting (5.21) into (5.40) and splitting the bracket as in the proof of Proposition 5.15 writes the equation for the adjustment at each state in the same transport form as the pricing equations: for $z \in \mathcal{S}$,

$$\Big(\frac{\partial}{\partial t} + \widetilde{\mathcal{A}}\Big) g^{\mathrm{cds}}(t,x,z) - c_z(x)\, g^{\mathrm{cds}}(t,x,z) + \sum_{j\,:\,z_j=0} g^{\mathrm{cds}}(t, x+w_j, z^j)\, x_j + \Phi(t,x,z) = 0, \tag{6.2}$$

with $g^{\mathrm{cds}}(T,x,z) = 0$ and the killing rate c_z of (5.25). This is the form (5.27) of the pricing equations, with the successor terms of g^{cds} in place of those of F_i^{cds} and with Φ as source, so that one scheme serves both systems.

Proposition 5.17 fixes $F_i^{\mathrm{cds}}(\cdot,\cdot,z) = L_i(z)$ on the states with $z_i = 1$, so each pricing function is unknown on the 2^N states with $z_i = 0$. The adjustment vanishes on a set of the same kind.

Lemma 6.1 (where the adjustment vanishes)**.** *Under the hypotheses of Proposition 5.19, let*

$$\mathcal{Z}_0 := \{z \in \mathcal{S} : z_{N+1} = 1\} \cup \{z \in \mathcal{S} : z_1 = \cdots = z_N = 1\}.$$

Then $g^{\mathrm{cds}}(t,x,z) = 0$ *for every* $(t,x) \in [0,T] \times \mathbb{R}_+^{N+1}$ *and every* $z \in \mathcal{Z}_0$.

Proof. The set $\mathcal{Z}_0$ is closed under $z \mapsto z^j$: setting a further component to 1 undoes neither $z_{N+1} = 1$ nor $z_1 = \cdots = z_N = 1$. On $\mathcal{Z}_0$ the source (5.39) vanishes, through the factor $1 - z_{N+1}$ in the first case and, in the second, because every summand inside the positive part carries a factor $1 - z_i = 0$. Proceed by downward induction on the number of surviving names. Let $z \in \mathcal{Z}_0$ and assume $g^{\mathrm{cds}}(\cdot,\cdot,z^j) = 0$ for every j with $z_j = 0$, which is vacuous when no name survives. Then $u := g^{\mathrm{cds}}(\cdot,\cdot,z)$ solves (6.2) with no source and a zero terminal datum. The representation part of the proof of Proposition 5.15 uses of u only that it is a bounded classical solution of a problem of this form, and it gives $u = 0$. □

The complement of $\mathcal{Z}_0$ has $2^{N+1} - 2^N - 1 = 2^N - 1$ elements, the two sets in the definition having 2^N and 2 elements and one in common. At $N = 2$ the adjustment is therefore to be computed at the three states $(0,0,0)$, $(1,0,0)$ and $(0,1,0)$, and each of F_1^{cds}, F_2^{cds} at four.

The computational domain

The equations are posed on $\mathbb{R}^{N+1}_+$ and are solved on the box $[0,\bar{x}_1]\times\cdots\times[0,\bar{x}_{N+1}]$, with

$$\bar{x}_i := \frac{\kappa_i}{\nu_i} + 9\,s_i + \max_j w_{ij}, \qquad s_i := \frac{\sigma_i}{\nu_i}\sqrt{\frac{\kappa_i}{2}}. \tag{6.3}$$

The constant $s_i^2 = \sigma_i^2\kappa_i/(2\nu_i^2)$ is the limit as $t\to\infty$ of the variance of $\widetilde{X}_i(t)$. Itô's formula applied to $\widetilde{X}_i$ and to $\widetilde{X}_i^2$ gives the linear equations

$$\frac{d}{dt}\mathbb{E}[\widetilde{X}_i(t)] = \kappa_i - \nu_i\mathbb{E}[\widetilde{X}_i(t)], \qquad \frac{d}{dt}\mathbb{E}[\widetilde{X}_i(t)^2] = (2\kappa_i+\sigma_i^2)\,\mathbb{E}[\widetilde{X}_i(t)] - 2\nu_i\,\mathbb{E}[\widetilde{X}_i(t)^2],$$

whose stationary point has mean κ_i/ν_i and variance s_i^2. The margin $\max_j w_{ij}$ keeps inside the box, after the shift of any single default, every point whose i-th coordinate lies below $\kappa_i/\nu_i + 9s_i$. The box enters the computed values through the grid, so configurations that are compared with each other are solved on one box: the box (6.3) of the largest contagion matrix among them, whose margin covers the others.

The discrete generator

Let $n_x \geqslant 2$ be the number of grid points on each axis, let $h_i := \bar{x}_i/(n_x-1)$, let $\mathcal{X}_h$ be the set of nodes x with $x_i = l_i h_i$, $l_i = 0,\dots,n_x-1$, and let e_i be the i-th unit vector of $\mathbb{R}^{N+1}$. Under (5.11) the generator (5.22) has no mixed derivatives,

$$\widetilde{\mathcal{A}}f(x) = \sum_{i=1}^{N+1}\Big(\mu_i(x)\,\frac{\partial f}{\partial x_i}(x) + \tfrac{1}{2}\sigma_i^2 x_i\,\frac{\partial^2 f}{\partial x_i^2}(x)\Big), \qquad \mu_i(x) = \kappa_i - \nu_i x_i,$$

and it is discretized one axis at a time: for a function u on $\mathcal{X}_h$,

$$\widetilde{\mathcal{A}}_h u(x) := \sum_{i=1}^{N+1}\Big(q_i^-(x)\,u(x-h_ie_i) + q_i^0(x)\,u(x) + q_i^+(x)\,u(x+h_ie_i)\Big). \tag{6.4}$$

At a node with $0 < x_i < \bar{x}_i$ the coefficients on axis i are those of central differences,

$$q_i^{\pm}(x) = \frac{\sigma_i^2 x_i}{2h_i^2} \pm \frac{\mu_i(x)}{2h_i}, \qquad q_i^0(x) = -\frac{\sigma_i^2 x_i}{h_i^2}, \tag{6.5}$$

wherever $h_i|\mu_i(x)| \leqslant \sigma_i^2 x_i$, and those of the upwind difference of the drift elsewhere,

$$\begin{gathered} q_i^+(x) = \frac{\sigma_i^2 x_i}{2h_i^2} + \frac{(\mu_i(x))^+}{h_i}, \qquad q_i^-(x) = \frac{\sigma_i^2 x_i}{2h_i^2} + \frac{(\mu_i(x))^-}{h_i}, \\ q_i^0(x) = -\frac{\sigma_i^2 x_i}{h_i^2} - \frac{|\mu_i(x)|}{h_i}. \end{gathered} \tag{6.6}$$

The condition $h_i|\mu_i(x)| \leqslant \sigma_i^2 x_i$ is the condition under which both central coefficients $q_i^{\pm}$ are non-negative, and the upwind coefficients are non-negative everywhere. The switch is made for that reason. Non-negative $q_i^{\pm}$, together with $c_z \geqslant 0$, are the *positive coefficient condition* of Forsyth and Labahn (2007, Condition 4.1) and Wang and Forsyth (2008, Condition 3.1), and the stencil is chosen as there, node by node: central differences wherever they satisfy the condition, forward or backward differences elsewhere (Forsyth and Labahn, 2007, Section 4.2). On a uniform grid their coefficients (Wang and Forsyth, 2008, Appendix A) are (6.5) and (6.6). At the node $x_i = l_i h_i$, $l_i \geqslant 1$, the central coefficients are non-negative exactly when

$$l_i\big(\sigma_i^2 + \nu_i h_i\big) \geqslant \kappa_i \qquad \text{and} \qquad l_i\big(\nu_i h_i - \sigma_i^2\big) \leqslant \kappa_i. \tag{6.7}$$

The first condition fails near $x_i = 0$, where the diffusion coefficient vanishes while the drift tends to $\kappa_i > 0$, if $\sigma_i^2 + \nu_i h_i < \kappa_i$; the second fails away from it only on a grid with $\nu_i h_i > \sigma_i^2$.

At the two ends of each axis the stencil is one-sided. At $x_i = 0$ the diffusion coefficient is zero and the drift $\mu_i = \kappa_i$ points into the box; the forward difference $q_i^+ = \kappa_i/h_i$, $q_i^0 = -\kappa_i/h_i$, $q_i^- = 0$ is used, and no boundary condition is needed. Where the diffusion coefficient vanishes and the drift is at least the derivative of that coefficient — here $\kappa_i \geqslant \sigma_i^2/2$, which is (5.12) — the equation itself is the boundary condition (Forsyth and Labahn, 2007, Section 3.1), and a one-sided difference enforces it, as in Forsyth and Labahn (2007, Section 4.4) and Wang and Forsyth (2008, Section 3.1). For the square-root diffusion in one variable, the solution given by the stochastic representation satisfies this boundary condition (Ekström and Tysk, 2011, Theorem 2.3 and the Example after it), which, when zero is not reached, as here, is not needed to identify the solution (Ekström and Tysk, 2011, Section 1). At $x_i = \bar{x}_i$ the drift $\mu_i(x) = \kappa_i - \nu_i \bar{x}_i$ is negative, since $\bar{x}_i > \kappa_i/\nu_i$, and it too points into the box; the backward difference $q_i^- = -\mu_i(x)/h_i$, $q_i^0 = \mu_i(x)/h_i$, $q_i^+ = 0$ is used, and the second derivative, which would require a node beyond the box, is dropped. That the second derivative is negligible for large values of the variable is the assumption from which Forsyth and Labahn (2007, Section 3.1) and d'Halluin et al. (2005, Section 2.1) derive their far boundary conditions. With it dropped, the diffusion coefficient along axis i vanishes on the face and the drift points into the box, so that the face is a boundary that needs no condition, differenced backward as in Wang and Forsyth (2008, Assumption 2.1 and Section 3.1). In every case

$$q_i^- + q_i^0 + q_i^+ = 0, \qquad q_i^{\pm} \geqslant 0, \tag{6.8}$$

so that $\widetilde{\mathcal{A}}_h$, like $\widetilde{\mathcal{A}}$, annihilates constants.

The contagion shift

The jump part of (5.21) evaluates the successor function at $x+w_j$, which is in general not a node. It is read by multilinear interpolation on the cell of $\mathcal{X}_h$ containing that point, each coordinate being first clamped to $[0,\bar{x}_i]$:

$$\mathcal{I}_h u(y)=\sum_v \lambda_v(y)\,u(v),\qquad \lambda_v(y)\geqslant 0,\qquad \sum_v \lambda_v(y)=1,$$

the sum running over the 2^{N+1} corners v of the cell and the weights $\lambda_v(y)$ being products of the fractional positions of y along each axis. The clamping moves a point that the shift carries beyond $\bar{x}_i$ back to the face of the box, and by (6.3) this can happen only from the upper margin of the box. A coupling term read by interpolation, with weights in $[0,1]$ summing to one, at a jumped point truncated at the edge of the domain, is that of the regime-switching systems of Huang et al. (2011, Section 2 and eqs. (3.6)–(3.9)), whose transitions move the underlying; d'Halluin et al. (2005, eqs. (3.8)–(3.14)) read a jump integral off the grid by the same interpolation.

Time stepping

Time is discretized backwards from maturity in n_t steps, $t_k:=T-k\Delta t$, $k=0,\dots,n_t$, $\Delta t:=T/n_t$. For a state z at which a value function — some $F_i^{\mathrm{cds}}(\cdot,\cdot,z)$ or $g^{\mathrm{cds}}(\cdot,\cdot,z)$ — is unknown, let u_z^k be its approximation on $\mathcal{X}_h$ at t_k; the known states are set once, to $L_i(z)$ by Proposition 5.17 and to zero on $\mathcal{Z}_0$ by Lemma 6.1. Every unknown state is advanced by one explicit Euler step of its transport form, (5.27) for the prices and (6.2) for the adjustment:

$$\begin{aligned}u_z^{k+1}(x)=u_z^k(x)+\Delta t\Big(&\widetilde{\mathcal{A}}_h u_z^k(x)-c_z(x)\,u_z^k(x)\\&+\sum_{j\,:\,z_j=0}\mathcal{I}_h u_{zj}^k(x+w_j)\,x_j+\vartheta_z^k(x)\Big),\qquad x\in\mathcal{X}_h,\end{aligned}\tag{6.9}$$

starting from $u_z^0=0$, which is the terminal datum of every unknown state. The source ϑ_z^k is $-\varepsilon_i$ for F_i^{cds}, the compensator term of (5.26) vanishing where $z_i=0$; for g^{cds} it is $\Phi(t_k,x,z)$ computed by (5.39) from the prices u^k, the shifted argument $x+w_{N+1}$ being read through $\mathcal{I}_h$.

In (6.9) every state reads the others at the previous time level only, so all the unknown states are advanced together and no ordering of the lattice is needed. Huang

et al. (2011, Section 3.1) evaluate the coupling between the regimes of their system at the previous time level in the same way, in a scheme implicit within each regime. An implicit step requires a linear system at every step, as does the Crank–Nicolson step of Bo et al. (2019, Appendix B). Here it would also read the successors of each state at the new time level, so that the states would have to be solved one after the other in the order of Proposition 5.14, from the states with the most defaults to the state with none. The explicit step avoids both, at the price of a restriction on Δt. Let

$$\rho := \max_{x \in \mathcal{X}_h} \sum_{i=1}^{N+1} |q_i^0(x)| + \max_{z \in \mathcal{S},\, x \in \mathcal{X}_h} c_z(x), \tag{6.10}$$

and let n_t be the smallest integer with $T/n_t \leqslant 0.9/\rho$.

With $\Delta t \rho \leqslant 1$, the right-hand side of (6.9) is a combination of old values with non-negative coefficients, plus $\Delta t \vartheta_z^k(x)$: the coefficient of $u_z^k(x)$ is $1 + \Delta t(\sum_i q_i^0(x) - c_z(x)) \geqslant 1 - \Delta t \rho \geqslant 0$, those of the neighbouring nodes are $\Delta t q_i^{\pm}(x) \geqslant 0$, and those of the interpolation corners are $\Delta t x_j \lambda_v(x + w_j) \geqslant 0$. By (6.8) and $c_z(x) = \sum_{j : z_j = 0} x_j$ these coefficients sum to $1 - \Delta t c_z(x) + \Delta t \sum_{j : z_j = 0} x_j = 1$. Two properties follow. The step is monotone: for a given source, ordered data, node by node and state by state, give ordered values, and non-negative data with a non-negative source give non-negative values. And it is bounded in the maximum norm over all states, the new value being a convex combination of old values plus $\Delta t \vartheta_z^k$. The first property is what makes the computed adjustment non-negative, as (5.34) is: its source Φ is non-negative, and so is its terminal datum. Both properties are obtained from the positive coefficient condition, as in Forsyth and Labahn (2007, Lemmas 5.1 and 5.3), and the bound across states from interpolation weights in $[0, 1]$ that sum to one, as in Huang et al. (2011, Proposition 3.1) and d'Halluin et al. (2005, Theorem 3.1). Their proofs are for stepping fully implicit in the diffusion, which needs no restriction on the time step; the explicit step has both properties under (6.10).

For schemes that satisfy the positive coefficient condition, convergence to the viscosity solution is proved in one space variable with fully implicit time stepping (Wang and Forsyth, 2008, Theorem 4.1), and is stated without proof for several variables (Wang and Forsyth, 2008, Section 8) and for systems coupled across states (Huang et al., 2011, Remark 3.3); it is not established for (6.9). The discretization error is measured rather than bounded: by refining the grid, by widening the box, at $N = 1$ against the closed form of (5.44), and against a Monte Carlo simulation, at $N = 1$ and, when the reference entities do not act on each other's intensities, at $N = 2$. After n_t steps, $u_{\mathbf{0}}^{n_t}$ approximates $g^{\mathrm{cds}}(0, \cdot, \mathbf{0})$ on $\mathcal{X}_h$, and the adjustment (6.1) is read at χ by multilinear interpolation. It is reported in basis points (bp) of the

unit notional, $1\,\mathrm{bp} = 10^{-4}$. Differences, ratios and percentages are computed from unrounded values, and may differ in the last digit from the same arithmetic on the printed ones.

6.2 The benchmark without contagion

With the contagion matrix $w = (w_{ij})$ equal to zero, the jump term of (5.10) vanishes and each intensity is a square-root diffusion driven by its own Brownian motion. The default times are then independent, as at $N = 1$ (Section 5.5, after Proposition 5.21), and so are the pairs (X_i, τ_i), $i = 1, \dots, N+1$. For $i \leqslant N$ and a constant loss rate L_i, let

$$\varphi_i(t, x_i) := \mathbb{E}\Big[\int_t^T \big(L_i \widetilde{X}_i(s) - \varepsilon_i\big) e^{-\int_t^s \widetilde{X}_i(u)\,du}\, ds\Big], \tag{6.11}$$

with $\widetilde{X}_i$ started at x_i at time t. This is the price (4.11) of a CDS on name i alone, with zero interest rate, hazard process $\widetilde{X}_i$ and the loss rate L_i in the role of the loss given default δ_R.

Proposition 6.2 (the benchmark)**.** *Let the hypotheses of Proposition 5.19 hold, let $w = 0$ and let $L_1, \dots, L_{N+1}$ be constants. Then:*

(i) for $i \leqslant N$ and every z with $z_i = 0$, $F_i^{\mathrm{cds}}(t, x, z) = \varphi_i(t, x_i)$ on $[0, T] \times \mathbb{R}_+^{N+1}$;

(ii) the adjustment at time zero is

$$\begin{aligned} \mathrm{CVA}_{\bar{N}}(0, T) &= L_{N+1} \int_0^T \mathbb{E}^{\mathbb{Q}}\big[(V_s)^+\big]\, f_{N+1}(s)\, ds, \\ V_s &:= \sum_{i=1}^{N} b_i\, \mathbf{1}_{\{\tau_i > s\}}\, \varphi_i\big(s, X_i(s)\big), \end{aligned} \tag{6.12}$$

where f_{N+1} is the density of τ_{N+1}.

Proof. (i) Fix $i \leqslant N$ and proceed by downward induction on the number of surviving names, over the states with $z_i = 0$. If i is the only survivor, the killing rate (5.25) is x_i, the only successor is z^i, where $F_i^{\mathrm{cds}} = L_i$ by Proposition 5.17, and the source (5.26) is $L_i x_i - \varepsilon_i$. The representation (5.28) then involves $\widetilde{X}_i$ alone and equals $\varphi_i(t, x_i)$. In particular φ_i, bounded by $L_i + \varepsilon_i T$, is a classical solution of

$$\Big(\frac{\partial}{\partial t} + \widetilde{\mathcal{A}}\Big)\varphi_i - x_i\, \varphi_i + L_i x_i - \varepsilon_i = 0,$$

on which $\widetilde{\mathcal{A}}$ acts through its i-th term alone, by (5.11). Let now z have further survivors and assume the claim at every z^j with $j \neq i$ and $z_j = 0$. Substitute φ_i

into (5.27) at z. With $w = 0$ each such j contributes $\varphi_i(t, x_i)x_j$ to the successor sum, which cancels the term $-x_j\varphi_i(t, x_i)$ of the killing term; the successor $j = i$ contributes $L_i x_i$; what is left is the equation displayed above. Hence φ_i solves (5.27) at z, and the representation part of the proof of Proposition 5.15, which holds for every bounded classical solution, gives $F_i^{\mathrm{cds}}(\cdot, \cdot, z) = \varphi_i$.

(ii) With $w = 0$ the intensities do not jump, so $X(\tau_{N+1}) = X(\tau_{N+1}-)$, and $1 - H_i(\tau_{N+1}-) = \mathbf{1}_{\{\tau_i > \tau_{N+1}\}}$ almost surely by (5.9). By (5.33) with $w_{N+1} = 0$ and by (i), the exposure at the counterparty's default is $\varepsilon_{\bar{N}}(\tau_{N+1}, T) = V_{\tau_{N+1}}$, and (5.34) at $t = 0$ reads $\mathrm{CVA}_{\bar{N}}(0, T) = \mathbb{E}^{\mathbb{Q}}[L_{N+1}\mathbf{1}_{\{\tau_{N+1} \leqslant T\}} \left(V_{\tau_{N+1}}\right)^+]$. The process (V_s) is a functional of $(X_i, \tau_i)_{i \leqslant N}$ and is independent of τ_{N+1}; conditioning on τ_{N+1}, as in the passage from (3.4) to (3.5), gives (6.12). □

Equation (6.12) is the adjustment under independence (3.5), with zero interest rate, the loss rate L_{N+1} in the role of the loss given default δ_S, and the value of the netting set in place of a single position. At $N = 1$ and $b_1 = 1$, (5.47) with $w_{12} = 0$ is the conditionally independent formula (4.12), with $\gamma^S = \widetilde{X}_2$, $\gamma^R = \widetilde{X}_1$, $\delta_S = L_2$ and $\widetilde{V}_t = \varphi_1(t, \widetilde{X}_1(t))$; the weight w_{21} does not enter (5.47). With L_i constant, the adjustment at $w \neq 0$ differs from (6.12) through contagion alone. The comparison of the two is not the inequality $\mathrm{CVA} > \mathrm{CVA}^{\mathrm{indep}}$ of (3.8), in which $\mathrm{CVA}^{\mathrm{indep}}$ keeps the law of the counterparty's default time and the expected exposure of one model and drops only their dependence. Setting $w = 0$ changes these as well: the weights raise the intensities of the surviving names at each default, the counterparty's among them, and the prices before any default anticipate those jumps. What is compared is the adjustment of the model without contagion with that of the model with it.

Part (i) uses of L_i only that $L_i(z^i)$ takes the same value at every z with $z_i = 0$. With a state-dependent loss rate that violates this, the default times are still independent at $w = 0$, but the price of a contract at the counterparty's default is computed at the state z^{N+1}, where the loss rates are those of (5.14) read after that default, and differs from its price at the same point x while the counterparty is alive. The exposure then depends on the counterparty's default through the loss rates alone: this is the second channel of Remark 5.11, at work with no dependence between the default times. Varying w at constant loss rates isolates the first channel, and varying the loss rates at $w = 0$ isolates the second.

The premium rates are fixed by $\varphi_i(0, \chi_i) = 0$ with $L_i = 0.60$, so that each contract is entered at par when there is no contagion and the loss rates are constant. By (5.46) applied to $\widetilde{X}_i$,

$$\varepsilon_i = \frac{L_i\big(1 - \mathbb{E}\big[e^{-\int_0^T \widetilde{X}_i(u)\,du}\big]\big)}{\int_0^T \mathbb{E}\big[e^{-\int_0^s \widetilde{X}_i(u)\,du}\big]\,ds}, \tag{6.13}$$

with $\widetilde{X}_i(0) = \chi_i$: the fair spread (2.11) with zero interest rate and survival function $s \mapsto \mathbb{E}[e^{-\int_0^s \widetilde{X}_i(u)\,du}]$. The expectations are known in closed form (McNeil et al., 2015, Section 10.6.2, eqs. (10.69)–(10.70)), and the integral in the denominator is computed by the trapezoidal rule on 400 nodes. The rates are held fixed in every configuration, so that a change in w or in the loss rates changes the value of the contracts and not their terms.

6.3 Validation at $N = 1$

Name 1, the reference entity, and name 2, the counterparty, have $(\kappa_1, \kappa_2) = (0.020, 0.030)$, $(\nu_1, \nu_2) = (1.00, 1.20)$ and $(\sigma_1, \sigma_2) = (0.150, 0.180)$, start at their long-run levels, $\chi_i = \kappa_i/\nu_i$, and satisfy (5.12). The maturity is $T = 5$, the loss rates are $L_1 = L_2 = 0.60$, and (6.13) gives $\varepsilon_1 = 119.08$ bp. With contagion, $w_{12} = 0.010$ and $w_{21} = 0.008$: the default of the counterparty raises the intensity of the reference entity by half of its long-run level.

The grid and the operators

At $n_x = 41$, 61, 81 and 121 points per axis the adjustment $g^{\mathrm{cds}}(0, \chi, (0, 0))$ is 3.4732, 3.4668, 3.4630 and 3.4626 bp. It decreases monotonically under refinement, by 0.0064, 0.0038 and 0.0004 bp, but not at a steady rate: at second order in h_i the ratios of successive changes would be 0.35 and 0.71, and they are 0.59 and 0.11. The last change is therefore no estimate of the error at 81 points. The whole sequence spans 0.0106 bp, and the change from 61 to 81 points is 0.0038 bp, 0.11% of the value; both lie well below the sampling standard error of the simulation below. The validation uses $n_x = 81$.

Both names have $\sigma_i^2 > \kappa_i$, and $\nu_i h_i \leqslant \sigma_i^2$ on every grid of this section, so that (6.7) holds at every interior node: no node is on the upwind branch, and the scheme run is central in the interior. Two parts of the scheme are checked in isolation. On a single axis, with the drift of name 1 and a test grid of 41 points on $[0, 0.5]$, $\widetilde{\mathcal{A}}_h$ applied to $f(x) = x$ returns the drift $\mu(x)$ at every interior node, each first difference being exact on linear functions. At the volatility 0.15 of name 1, where the central coefficients (6.5) are in use, the largest deviation is 3.1×10^{-15}; at the test volatility 0.02, at which (6.7) fails at every interior node and the upwind coefficients (6.6) are in use, it is 2.7×10^{-15}. And the interpolation $\mathcal{I}_h$ agrees with an independent implementation, SciPy's `RegularGridInterpolator`, the largest difference being 4.4×10^{-16}.

States with a known value

The constant $L_1(z)$ of Proposition 5.17 solves the step (6.9) exactly: $\widetilde{\mathcal{A}}_h$ annihilates constants by (6.8), $\mathcal{I}_h$ reproduces them, and the killing, successor and compensator terms cancel node by node, as in the proof of Proposition 5.17. Marching the states with $z_1 = 1$ instead of setting them therefore tests the implementation of the source (5.26). Marched from their terminal data, they return 0.600000 at $(1, 0)$ and at $(1, 1)$ with constant loss rates; with the state-dependent loss rate $L_1(z) = 0.60 + 0.08\, z_2$, for which the compensator term does not vanish, they return 0.600000 and 0.680000, the values of $L_1((1, 0))$ and $L_1((1, 1))$. The three states of $\mathcal{Z}_0$, marched in the same way, return an adjustment of 0.00000 bp to five decimals, as Lemma 6.1 requires.

At $w = 0$, Proposition 6.2 gives $F_1^{\mathrm{cds}}(\cdot, \cdot, (0, 0)) = F_1^{\mathrm{cds}}(\cdot, \cdot, (0, 1)) = \varphi_1$, and ε_1 makes $\varphi_1(0, \chi_1) = 0$. The scheme returns -3.58×10^{-7} at both states, the two values agreeing to within 10^{-8}.

Closed form and simulation, with contagion

With the weights above, the price after the counterparty's default is still φ_1: (5.44) does not involve w, the shift being immaterial against the constant successor $(1, 1)$, as in the proof of Corollary 5.20. The scheme returns $F_1^{\mathrm{cds}}(0, \chi, (0, 1)) = -4.18 \times 10^{-7}$, against the exact value 0. The price at $(0, 0)$ and the adjustment have no closed form, and are checked against a Monte Carlo simulation of the model.

The simulation is the algorithm of Section 3.5, with the contagion model in place of the common factor. On a uniform grid of n_{MC} steps over $[0, T]$, the intensities follow the Euler scheme of (5.10), the value being set to zero whenever a step makes it negative. Each name i receives an independent unit exponential threshold ζ_i and defaults at the end of the first step at which its integrated intensity $\int_0^t X_i(s)\, ds$, accumulated along the path, reaches ζ_i; at the default of j the whole intensity vector is shifted by w_j, as in (5.6). In the construction of the model by Bo et al. (2019, Appendix A, proof of Lemma 2.1) the thresholds are drawn afresh: at the start and after each default, every surviving name receives a new unit exponential threshold and defaults when its intensity, integrated from the last default, reaches it. The two constructions give the same law. Up to the first default they coincide. Conditionally on the Brownian motions and on the time and the name of the first default, each surviving name i has ζ_i larger than its intensity integrated up to that default, and by the memoryless property of the exponential distribution and the independence of the thresholds the excesses are independent unit exponentials, which is the law of the thresholds drawn afresh. The argument repeats at each later default.

The price $F_1^{\mathrm{cds}}(0, \chi, (0,0))$ is estimated by the sample mean of the dividend process (5.15) at T, and the adjustment by the sample mean of

$$L_2 \, \mathbf{1}_{\{\tau_2 \leqslant T,\, \tau_2 < \tau_1\}} \, \left(b_1 \varphi_1(\tau_2, X_1(\tau_2))\right)^+ ,$$

with $b_1 = 1$, $X_1(\tau_2)$ read after the shift and φ_1 evaluated through the closed-form expectations used in (6.13); this is (5.34) with the price after the counterparty's default replaced by its closed form. Both estimators use $M = 200\,000$ paths.

Table 6.1: The price and the adjustment at $(0,0)$ at $N = 1$, with contagion: finite differences against the Monte Carlo simulation at three step counts, on shared paths. The $\pm$ is one sampling standard error, with $M = 200\,000$ paths.

	$F_1^{\mathrm{cds}}(0, \chi, (0,0))$	$g^{\mathrm{cds}}(0, \chi, (0,0))$ (bp)
Finite differences, $n_x = 81$	5.2868×10^{-4}	3.4630
Monte Carlo, $n_{\mathrm{MC}} = 250$	$6.9647 \times 10^{-4} \pm 4.14 \times 10^{-4}$	3.4861 ± 0.0362
Monte Carlo, $n_{\mathrm{MC}} = 1000$	$6.8406 \times 10^{-4} \pm 4.14 \times 10^{-4}$	3.4835 ± 0.0361
Monte Carlo, $n_{\mathrm{MC}} = 4000$	$6.9213 \times 10^{-4} \pm 4.14 \times 10^{-4}$	3.4718 ± 0.0359

Table 6.1 gives the results. The estimates of the price move by less than one standard error across the three step counts, and the finite-difference value lies within one standard error of each of them. The standard error measures the sampling error alone; the simulation also discretizes time, in the Euler step and in placing each default at the end of the step in which the threshold is crossed. The three runs share their thresholds and their Brownian paths, the increments being drawn on the grid of 4000 steps and summed over the coarser steps, so that the difference between two step counts is estimated path by path, with a standard error of its own. For the adjustment it is 0.0026 ± 0.0084 bp from 250 to 1000 steps and 0.0117 ± 0.0056 bp from 1000 to 4000, both well below the sampling standard error of each estimate; against its own standard error the second is 2.1 standard errors from zero, the first 0.3. A bias of first order in the time step would make the first change four times the second; here the first is the smaller, and at this number of paths the dependence of the time-discretization error on n_{MC} is not resolved. The finite-difference value lies 0.64, 0.57 and 0.25 standard errors from the estimates with 250, 1000 and 4000 steps.

The direction of the effect

At $n_x = 61$, on the box of the largest weight, $w_{12} = 0.040$, and with $w_{21} = 0.008$, the adjustment for $w_{12} = 0$, 0.005, 0.010, 0.020 and 0.040 is 1.6526, 2.4323, 3.4571, 6.2848 and 12.4974 bp, non-decreasing as Proposition 5.21 requires for $b_1 > 0$. The value

at $w_{12} = 0$ is the benchmark (6.12), since w_{21} does not enter (5.47). At $w_{12} = 0.010$ the adjustment is 2.09 times the benchmark.

With the state-dependent loss rate $L_1(z) = 0.60 + 0.08\, z_2$ and the same weights, the adjustment at $n_x = 81$ rises from 3.4630 to 6.2343 bp. The only loss rate that changes on the states reached by (5.47) is $L_1((1,1))$, from 0.60 to 0.68, and by (5.48) it raises the price after the counterparty's default by $0.08\,(1 - \mathbb{E}[e^{-\int_t^T \tilde{X}_1(u)\,du}])$, the change in the loss rate times the probability that the reference entity defaults over the remaining life of the contract. That price is the exposure which (5.47) truncates, and the premium, fixed at par for $L_1 = 0.60$, does not change with it.

6.4 The experiment at $N = 2$

Names 1 and 2 are the reference entities and name 3 is the counterparty. Their parameters are in Table 6.2. The speeds of mean reversion are equal, so that the names differ in level and volatility only, and the counterparty has the highest long-run intensity. Every name starts at its long-run level, the maturity is $T = 5$, and (5.12) holds for every name. Name 1 has the parameters of the reference entity of Section 6.3, and the same premium rate.

Table 6.2: Parameters at $N = 2$, the premium rates (6.13), and the box (6.3) with its grid spacing at $n_x = 33$ for $w = \bar{w}$ of (6.14). Names 1 and 2 are the reference entities, name 3 the counterparty.

	κ_i	ν_i	σ_i	χ_i	ε_i (bp)	$\bar{x}_i$	h_i
Name 1	0.020	1.00	0.150	0.020	119.08	0.1650	0.00516
Name 2	0.025	1.00	0.160	0.025	148.69	0.1950	0.00609
Name 3	0.030	1.00	0.180	0.030	—	0.2364	0.00739

The contagion matrix

The reference contagion matrix, with w_{ij} in row i and column j, is

$$\bar{w} = \begin{pmatrix} 0 & 0.006 & 0.010 \\ 0.005 & 0 & 0.009 \\ 0.008 & 0.007 & 0 \end{pmatrix} = w^{\mathrm{E}} + w^{\mathrm{C}} + w^{\mathrm{I}}, \tag{6.14}$$

where w^{E} keeps the entries $w_{13} = 0.010$ and $w_{23} = 0.009$, the impact of the counterparty's default on the reference entities; w^{C} keeps $w_{31} = 0.008$ and $w_{32} = 0.007$, the impact of a reference entity's default on the counterparty; and w^{I} keeps $w_{12} = 0.006$

and $w_{21} = 0.005$, the contagion between the two reference entities. The default of the counterparty raises the intensity of name 1 by half of its long-run level and that of name 2 by 36% of it. The blocks w^{E} and w^{C}, which link each reference entity to the counterparty, are given larger entries than w^{I}, which links the reference entities to each other.

Where the weights enter

By Lemma 6.1 the adjustment is unknown at $(0,0,0)$, $(1,0,0)$ and $(0,1,0)$. At $(1,0,0)$ both successors lie in $\mathcal{Z}_0$, and (6.2) reads

$$\Big(\frac{\partial}{\partial t} + \widetilde{\mathcal{A}}\Big) g^{\mathrm{cds}} - (x_2 + x_3)\, g^{\mathrm{cds}} + L_3\big((1,0,1)\big)\,\big(b_2 F_2^{\mathrm{cds}}\big(t, x + w_3, (1,0,1)\big)\big)^+ x_3 = 0, \tag{6.15}$$

with $g^{\mathrm{cds}} = g^{\mathrm{cds}}(t,x,(1,0,0))$, and symmetrically at $(0,1,0)$. At $(0,0,0)$ the successor $(0,0,1)$ lies in $\mathcal{Z}_0$, and

$$\begin{aligned}\Big(\frac{\partial}{\partial t} + \widetilde{\mathcal{A}}\Big) g^{\mathrm{cds}} &- (x_1 + x_2 + x_3)\, g^{\mathrm{cds}} \\ &+ g^{\mathrm{cds}}\big(t, x + w_1, (1,0,0)\big) x_1 + g^{\mathrm{cds}}\big(t, x + w_2, (0,1,0)\big) x_2 \\ &+ \Phi\big(t, x, (0,0,0)\big) = 0,\end{aligned} \tag{6.16}$$

with $g^{\mathrm{cds}} = g^{\mathrm{cds}}(t,x,(0,0,0))$ and, by (5.39),

$$\begin{aligned}\Phi\big(t,x,(0,0,0)\big) = L_3\big((0,0,1)\big)\Big(&b_1 F_1^{\mathrm{cds}}\big(t, x + w_3, (0,0,1)\big) \\ &+ b_2 F_2^{\mathrm{cds}}\big(t, x + w_3, (0,0,1)\big)\Big)^+ x_3.\end{aligned}$$

The prices in the two sources are read at states with $z_3 = 1$, and there they do not depend on x_3: the killing rate (5.25) and the source (5.26) involve the surviving names only, the counterparty is not among them, and under (5.11) the other components of $\widetilde{X}$ do not depend on its starting point, so that the representation (5.28) involves (x_1, x_2) alone, by downward induction over the states with $z_3 = 1$. At $(0,0,1)$ the price of the first contract solves

$$\Big(\frac{\partial}{\partial t} + \widetilde{\mathcal{A}}\Big) F_1^{\mathrm{cds}} - (x_1 + x_2)\, F_1^{\mathrm{cds}} + L_1\big((1,0,1)\big) x_1 + F_1^{\mathrm{cds}}\big(t, x + w_2, (0,1,1)\big) x_2 - \varepsilon_1 = 0, \tag{6.17}$$

with $F_1^{\mathrm{cds}} = F_1^{\mathrm{cds}}(t,x,(0,0,1))$, and $F_1^{\mathrm{cds}}(\cdot,\cdot,(0,1,1))$, where name 1 is the only survivor, depends on x_1 alone, so that the shift w_2 acts on it as $x_1 \mapsto x_1 + w_{12}$; symmetrically for the second contract.

The three blocks of (6.14) therefore enter at three different places.

- w^{E} enters the sources of (6.15) and (6.16), through the shift $x + w_3$ at which the prices are read: at every state in which the counterparty is alive, the exposure at its default is priced at intensities raised by w_{13} and w_{23}. This is (5.33), and w^{E} is the only block that moves the intensities at the instant of the counterparty's default.

- w^{I} enters one default later. It enters the prices in the source of (6.16), through the successor term of (6.17): a default of name 2 after the counterparty's raises the intensity of name 1 by w_{12}, and conversely. And it enters the successor terms of (6.16), through the second component of w_1 and the first of w_2: the default of one reference entity raises the intensity of the other, whose contract is then the whole netting set.

- w^{C} enters only the successor terms of (6.16), through the third components of w_1 and w_2: after the default of one reference entity the counterparty's intensity, which is the rate of the trigger in (6.15), is raised while a single contract remains. It does not enter the prices in the sources, which do not depend on x_3, and in the source of (6.16) the counterparty's intensity is read unshifted.

Loss rates and portfolios

The loss rates are either constant, $L_i = 0.60$ for $i = 1, 2, 3$, or state-dependent for the reference entities,

$$L_i(z) = 0.60 + 0.08 \sum_{j \neq i} z_j, \quad i = 1, 2, \qquad L_3 = 0.60, \tag{6.18}$$

which satisfies $L_i(z^j) > L_i(z)$ for $i = 1, 2$, $j \neq i$ and $z_j = 0$, the ordering of Remark 5.11; at $N = 1$ the same rule is the rate of Section 6.3. The counterparty's rate is held constant because in (5.39) it multiplies the exposure: a state-dependent one would raise the adjustment after the default of a reference entity through the counterparty's own loss rate, which is not a dependence of the exposure on the counterparty's default. Two portfolios are held, with weight vector $b = (b_1, b_2)$ equal to $(1, 1)$, protection bought from the counterparty on both reference entities, and to $(1, -1)$, protection bought on name 1 and sold on name 2.

The grid

At $w = \bar{w}$ with constant loss rates, the adjustment of $b = (1, 1)$ is 7.8273, 7.8133 and 7.8069 bp at $n_x = 25$, 33 and 41. The successive changes, 0.0140 and 0.0065 bp, are in the ratio 0.462; for these spacings, convergence at second order in h_i gives 0.463, and at first order 0.600. The rate is measured, not guaranteed: the error bounds obtained for monotone schemes range from order 1/27 to order 1/2 in the mesh size, and the higher rates observed when central differencing is used at most nodes cannot be guaranteed (Wang and Forsyth, 2008, Remark 4.1 and Section 9). Extrapolation at second order from 33 and 41 points puts the limit at 7.7954 bp, and the error of the value at 33 points at 0.0179 bp, 0.23% of it. Refining the grid in a fixed box does not reveal a box too narrow for the shift, and the box is checked separately: widened from 9 to 13 long-run standard deviations in (6.3), with $n_x = 44$ so that the spacings h_i change by 1.5 to 2.2%, it gives 7.8133 bp, the value on the box of Table 6.2 to four decimals. This shows that the box is wide enough for the shift; how close the two values come is no measure of the error, which the refinement measures. Every name has $\sigma_i^2 > \kappa_i$, and $\nu_i h_i \leqslant \sigma_i^2$ on every grid solved at $N = 2$, so that, as at $N = 1$, no interior node is on the upwind branch. The grid has $n_x = 33$ points per axis on the box of $\bar{w}$, and the bound (6.10) gives $n_t = 2234$ time steps for every contagion matrix solved on it.

A simulation check

The simulation of Section 6.3 evaluates the price after the counterparty's default by the closed form (5.44). At $N = 2$ that price, at $(0, 0, 1)$, solves (6.17), whose successor term reads F_1^{cds} at $(0, 1, 1)$. There name 1 is the only survivor: the killing rate is x_1, the one successor $(1, 1, 1)$ carries the constant L_1, and with constant loss rates $F_1^{\mathrm{cds}}(\cdot, \cdot, (0, 1, 1)) = \varphi_1$ for every w, as in the first step of the proof of Proposition 6.2 (i). The successor term of (6.17) is therefore $\varphi_1(t, x_1 + w_{12})\, x_2$. With $w_{12} \neq 0$, (6.17) carries it as (5.45) carries its own and reduces no further: a simulation of (5.34) would need the price at every simulated default of the counterparty, for instance by a second simulation or by the solution of (6.17), and in the second case it would no longer be independent of the scheme. With $w_{12} = 0$ the successor term is $\varphi_1(t, x_1)\, x_2$, which cancels the term $-x_2\, \varphi_1(t, x_1)$ of the killing term, and the inductive step of the same proof gives $F_1^{\mathrm{cds}}(\cdot, \cdot, (0, 0, 1)) = \varphi_1$, whatever the weights w_{13}, w_{23}, w_{31} and w_{32}; symmetrically for the second contract when $w_{21} = 0$. For a contagion matrix with $w^{\mathrm{I}} = 0$ and constant loss rates the exposure (5.33) is therefore

in closed form,

$$\varepsilon_{\bar{N}}(\tau_3, T) = \sum_{i=1}^{2} b_i \, \mathbf{1}_{\{\tau_i > \tau_3\}} \, \varphi_i\big(\tau_3, X_i(\tau_3)\big),$$

with $X_i(\tau_3)$ read after the shift, and the simulation of Section 6.3 estimates the adjustment by the sample mean of $L_3 \, \mathbf{1}_{\{\tau_3 \leqslant T\}} \, (\varepsilon_{\bar{N}}(\tau_3, T))^+$. It checks the scheme in three dimensions, over the eight default states, and with the positive part acting on two contracts.

The check is run at $w = 0$ and at $w = w^{\mathrm{E}} + w^{\mathrm{C}}$, which carries 4.4507 of the 4.6945 bp by which contagion raises the adjustment of $b = (1, 1)$ (Table 6.8), for both portfolios. The simulation uses $M = 2\,000\,000$ paths at $n_{\mathrm{MC}} = 1000$ and 4000, the paths shared between the two step counts as in Section 6.3 and between the two matrices. Table 6.3 gives the results.

Table 6.3: The adjustment at $N = 2$ for contagion matrices without w^{I}, in bp, constant loss rates: finite differences, $n_x = 33$ on the box of $\bar{w}$, against the simulation with $M = 2\,000\,000$ paths at two step counts, on shared paths. The $\pm$ is one sampling standard error.

			Monte Carlo	
w	Portfolio	Finite differences	$n_{\mathrm{MC}} = 1000$	$n_{\mathrm{MC}} = 4000$
0	$b = (1, 1)$	3.1188	3.1053 ± 0.0124	3.0991 ± 0.0124
	$b = (1, -1)$	3.0833	3.0863 ± 0.0110	3.0836 ± 0.0110
$w^{\mathrm{E}} + w^{\mathrm{C}}$	$b = (1, 1)$	7.5695	7.5435 ± 0.0205	7.5416 ± 0.0204
	$b = (1, -1)$	3.3147	3.3182 ± 0.0114	3.3158 ± 0.0114

All eight estimates lie within 1.6 standard errors of the finite-difference values. For $b = (1, 1)$ they lie below them at both matrices and both step counts, by 1.1 to 1.6 standard errors; a grid error of that sign is what the extrapolation above finds at $w = \bar{w}$, where the value at 33 points exceeds its limit by 0.0179 bp. The difference between the two step counts, estimated path by path, is at most 0.0062 bp; that largest one, for $b = (1, 1)$ at $w = 0$, is 2.6 of its own standard errors from zero, and the other three are within 1.3. With the same paths at both matrices, the contagion increment is estimated path by path as well: at 4000 steps it is 4.4425 ± 0.0103 bp for $b = (1, 1)$ and 0.2322 ± 0.0019 bp for $b = (1, -1)$, against 4.4507 and 0.2314 bp by finite differences, 0.80 and 0.38 standard errors apart.

6.5 CVA under contagion against the $w = 0$ benchmark

Table 6.4 and Figure 6.1 give the adjustment (6.1) with and without contagion, for both specifications of the loss rates and both portfolios. By Proposition 6.2, the first row is the adjustment under independence (6.12).

Table 6.4: The adjustment $g^{\text{cds}}(0, \chi, (0, 0, 0))$ at $N = 2$, in bp, with its change relative to the benchmark of the first row. State-dependent loss rates are those of (6.18). Grid of $n_x = 33$ points per axis, on the box of $\bar{w}$ for every row.

		$b = (1, 1)$		$b = (1, -1)$	
w	Loss rates	CVA	change	CVA	change
0	constant	3.1188	—	3.0833	—
$\bar{w}$	constant	7.8133	+150.5%	3.3554	+8.8%
0	state-dependent	8.3679	+168.3%	3.1763	+3.0%
$\bar{w}$	state-dependent	15.9817	+412.4%	3.6369	+18.0%

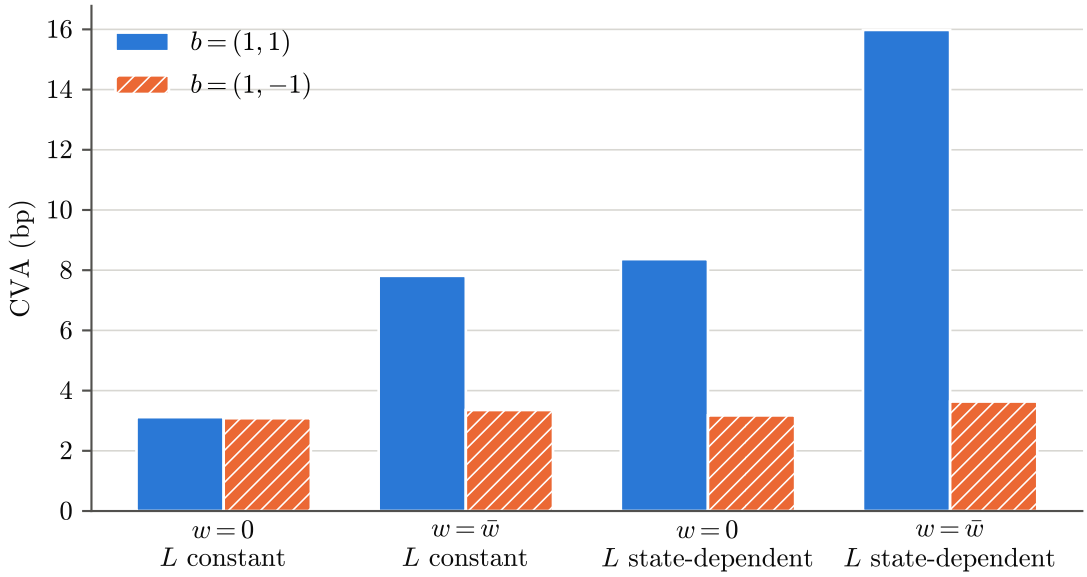


Figure 6.1: The adjustment of Table 6.4, with contagion ($w = \bar{w}$) and without ($w = 0$), for constant and for state-dependent loss rates.

Contagion

For $b = (1, 1)$ and constant loss rates, contagion raises the adjustment from 3.1188 to 7.8133 bp, by 150.5%, to 2.5 times its value without contagion. Of the three

blocks, w^{E} is the only one that moves the intensities at the counterparty's default (Section 6.4): with it, both contracts are priced at intensities raised by w_{13} and w_{23}. At $N = 1$ the price after the counterparty's default is non-decreasing in the intensity of the reference entity, as the proof of Proposition 5.21 shows. At $N = 2$ this is not proved, and it is checked on the grid: at $w = \bar{w}$, F_1^{cds} and F_2^{cds} at $(0, 0, 1)$ are non-decreasing in both intensities, at every node and every time level of the scheme, up to rounding errors of order 10^{-19}. The exposure at the counterparty's default is therefore at least the value of the contracts at the intensities just before that default: it increases in the event in which the counterparty defaults, which is wrong way risk in the sense of Definition 3.1. The direction of the effect is a theorem at $N = 1$, for a single weight and a bought position (Proposition 5.21); at $N = 2$ it is a property of the configurations computed here.

Severity

With state-dependent loss rates and no contagion the default times are independent, and the adjustment rises from 3.1188 to 8.3679 bp, by 168.3%. At the counterparty's default the loss rates of the surviving reference entities rise from 0.60 to 0.68, by (6.18), against premium rates fixed at par for 0.60. As at $N = 1$, this raises the price of each contract after the counterparty's default by the increase in its loss rate times the probability that its reference entity defaults over the remaining life of the contract, the default of the other reference entity raising the loss rate further, to 0.76. The second channel of Remark 5.11 acts here without any dependence between the default times, and for $b = (1, 1)$, at these parameters, it is the larger of the two.

Both channels

With both channels the adjustment is 15.9817 bp, 412.4% above the benchmark. The joint increase, 12.8629 bp, exceeds the sum of the two separate increases, 9.9437 bp, by 29%. At $N = 1$ one reason is visible in (5.48): the default leg of the price after the counterparty's default is the product of the loss rate and of the probability that the reference entity defaults before maturity. Contagion raises the probability, the state-dependent loss rate raises the loss rate, and raising both raises the product by more than the sum of the two separate increases, by the product of the two increments, which belongs to neither channel alone.

The opposite-sign portfolio

For $b = (1, -1)$ the same three changes are +8.8%, +3.0% and +18.0%, and contagion is the larger of the two channels. At the counterparty's default the shift w_3 raises both prices, F_1^{cds} and F_2^{cds}; in the netted exposure $F_1^{\text{cds}} - F_2^{\text{cds}}$ the two increases largely offset, and so do the two increases in the loss rates.

The counterparty's loss rate

In (6.18) the rate of the counterparty is constant. Table 6.5 gives the adjustment when it follows the rule of the reference entities as well, $L_3(z) = 0.60 + 0.08\,(z_1 + z_2)$. The prices F_1^{cds} and F_2^{cds} do not involve L_3, and in the source of (6.16) the factor $L_3((0, 0, 1))$ is still 0.60. The only factor the rule changes is $L_3((1, 0, 1))$ in (6.15), from 0.60 to 0.68, and its counterpart at $(0, 1, 0)$: it acts after the default of a reference entity, on a netting set reduced to one contract.

Table 6.5: The adjustment $g^{\text{cds}}(0, \chi, (0, 0, 0))$ at $N = 2$, in bp, with the loss rates of (6.18) and with the counterparty's rate replaced by $L_3(z) = 0.60 + 0.08\,(z_1 + z_2)$; their difference, and its share of the increase over constant loss rates when all three rates are state-dependent. Grid of $n_x = 33$ points per axis, on the box of $\bar{w}$ for every row.

b	w	$L_3 = 0.60$	L_3 state-dependent	difference	share
$(1, 1)$	0	8.3679	8.4397	+0.0718	1.3%
$(1, 1)$	$\bar{w}$	15.9817	16.1220	+0.1403	1.7%
$(1, -1)$	0	3.1763	3.2163	+0.0400	30.1%
$(1, -1)$	$\bar{w}$	3.6369	3.7132	+0.0763	21.3%

For $b = (1, 1)$ the rate of the counterparty adds 0.0718 bp to the adjustment without contagion and 0.1403 bp with it, 1.3% and 1.7% of the increase over constant loss rates when all three are state-dependent.

For $b = (1, -1)$ it adds 0.0400 and 0.0763 bp, of the same order, but 30.1% and 21.3% of the increase. Without contagion the rates of (6.18) raise the adjustment of this portfolio by 0.0930 bp, against 5.2491 bp for $b = (1, 1)$: the increases in the loss rates of the two contracts largely offset in $F_1^{\text{cds}} - F_2^{\text{cds}}$. In (6.15) the increase of L_3 multiplies the exposure of a single contract, which no other contract offsets.

6.6 Netting under contagion

The gross adjustment (5.36) is the sum of the adjustments of the contracts taken one at a time, each a netting set of its own. By Proposition 5.19 applied to each contract,

it is computed by the scheme of Section 6.1 with the source (5.39) replaced by

$$L_{N+1}(z^{N+1}) \sum_{i=1}^{N} \big(b_i(1-z_i)\, F_i^{\mathrm{cds}}\big(t, x+w_{N+1}, z^{N+1}\big)\big)^+ (1-z_{N+1})\, x_{N+1},$$

the positive part acting on each contract instead of on the netting set. Table 6.6 and Figure 6.2 compare the two, with constant loss rates. In every row the netted adjustment lies below the gross one, as (5.37) requires.

Table 6.6: Netted adjustment (6.1) and gross adjustment (5.36) at $N = 2$, in bp, constant loss rates, and the reduction due to netting, $(\mathrm{CVA}^{\mathrm{gross}}_{\bar N} - \mathrm{CVA}_{\bar N})/\mathrm{CVA}^{\mathrm{gross}}_{\bar N}$. Grid of $n_x = 33$ points per axis, on the box of $\bar w$.

w	Portfolio	Netted	Gross	Reduction
0	$b = (1, 1)$	3.1188	4.2506	26.6%
	$b = (1, -1)$	3.0833	4.3128	28.5%
$\bar w$	$b = (1, 1)$	7.8133	8.4462	7.5%
	$b = (1, -1)$	3.3554	5.0440	33.5%

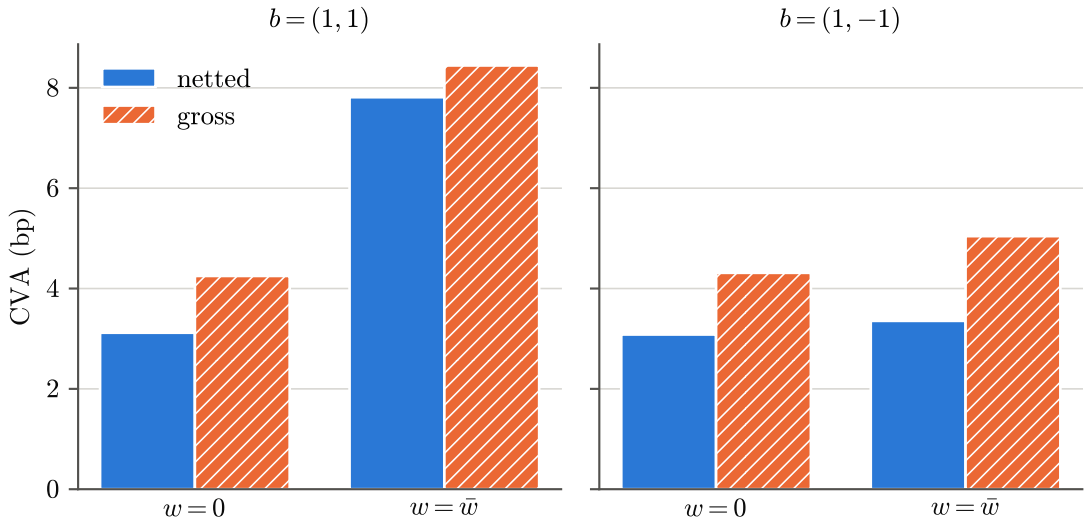


Figure 6.2: Netted and gross adjustment of Table 6.6, without contagion ($w = 0$) and with it ($w = \bar w$), for the two portfolios.

By (5.37) and the condition for equality stated with it, netting reduces the adjustment only on the part of the triggering event on which the values $b_i S_i(\tau_3, T)$ of the contracts have opposite signs. Without contagion, by Proposition 6.2, these values are $b_i \varphi_i(\tau_3, X_i(\tau_3))$, with X_1 and X_2 independent, and each takes both signs: the contracts were entered at par, and by (2.12) a position is worth a positive amount

to the protection buyer exactly when the spread of its reference entity has widened since inception. The values have opposite signs on part of the triggering event for either portfolio, and netting removes 26.6% and 28.5% of the gross adjustment.

With contagion, the shift w_3 raises both intensities at the counterparty's default, and with them, on the grid of Section 6.5, both prices F_1^{cds} and F_2^{cds}. For $b = (1, 1)$ it moves the two values in the same direction, towards a common positive sign, and the reduction due to netting falls to 7.5%: the netted adjustment rises by 150.5%, the gross one by 98.7%. For $b = (1, -1)$ it moves $b_1 F_1^{\text{cds}}$ up and $b_2 F_2^{\text{cds}} = -F_2^{\text{cds}}$ down, towards opposite signs, and the reduction rises to 33.5%: the netted adjustment rises by 8.8%, the gross one by 17.0%.

6.7 Sensitivity in the contagion weights

The level of contagion

The whole matrix is scaled, $w = \eta\,\bar{w}$ with $\eta = 0, 0.5, \ldots, 3$, at constant loss rates. At $\eta = 3$ the default of the counterparty raises the intensity of name 1 by 0.030, one and a half times its long-run level. Table 6.7 and Figure 6.3 give the adjustment. The seven values are computed on one grid, $n_x = 33$ points per axis on the box of $3\bar{w}$, so that no step between consecutive values of η carries a change of grid; the bound (6.10) gives $n_t = 2044$. On this box the values at $\eta = 1$ are 7.8039 bp for $b = (1, 1)$ and 3.3702 bp for $b = (1, -1)$, against 7.8133 and 3.3554 bp on the box of $\bar{w}$ in Table 6.4.

Table 6.7: The adjustment at $N = 2$, in bp, for $w = \eta\,\bar{w}$ and constant loss rates, with the step from the previous value of η and the gap between the two portfolios. Grid of $n_x = 33$ points per axis, on the box of $3\bar{w}$ for every row.

	$b = (1, 1)$		$b = (1, -1)$		
η	CVA	step	CVA	step	gap
0.0	3.1300	—	3.0998	—	0.0302
0.5	5.1147	+1.9847	3.2167	+0.1169	1.8980
1.0	7.8039	+2.6892	3.3702	+0.1535	4.4336
1.5	11.1235	+3.3196	3.5588	+0.1886	7.5647
2.0	14.7695	+3.6460	3.7801	+0.2213	10.9893
2.5	18.5186	+3.7491	4.0239	+0.2438	14.4947
3.0	22.2846	+3.7661	4.2813	+0.2574	18.0033

Both sequences are non-decreasing in η on the grid computed. At $N = 1$ the monotonicity of the adjustment in the one weight that reaches it is Proposition 5.21;

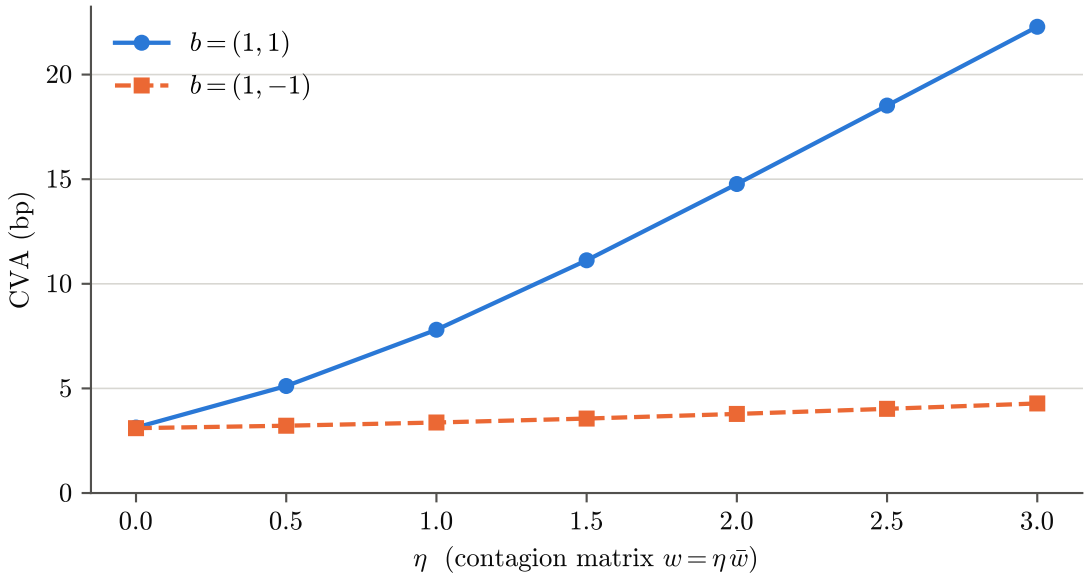


Figure 6.3: The adjustment of Table 6.7 as a function of the scale η of the contagion matrix, $w = \eta\,\bar{w}$, constant loss rates. All seven points are computed on one grid, $n_x = 33$ points per axis on the box of $3\bar{w}$.

at $N = 2$, with the six weights scaled together, it is a property of the configurations computed. For $b = (1, 1)$ the steps increase at every value of η, by less and less, from 1.9847 to 3.7661 bp, and the adjustment at $\eta = 3$ is 22.2846 bp. For $b = (1, -1)$ it rises from 3.0998 to 4.2813 bp, its steps increasing as well, and the gap between the two portfolios widens from 0.0302 to 18.0033 bp: the shift of the prices at the counterparty's default adds to both values of $b = (1, 1)$ and is largely offset between the two values of $b = (1, -1)$, as in Section 6.5.

The three blocks

The blocks of (6.14) are switched on one at a time and in combination, for $b = (1, 1)$ and constant loss rates, on the grid of 33 points and the box of $\bar{w}$. Table 6.8 gives the adjustment.

The increments are ordered as the places at which the blocks enter the equations of Section 6.4. The block w^{E}, which moves the intensities at the counterparty's default, accounts alone for 4.4195 of the 4.6945 bp. The block w^{I}, which acts only after a further default, adds 0.1345 bp alone. The block w^{C}, which acts only after a reference entity has defaulted, and then only on the trigger of a netting set reduced to one contract, adds 0.0167 bp.

The increments are not additive. Taken one at a time they sum to 4.5708 bp,

Table 6.8: The adjustment at $N = 2$ for $b = (1, 1)$ and constant loss rates, in bp, with the blocks of (6.14) switched on alone and in combination, and the increment over the benchmark $w = 0$. Grid of $n_x = 33$ points per axis, on the box of $\bar{w}$ for every row.

w	CVA	increment	change
0	3.1188	—	—
w^{E}	7.5383	+4.4195	+141.7%
w^{C}	3.1355	+0.0167	+0.5%
w^{I}	3.2533	+0.1345	+4.3%
$w^{\mathrm{E}} + w^{\mathrm{C}}$	7.5695	+4.4507	+142.7%
$\bar{w} = w^{\mathrm{E}} + w^{\mathrm{C}} + w^{\mathrm{I}}$	7.8133	+4.6945	+150.5%

against a joint increment of 4.6945 bp, 2.7% more. The blocks w^{E} and w^{C} are nearly additive between themselves, 4.4363 bp summed against 4.4507 bp jointly, 0.3% apart; the rest of the difference lies with w^{I}, which adds 0.1345 bp alone and 0.2438 bp on top of $w^{\mathrm{E}} + w^{\mathrm{C}}$, 1.8 times as much. The equations of Section 6.4 locate the interaction. In (6.17) the shift $x_1 \mapsto x_1 + w_{12}$ of w^{I} is reached at the rate x_2 at which name 2 defaults after the counterparty, and in the source of (6.16) that price is read at $x + w_3$, where x_2 has been raised by the entry w_{23} of w^{E}; symmetrically for the second contract. The block w^{E} raises the rate at which w^{I} acts, and the contribution of w^{I} depends on whether w^{E} is present.

Chapter 7

Conclusions

7.1 Contagion in the adjustment

Under contagion the exposure is priced after the jump that the default of the counterparty causes. At that default the intensities move from x to $x + w_{N+1}$ and the default state from z to z^{N+1}. Which contracts are alive is read before the jump, and what they are worth after it (5.33). The exposure therefore depends on the counterparty's default through the state, and not only through its date. The step that gives the formula under independence, conditioning on that date and integrating an expected exposure against the law of the default time, is unavailable (Remark 5.18). In its place the adjustment is the solution of a Cauchy problem of its own over the default states, driven by the pricing functions of the contracts (Proposition 5.19).

Both objects are in the reference paper. It writes the exposure in the form (5.33) (Bo and Ceci, 2020, Lemma 3.6), and it reaches the Cauchy problem from its hedging problem, for the conditional expectation of the loss at the counterparty's default under its measure for hedging (Bo and Ceci, 2020, Sections 3.5 and 4.1). With zero risk premia that measure is the pricing measure, and before the counterparty's default the expectation is the adjustment. What is not in the reference paper is the derivation of the identification from the definition of the adjustment alone, without the hedging problem.

At one reference entity the direction of the effect is a theorem. The adjustment of a bought position is non-decreasing in the weight w_{12} by which the default of the counterparty raises the intensity of the reference entity, and that of a sold position is non-increasing (Proposition 5.21). The adjustment computed without contagion is therefore a lower bound for the protection buyer and an upper bound for the protection seller. The proof shows where the effect comes from. After the counterparty's default the price of the contract is non-decreasing in the intensity

of the reference entity (5.48), and contagion raises that intensity at the instant the counterparty defaults. The protection gains value in the event in which the counterparty fails to deliver it. This is wrong way risk in the sense of Definition 3.1, and it is a loss only for the party owed the protection. The proposition is not in the reference paper, and it gives the direction of the effect, not its size.

At two reference entities the size is computed, at the parameters of Table 6.2 and the contagion matrix (6.14), with the two channels of wrong way risk of Remark 5.11 measured separately. For protection bought from the counterparty on both reference entities, the adjustment without contagion and with constant loss rates is 3.1188 bp. Contagion alone raises it to 7.8133 bp, by 150.5%. Loss rates that rise with each default (6.18), with the default times independent, raise it to 8.3679 bp, by 168.3%. The two channels together raise it to 15.9817 bp, by 412.4% (Table 6.4), and the joint increase exceeds the sum of the two separate ones by 29%. At one reference entity one reason for the excess is visible in (5.48). After the counterparty's default, the default leg of the price is the product of a loss rate and of the probability that the reference entity defaults before maturity. Contagion raises the second factor, the state-dependent loss rates raise the first, and the product of the two increments belongs to neither channel alone.

The adjustment rises with the level of contagion at every value computed. With the reference matrix scaled by η from 0 to 3, it goes from 3.1300 to 22.2846 bp, on the box of $3\bar{w}$ common to the whole sweep (Table 6.7).

Almost all of the effect arises at the counterparty's default. Of the 4.6945 bp that the reference matrix adds for this portfolio, the block w^{E} alone adds 4.4195 (Table 6.8). It is the only block that moves the intensities at that default, which is where the exposure is priced. The blocks w^{C} and w^{I} act only after the default of a reference entity, and alone add 0.0167 and 0.1345 bp.

When protection is bought on one reference entity and sold on the other, the effect is much smaller: the same three increases are 8.8%, 3.0% and 18.0% (Table 6.4). The jump at the counterparty's default raises both prices. In the netted exposure the second contract enters with a negative weight, and the two increases largely offset.

Contagion also changes how much netting reduces the adjustment. The netted adjustment lies below the gross adjustment (5.36), computed contract by contract, and the two differ only on the part of the triggering event on which the values of the contracts have opposite signs (5.37). Without contagion the two values are independent and each takes both signs, and netting removes 26.6% and 28.5% of the gross adjustment of the two portfolios. With contagion the jump pushes both prices up. For the portfolio bought on both reference entities it moves the two values

towards a common positive sign, and the reduction falls to 7.5%. For the other it moves them towards opposite signs, and the reduction rises to 33.5% (Table 6.6). The comparison of the netted with the gross adjustment is not in the reference paper, which defines the netted exposure but not the gross adjustment.

These numbers come from a numerical solution of the systems, which the reference paper does not treat. It closes on the remark that the probabilistic representation of its quantities makes it possible to approximate its hedging strategy by Monte Carlo simulation (Bo and Ceci, 2020, Section 4.3). Here the systems are solved by an explicit finite-difference scheme (Section 6.1), validated at one reference entity against the closed forms and a simulation (Section 6.3).

7.2 Simplifications and extensions

Three choices ensure that the gap between the adjustment with contagion and the adjustment without it is due to contagion alone. Each intensity has its own Brownian motion (5.11), so that the default times depend on each other only through the jumps. With the contagion weights at zero they are independent, and with constant loss rates the model then computes the adjustment under independence (Proposition 6.2), a case the reference paper does not treat. The loss rates are constant in the base case, which switches off the second channel of Remark 5.11. The state-dependent rates (6.18) are measured apart, and in them the rate of the counterparty is held constant, since it multiplies the exposure without making it depend on the counterparty's default. The premium rates are fixed at par in the model without contagion (6.13) and held fixed in every configuration, so that a change in the weights or in the loss rates changes the value of the contracts and not their terms. Within the contagion matrix, switching the blocks on one at a time locates where the effect arises (Table 6.8).

The other simplifications reduce the problem rather than isolate the effect. The adjustment is unilateral and uncollateralized, the interest rate and the risk premia are zero, and the systems, stated for any number of reference entities, are solved for one and two (Section 1.4).

The parameters of Chapter 6 are illustrative, and none is estimated from market data. Under martingale modelling the parameters of a pricing model are determined by calibration: the prices of traded securities computed by the model at time zero are required to coincide with those observed in the market (McNeil et al., 2015, Section 10.4.2). Hazard rate models are calibrated in this way to quoted CDS spreads (McNeil et al., 2015, Section 10.4.4). In the model of Chapter 5 those spreads depend on the

contagion weights. The price of a contract before any default depends on them, since the Cauchy problem it solves reads the prices after each default at the intensities shifted by w_j (Proposition 5.15): the spreads of the reference entities anticipate the jumps. Whether observed spreads determine the weights is not established here. In bottom-up models portfolio products can be priced consistently with observed single-name CDS spreads, and pricing and calibration pose computational challenges, particularly for large portfolios (McNeil et al., 2015, Section 17.1.2).

The contagion weights are varied rather than fixed. The evidence on default contagion runs both ways (Section 4.6), and in the two-firm example of Bo et al. (2019, Section 5) the weights, 0.2 in both directions, are chosen for a scenario in which the two firms belong to the same industrial sector, in an optimal allocation problem. In Chapter 6 the weights are scaled together by η from 0 to 3, up to a jump at the counterparty's default of one and a half times the long-run level of name 1 (Table 6.7), and switched on block by block (Table 6.8).

The reference paper drives the intensities with a common multidimensional Brownian motion through a full diffusion matrix (Bo and Ceci, 2020, Section 2.2), which (5.11) specializes to one Brownian motion per name (Remark 5.6). With a full matrix the intensities are dependent between defaults, and the default times are dependent even with the contagion weights at zero. Proposition 6.2 then no longer identifies the model without contagion with the adjustment under independence, and the comparison of Chapter 6 would measure contagion on top of the dependence generated by a common disturbance, present in both of the models compared.

The positions are uncollateralized. Under collateralization the amount available at the default of the counterparty is the collateral posted immediately before it. The distinction matters when the close-out amount jumps at that default, for instance through contagion; when the value of the contract does not jump there, collateral equal to its market value removes the adjustment (McNeil et al., 2015, Section 17.2.2). Both cases occur in the model of Chapter 5. Without contagion and with constant loss rates the price of each contract depends on the intensity of its own reference entity alone (Proposition 6.2 (i)). That intensity does not jump at the counterparty's default, and neither does the price. With contagion the price moves there from its value at (x, z) to its value at $(x + w_{N+1}, z^{N+1})$ (5.33), and the difference is the part of the exposure that collateral posted before the default does not cover. For a single CDS, in the example of Frey and Rösler (2014), collateral equal to the market value removes the collateralized adjustment in a model without contagion and leaves 35 bp in a version with contagion, the joint law of the default times being the same in the two (McNeil et al., 2015, Section 17.4.4, Table 17.3).

The reference paper also derives a strategy hedging the adjustment with a CDS written on the counterparty, held up to the earlier of the maturity and the counterparty's default (Bo and Ceci, 2020, Sections 1 and 3.2). That contract is the claim of (5.16) written on the name $N+1$. Its price F^{cds}_{N+1} solves the pricing system (5.23), and is not computed here. The hedge is taken under the real-world measure, and requires a measure for hedging besides the pricing measure (Bo and Ceci, 2020, Section 1); with the risk premia equal to zero, as here, both coincide with the real-world measure (Bo and Ceci, 2020, Section 3.3).

The systems are solved for at most two reference entities because the scheme carries a grid in $N+1$ dimensions at each of the 2^{N+1} default states (Section 1.4). Credit value adjustments are mostly computed by Monte Carlo simulation (McNeil et al., 2015, Section 17.4.4), which carries no grid. Under contagion, however, the exposure at the counterparty's default is a price read at the shifted intensities (5.33). A simulation of the adjustment needs that price at every simulated default of the counterparty, for instance by a second simulation or by the solution of the pricing systems (Section 6.4). At $N = 2$ the price is in closed form when the reference entities do not act on each other's intensities and the loss rates are constant, and the simulation of Chapter 6 is restricted to that case.

7.3 Limits of the results

The results of Section 7.1 hold in the configuration computed, and at two reference entities they are observed rather than proved. Their sizes belong to the parameters of Table 6.2, the matrix (6.14), the rule (6.18) with its step of 0.08, and two portfolios: they measure the effect there, and are not an estimate of contagion in any market. The weights are varied (Section 6.7). The step of the loss rates is not, and neither is the speed of mean reversion, $\nu_i = 1$ for every name, which in (5.10) sets how fast the excess created by a jump decays. Unlike the weights, it also enters the drift between defaults, so that varying it moves the benchmark of Proposition 6.2 as well. At two reference entities, for the block w^{E} and protection bought on both, the argument of Proposition 5.21 has two steps without a proof. The first is the monotonicity of the prices after the counterparty's default in the intensities of the reference entities, which at $w = \bar{w}$ holds at every node and every time level of the scheme (Section 6.5); the second is its passage through the successor terms of (6.16). The monotonicity in η, where the block w^{C} also raises the counterparty's intensity, is observed on the seven values of Table 6.7. The netting results concern two portfolios, contagion lowering the reduction for one and raising it for the other (Table 6.6). The scheme

of Section 6.1, finally, is monotone and bounded in the maximum norm under (6.10), but its convergence is not established. Its error is measured instead: at $N = 2$ the changes of the adjustment over three successive grids are in the ratio that second order in the grid spacing predicts (Section 6.4).

Bibliography

T. Aven. A theorem for determining the compensator of a counting process. *Scandinavian Journal of Statistics*, 12(1):69–72, 1985.

S. Azizpour, K. Giesecke, and G. Schwenkler. Exploring the sources of default clustering. *Journal of Financial Economics*, 129(1):154–183, 2018.

Basel Committee on Banking Supervision. Basel III: A global regulatory framework for more resilient banks and banking systems. Technical report, Bank for International Settlements, 2011.

T. Bielecki, M. Jeanblanc, and M. Rutkowski. Pricing and trading credit default swaps in a hazard process model. *The Annals of Applied Probability*, 18:2495–2529, 2008.

T. R. Bielecki and M. Rutkowski. *Credit Risk: Modeling, Valuation, and Hedging*. Springer, 2002.

T. Björk. *Arbitrage Theory in Continuous Time*. Oxford Finance Series. Oxford University Press, Oxford, 3 edition, 2009.

L. Bo and A. Capponi. Portfolio choice with market-credit risk dependencies. *SIAM Journal on Control and Optimization*, 56(4):3050–3091, 2018.

L. Bo and C. Ceci. Locally risk-minimizing hedging of counterparty risk for portfolio of credit derivatives. *Applied Mathematics & Optimization*, 82(2):799–850, 2020. doi: 10.1007/s00245-018-9549-y.

L. Bo, A. Capponi, and P. C. Chen. Credit portfolio selection with decaying contagion intensities. *Mathematical Finance*, 29(1):137–173, 2019. doi: 10.1111/mafi.12177.

D. Brigo and K. Chourdakis. Counterparty risk for credit default swaps: Impact of spread volatility and default correlation. *International Journal of Theoretical and Applied Finance*, 12:1007–1026, 2009.

D. Brigo, M. Morini, and A. Pallavicini. *Counterparty Credit Risk, Collateral and Funding*. Wiley, 2013.

D. Brigo, A. Capponi, and A. Pallavicini. Arbitrage-free bilateral counterparty risk valuation under collateralization and application to credit default swaps. *Mathematical Finance*, 24:125–146, 2014.

M. Brunnermeier, L. Clerc, and M. Scheicher. Assessing contagion risks in the CDS markets. Financial Stability Review, Technical Report 17, Banque de France, 2013.

C. Ceci. Lecture notes for the course risk management and capital requirements. Lecture notes, Master's Degree in Finance and Insurance (FINASS), Sapienza University of Rome, 2025. A.Y. 2025/2026.

G. Cesari, J. Aquilina, N. Charpillon, Z. Filipović, G. Lee, and I. Manda. *Modelling, Pricing and Hedging Counterparty Credit Exposure: A Technical Guide*. Springer, 2009.

P. Collin-Dufresne, R. S. Goldstein, and J. Hugonnier. A general formula for valuing defaultable securities. *Econometrica*, 72(5):1377–1407, 2004.

P. Collin-Dufresne, R. S. Goldstein, and J. Helwege. Is credit event risk priced? modeling contagion via the updating of beliefs. Technical report, National Bureau of Economic Research, 2010.

R. Cont. Credit default swaps. In R. Cont, editor, *Encyclopedia of Quantitative Finance*, pages 422–424. Wiley, 2010.

D. R. Cox and D. Oakes. *Analysis of Survival Data*. Chapman & Hall, London, 1984.

J. C. Cox, J. E. Ingersoll, and S. A. Ross. A theory of the term structure of interest rates. *Econometrica*, 53(2):385–407, 1985.

M. H. A. Davis and V. Lo. Infectious defaults. *Quantitative Finance*, 1(4):382–387, 2001.

F. Delbaen and W. Schachermayer. *The Mathematics of Arbitrage*. Springer, 2006.

Y. d'Halluin, P. A. Forsyth, and K. R. Vetzal. Robust numerical methods for contingent claims under jump diffusion processes. *IMA Journal of Numerical Analysis*, 25:87–112, 2005.

D. Duffie and R. Kan. A yield-factor model of interest rates. *Mathematical Finance*, 6(4):379–406, 1996.

D. Duffie, J. Pan, and K. Singleton. Transform analysis and asset pricing for affine jump diffusions. *Econometrica*, 68:1343–1376, 2000.

E. Ekström and J. Tysk. Boundary conditions for the single-factor term structure equation. *The Annals of Applied Probability*, 21(1):332–350, 2011. doi: 10.1214/10-AAP698.

R. J. Elliott, M. Jeanblanc, and M. Yor. On models of default risk. *Mathematical Finance*, 10:179–195, 2000.

H. Föllmer and M. Schweizer. Hedging of contingent claims under incomplete information. In M. H. A. Davis and R. J. Elliot, editors, *Applied Stochastic Analysis*, pages 389–414. Gordon and Breach, London, 1991.

H. Föllmer and D. Sondermann. Hedging of non-redundant contingent claims. In W. Hildenbrand and A. Mas-Colell, editors, *Contributions to Mathematical Economics*, pages 205–223. North-Holland, Amsterdam, 1986.

P. A. Forsyth and G. Labahn. Numerical methods for controlled Hamilton–Jacobi–Bellman PDEs in finance. *Journal of Computational Finance*, 11(2):1–44, 2007.

R. Frey and J. Backhaus. Pricing and hedging of portfolio credit derivatives with interacting default intensities. *International Journal of Theoretical and Applied Finance*, 11(6):611–634, 2008.

R. Frey and L. Rösler. Contagion effects and collateralized credit value adjustments for credit default swaps. *International Journal of Theoretical and Applied Finance*, 17(7):1450044, 2014.

P. Glasserman. *Monte Carlo Methods in Financial Engineering*. Springer, 2003.

J. Gregory. *Counterparty Credit Risk: The New Challenge for Global Financial Markets*. Wiley, Chichester, 2010.

D. Heath and M. Schweizer. Martingales versus PDEs in finance: An equivalence result with examples. *Journal of Applied Probability*, 37:947–957, 2000.

A. Herbertsson. Pricing synthetic CDO tranches in a model with default contagion using the matrix-analytic approach. *Journal of Credit Risk*, 4(4):3–35, 2008.

Y. Huang, P. A. Forsyth, and G. Labahn. Methods for pricing American options under regime switching. *SIAM Journal on Scientific Computing*, 33(5):2144–2168, 2011. doi: 10.1137/110820920.

J. C. Hull and A. White. CVA and wrong-way risk. *Financial Analysts Journal*, 68: 58–69, 2012.

J. C. Hull and A. White. Collateral and credit issues in derivative pricing. Preprint, Rotman School of Management, 2013.

S. Janson, S. M'Baye, and P. Protter. Absolutely continuous compensators. *International Journal of Theoretical and Applied Finance*, 14:335–351, 2011.

R. A. Jarrow and S. M. Turnbull. Pricing derivatives on financial securities subject to credit risk. *Journal of Finance*, 50(1):53–85, 1995.

R. A. Jarrow and F. Yu. Counterparty risk and the pricing of defaultable securities. *Journal of Finance*, 56(5):1765–1799, 2001.

R. A. Jarrow, D. Lando, and S. M. Turnbull. A markov model for the term structure of credit risk spreads. *Review of Financial Studies*, 10:481–523, 1997.

I. Karatzas and S. E. Shreve. *Brownian Motion and Stochastic Calculus*. Springer, 1988.

S. Kusuoka. A remark on default risk models. *Advances in Mathematical Economics*, 1:69–81, 1999.

D. Lamberton and B. Lapeyre. *Introduction to Stochastic Calculus Applied to Finance*. Chapman & Hall, London, 1996.

D. Lando. Cox processes and credit risky securities. *Review of Derivatives Research*, 2:99–120, 1998.

D. Lando and M. S. Nielsen. Correlation in corporate defaults: Contagion or conditional independence? *Journal of Financial Intermediation*, 19:355–372, 2010.

A. J. McNeil, R. Frey, and P. Embrechts. *Quantitative Risk Management: Concepts, Techniques and Tools*. Princeton Series in Finance. Princeton University Press, Princeton, NJ, revised edition, 2015.

P. Protter. *Stochastic Integration and Differential Equations*. Springer, New York, 2 edition, 2005.

P. J. Schönbucher. *Credit Derivatives Pricing Models*. Wiley, 2003.

M. Schweizer. A guided tour through quadratic hedging approaches. In E. Jouini, J. Cvitanic, and M. Musiela, editors, *Option Pricing, Interest Rates and Risk Management*, pages 538–574. Cambridge University Press, Cambridge, 2001.

R. M. Stulz. Credit default swaps and the credit crisis. *Journal of Economic Perspectives*, 24(1):73–92, 2010.

J. Wang and P. A. Forsyth. Maximal use of central differencing for Hamilton–Jacobi–Bellman PDEs in finance. *SIAM Journal on Numerical Analysis*, 46(3):1580–1601, 2008. doi: 10.1137/060675186.

F. Yu. Correlated defaults in intensity-based models. *Mathematical Finance*, 17: 155–173, 2007.